MATH 149: Precalculus Lecture Notes

The University of Akron, College of Engineering and Polymer Science, Department of Mathematics

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3.7: Polynomial and Rational Inequalities

Learning goals.

Solve polynomial and rational inequalities; identify critical values; use sign charts; state answers in interval notation.

Recall that multiplying or dividing a linear inequality by a negative number reverses the inequality sign. For nonlinear inequalities, a reliable method is to study the *sign* of an expression on intervals.

Example. Solve the linear inequality
$$-2x + 3 \geq 7.$$

Solution.
$$\begin{aligned} -2x + 3 &\geq 7, \\ -2x &\geq 4, \\ x &\leq -2. \end{aligned}$$

Remember: For linear inequalities, when you multiply or divide by a **negative number**, you must reverse the direction of the inequality sign.

Key idea. Move everything to one side, factor when possible, identify all critical values, and determine the sign on each interval.

Caveat: Nonlinear inequalities cannot be solved directly!

Sign-chart method

Step 1. Rewrite the inequality in the form $f(x) > 0$, $f(x) \geq 0$, $f(x) < 0$, or $f(x) \leq 0$.

Step 2. Find critical values: zeros of the numerator/polynomial and values excluded from the domain.

Step 3. Put the critical values on a number line.

Step 4. Determine the sign of f on each interval.

Step 5. Select the intervals satisfying the original inequality. Include zeros only for $\leq$ or $\geq$; never include denominator zeros.

Example 1. Solve $x^4 + 6x^2 \leq 7x^3$.

Solution. Move all terms left: $x^4 - 7x^3 + 6x^2 \leq 0$. Factor:
$$x^2(x-1)(x-6) \leq 0.$$

Critical values are $0, 1, 6$. Since x^2 has even multiplicity, the sign does not change at 0. Testing intervals gives the solution $[1, 6]$ together with $x = 0$, hence $\{0\} \cup [1, 6]$.

Example 2. Solve
$$\frac{x^2(x+3)(x-5)}{(x+2)^2(x-4)} \geq 0.$$

Solution. Critical values are $-3, -2, 0, 4, 5$. The values -2 and 4 are excluded. Even multiplicities at -2 and 0 do not change sign. A sign chart yields

$$[-3, -2) \cup (-2, 0] \cup (0, 4) \cup [5, \infty),$$

with the adjacent intervals around 0 combined when appropriate after checking signs. (Instructor: verify signs explicitly on the board.)

Domain review. Common restrictions are: denominators cannot be zero, and even-indexed radicals require a nonnegative radicand. If an even root appears in a denominator, its radicand must be strictly positive.

Try it. Find the domain of $f(x) = \sqrt[6]{\dfrac{x^5 - x^3}{x+2}}$.

Solution. Require $\dfrac{x^3(x-1)(x+1)}{x+2} \geq 0$ and $x \neq -2$. Use a sign chart with critical values $-2, -1, 0, 1$.

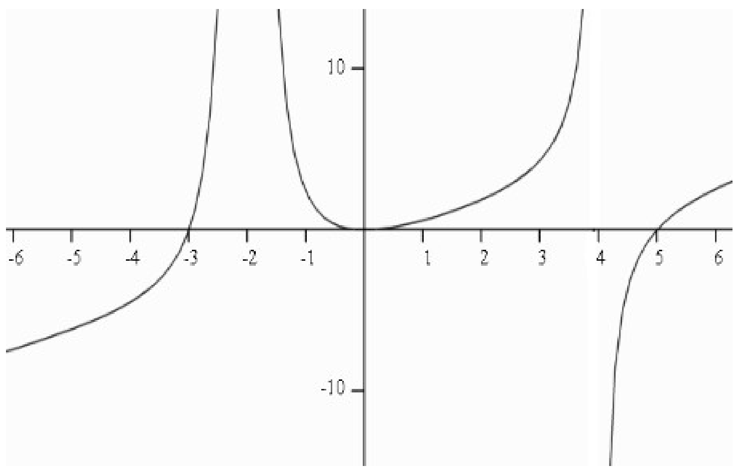


Figure 1: Graph of a rational inequality example

4.1: One-to-One and Inverse Functions

A function is **one-to-one** if distinct inputs produce distinct outputs. Graphically, a function is one-to-one exactly when it passes the **horizontal-line test**.

Example. Determine whether each of the following functions is one-to-one.

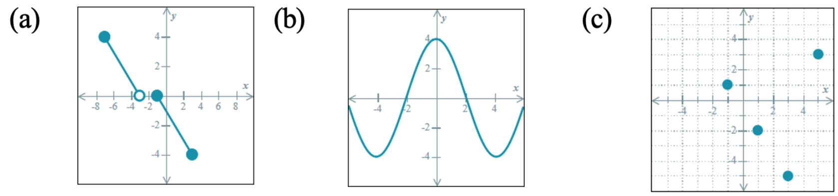


Figure 2: Graphs for determining whether the functions are one-to-one.

Solution.

1. The function is **one-to-one**. Every horizontal line intersects the graph at most once.
2. The function is **not one-to-one**. Some horizontal lines intersect the graph more than once.
3. The function is **one-to-one**. No two points have the same y-coordinate, so every horizontal line intersects the graph at most once.

Each input goes to exactly one output!

Key idea. An inverse reverses the input-output relationship. If (a, b) lies on $y = f(x)$, then (b, a) lies on $y = f^{-1}(x)$.

$$\mathrm{Dom}(f) = \mathrm{Ran}(f^{-1}), \qquad \mathrm{Ran}(f) = \mathrm{Dom}(f^{-1}).$$

Caution. $f^{-1}(x)$ means the inverse function, not $1/f(x)$.

Finding an inverse algebraically

Step 1. Write $y = f(x)$. Replace $f(x)$ with y.

Step 2. Interchange x and y.

Step 3. Solve for y.

Step 4. Rename the result $f^{-1}(x)$. Replace y with $f^{-1}(x)$, and state its domain/range when requested.

Example. Find the inverse of $f(x) = x^2 - 3$ on $[0, \infty)$.

Solution. $x = y^2 - 3$ after interchanging variables, so $y = \sqrt{x+3}$. Therefore $f^{-1}(x) = \sqrt{x+3}$, with domain $[-3, \infty)$ and range $[0, \infty)$.

Verification. Recall: given two functions f and g, the composite function $f \circ g$ is defined by

$$(f \circ g)(x) = f(g(x)).$$

You must use composites to verify algebraically that functions are inverses.

If $f(x)$ is one-to-ine function, then $f^{-1}(x)$ exists. Functions f and g are inverses when

$$(f \circ g)(x) = x \quad \text{and} \quad (g \circ f)(x) = x$$

on the appropriate domains.

Try it. Find f^{-1} for $f(x) = \dfrac{x-3}{2x+1}$ and state the domain of the inverse.

Solution. Interchange and solve: $x = (y-3)/(2y+1)$, hence $y = -(x+3)/(2x-1) = (x+3)/(1-2x)$. Thus $f^{-1}(x) = \dfrac{x+3}{1-2x}$, $x \neq \frac{1}{2}$.

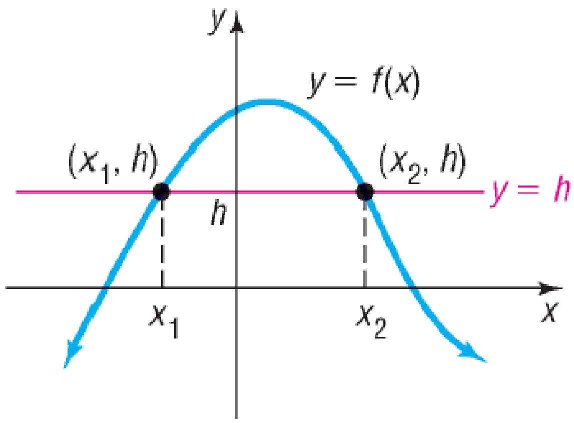


Figure 3: Horizontal-line test for a one-to-one function

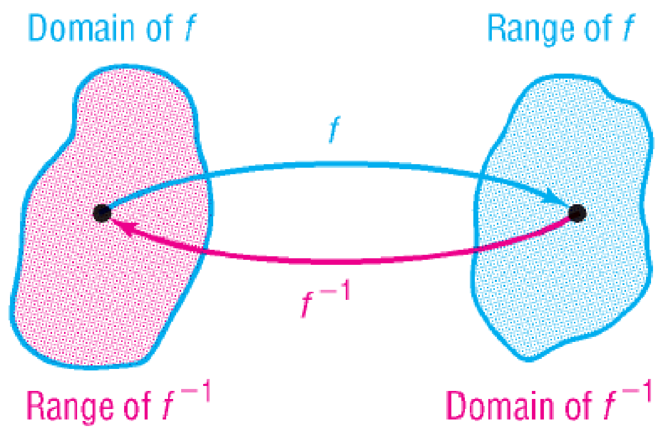


Figure 4: Domain and range relationship for inverse functions

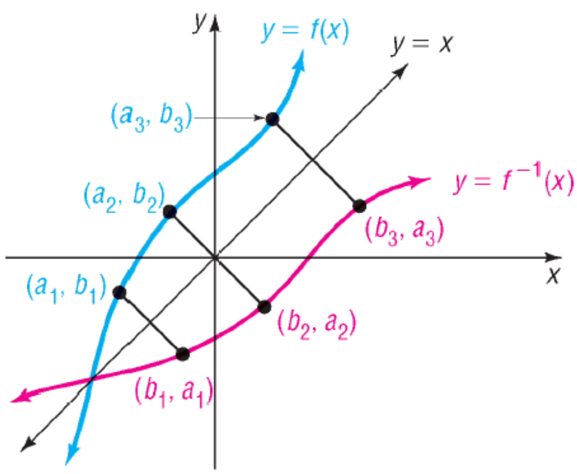


Figure 5: Graphs of a function and its inverse

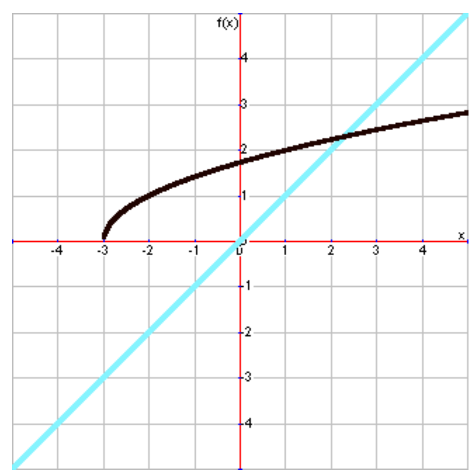


Figure 6: Graphical construction of an inverse function

Graphs of a function and its inverse.

The graph of a function f and its inverse f^{-1} are symmetric about the line $y = x$.

Can you draw the graph of the inverse for Figure 6?

4.2: Exponential Functions

An exponential function has the form

$$f(x) = a^x, \qquad a > 0, \quad a \neq 1.$$

For $a > 1$, the function is increasing; for $0 < a < 1$, it is decreasing.In either case,

$$\text{Dom}(f) = (-\infty, \infty), \quad \text{Ran}(f) = (0, \infty), \quad y = 0 \text{ is a horizontal asymptote.}$$

The graph passes through $(0, 1)$, $(1, a)$, and $(-1, 1/a)$.

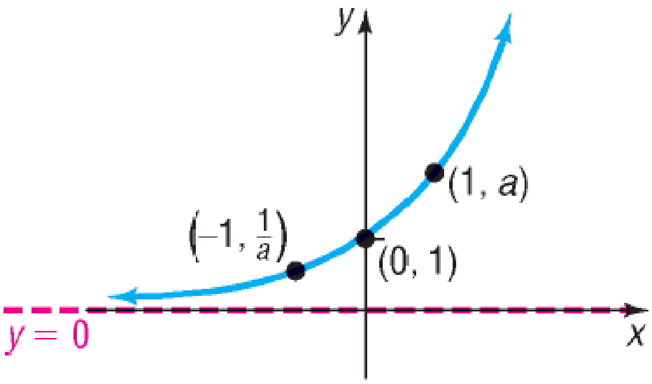


Figure 7: Exponential growth function

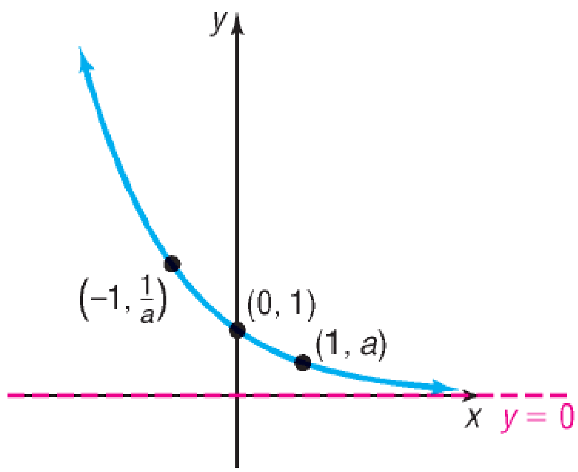


Figure 8: Exponential decay function

Properties of the Exponential Function $f(x) = a^x$, $a > 1$

1. The domain is the set of all real numbers:
$$x \in (-\infty, \infty).$$
2. The range is the set of positive real numbers:
$$y \in (0, \infty).$$
3. The y-intercept is 1, so the graph passes through the point
$$(0, 1).$$
There is no x-intercept.
4. The x-axis, $y = 0$, is the horizontal asymptote. In particular,
$$\lim_{x \to -\infty} a^x = 0.$$
There is no vertical asymptote.
5. The graph contains the points
$$(1, a), \qquad (0, 1), \qquad \left(-1, \frac{1}{a}\right).$$

6. The function is always increasing and its graph is a smooth, continuous curve with no gaps.

Exponent rules.

$$(x^m)(x^n) = x^{m+n}, \quad \frac{x^m}{x^n} = x^{m-n}, \quad (x^m)^n = x^{mn}, \quad x^0 = 1, \quad x^{-m} = \frac{1}{x^m}.$$

The number $e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n \approx 2.71828$ is the natural exponential base.

Transformations. For $g(x) = Aa^{B(x-h)} + k$, identify vertical scale/reflection, horizontal scale/reflection, shift h, vertical shift k, and horizontal asymptote $y = k$.

Recall graphical transformations of basic functions for $k > 0$:

$f(x)+k$	▶ Shift up k	$f(-x)$	▶ Reflect horizontally (over the y-axis)
$f(x)-k$	▶ Shift down k	$-f(x)$	▶ Reflect vertically (over the x-axis)
$f(x+k)$	▶ Shift left k	$kf(x)$	▶ Stretch/shrink vertically
$f(x-k)$	▶ Shift right k	$f(kx)$	▶ Stretch/shrink horizontally

- $kf(x)$: vertical scaling by a factor of $|k|$.
- $f(kx)$: horizontal scaling by a factor of $\frac{1}{|k|}$.

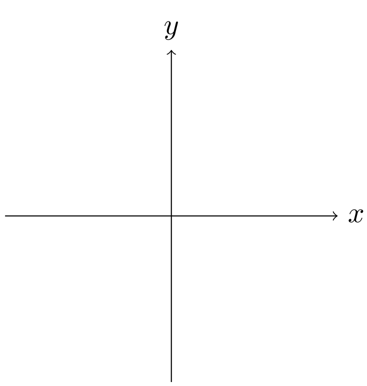
Example. Graph the following function by hand without the use of a calculator by identifying all transformations.

Determine the domain, range, and horizontal asymptote. Sketch the asymptote on the graph using a dashed line and label it. Label the point that would have been at $(0, 1)$ had there not been any transformations.

$$f(x) = 4^{x-3} - 2$$

Asymptote: ______________ **Domain:** ______________ **Range:** ______________

Basic function with asymptote



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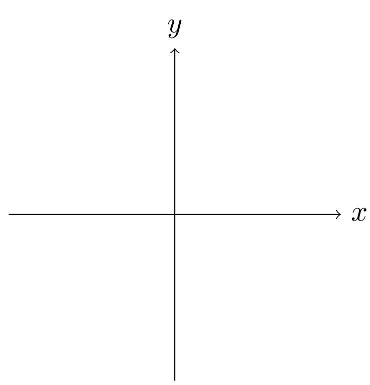

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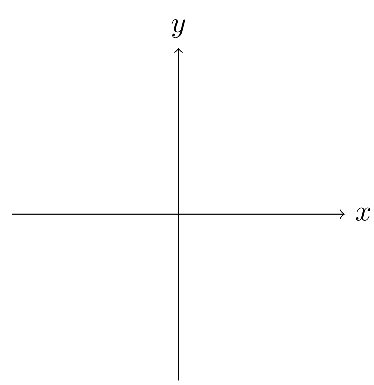


Solution. Begin with the basic exponential function

$$y = 4^x.$$

The transformation

$$4^x \longrightarrow 4^{x-3}$$

shifts the graph 3 units to the right. Thus, the point $(0,1)$ moves to

$$(3,1).$$

Next,

$$4^{x-3} \longrightarrow 4^{x-3} - 2$$

shifts the graph 2 units downward. Therefore,

$$(3,1) \longrightarrow (3,-1).$$

The horizontal asymptote $y = 0$ is also shifted downward 2 units. Hence, $y = -2$.
Therefore,

$$\text{Domain: } (-\infty, \infty), \qquad \text{Range: } (-2, \infty), \qquad \text{Horizontal asymptote: } y = -2.$$

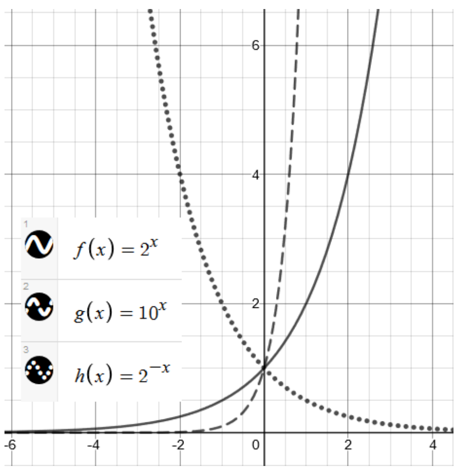


Figure 9: Comparison of exponential functions with different bases

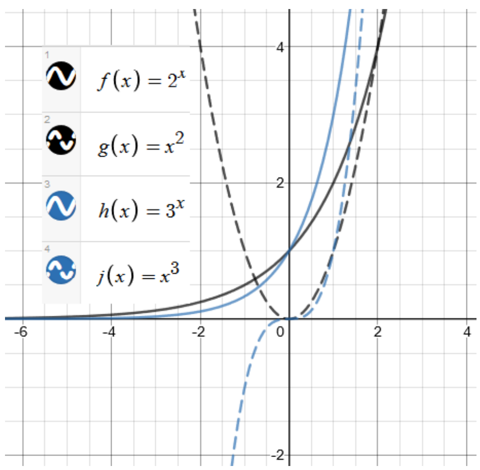


Figure 10: Comparison of exponential and polynomial growth

Recall: Basic Exponential Properties

Rule 1: To multiply identical bases, add the exponents:

$$(x^m)(x^n) = x^{m+n}.$$

Rule 2: To divide identical bases, subtract the exponents:

$$\frac{x^m}{x^n} = x^{m-n}.$$

Rule 3: When there are two or more exponents and only one base, multiply the exponents:

$$(x^m)^n = x^{mn}.$$

Rule 4: Any nonzero base raised to the zero power equals 1:

$$x^0 = 1, \qquad x \neq 0.$$

Rule 5: Negative exponents can be rewritten as positive exponents by taking the reciprocal:

$$x^{-m} = \frac{1}{x^m}, \qquad x \neq 0.$$

Solving exponential equations. If $a^u = a^v$ for $a > 0$, $a \neq 1$, then $u = v$.

Example. Solve each equation.

1. $\left(\frac{1}{8}\right)^{3x} = 4^{x+1}$.
2. $e^{x^2} = e^x \left(e^{x-1}\right)^2$.

Solution.

1. Rewrite both sides using base 2:

$$\begin{aligned} \left(2^{-3}\right)^{3x} &= \left(2^2\right)^{x+1}, \\ 2^{-9x} &= 2^{2x+2}. \end{aligned}$$

   Since the bases are equal,

$$\begin{aligned} -9x &= 2x + 2, \\ -11x &= 2, \\ x &= -\frac{2}{11}. \end{aligned}$$

2. Simplify the right-hand side:

$$\begin{aligned} e^{x^2} &= e^x \left(e^{x-1}\right)^2, \\ e^{x^2} &= e^x e^{2x-2}, \\ e^{x^2} &= e^{3x-2}. \end{aligned}$$

   Since the bases are equal,

$$\begin{aligned} x^2 &= 3x - 2, \\ x^2 - 3x + 2 &= 0, \\ (x-1)(x-2) &= 0. \end{aligned}$$

   Therefore,

$$x = 1 \quad \text{or} \quad x = 2.$$

Growth/decay model. A quantity changing by a fixed percentage per period can be modeled by

$$A(t) = A_0(1+r)^t \quad \text{(growth)}, \qquad A(t) = A_0(1-r)^t \quad \text{(decay)}.$$

Example. A town has population 3000 and grows by 4% per year. Find the population after 5 years.

Solution. $P(5) = 3000(1.04)^5 \approx 3650$.

Example. A principal of \$3,000 is invested at 4% interest, compounded annually. How much will the investment be worth after 5 years?

Solution. For annual compound interest,

$$A = P(1+r)^t.$$

Here,

$$P = 3000, \qquad r = 0.04, \qquad t = 5.$$

Therefore,

$$\begin{aligned} A &= 3000(1+0.04)^5 \\ &= 3000(1.04)^5 \\ &\approx \$3,649.96. \end{aligned}$$

Thus, the investment will be worth approximately \$3,649.96 after 5 years.

Example. The value $v(t)$ of a certain car t years old is given by the exponential function

$$v(t) = 35{,}000(0.95)^t.$$

What is the initial value of the car? What is its value after 10 years?

Solution. The initial value is found by setting $t = 0$:

$$\begin{aligned} v(0) &= 35{,}000(0.95)^0 \\ &= 35{,}000. \end{aligned}$$

Thus, the initial value of the car is

$$\$35{,}000.$$

After 10 years,

$$\begin{aligned} v(10) &= 35{,}000(0.95)^{10} \\ &\approx \$20{,}956.79. \end{aligned}$$

Thus, the value of the car after 10 years is approximately

$$\$20{,}956.79.$$

4.3: Logarithmic Functions

For $a > 0$, $a \neq 1$,

$$y = \log_a x \quad \iff \quad a^y = x.$$

The argument of a logarithm must be positive. Thus $\log_a x$ has domain $(0, \infty)$, range $\mathbb{R}$, x-intercept $(1, 0)$, and vertical asymptote $x = 0$.

We read $y = \log_a x$ as y is the logarithm to the base a of x.

Common logs:

$$\log x = \log_{10} x, \qquad \ln x = \log_e x.$$

Exponential and logarithmic functions with the same base are inverses:

$$\log_a(a^x) = x, \qquad a^{\log_a x} = x \quad (x > 0).$$

Example. Find the domain of

$$f(x) = \log_3 \left(\frac{(x-2)^2}{x+1} \right).$$

Solution. The argument of a logarithm must be positive. Therefore,

$$\frac{(x-2)^2}{x+1} > 0.$$

The critical values are

$$x = -1 \qquad \text{and} \qquad x = 2.$$

Since $(x-2)^2 > 0$ for $x \neq 2$, the fraction is positive when

$$x + 1 > 0,$$

so

$$x > -1.$$

However, $x = 2$ must be excluded because the argument of the logarithm is 0.

Therefore, the domain is

$$\boxed{(-1, 2) \cup (2, \infty)}.$$

Example. Rewrite $7^x = 15$ in logarithmic form.

Solution. $x = \log_7 15$.

Example. Find the exact value of each expression.

1. $\log_5 \left(\dfrac{1}{125} \right)$
2. $\ln(e^4)$

Solution.

1. Since

$$\frac{1}{125} = \frac{1}{5^3} = 5^{-3},$$

we have

$$\log_5 \left(\frac{1}{125} \right) = \log_5(5^{-3}) = -3.$$

2. Using the inverse relationship between $\ln x$ and e^x,

$$\ln(e^4) = 4.$$

Example. Evaluate each expression, rounded to four decimal places.

1. $\log(20\pi)$
2. $\ln(0.3)$

Solution. Using a calculator,

$$\log(20\pi) \approx 1.7982$$

and

$$\ln(0.3) \approx -1.2040.$$

Inverse of a Logarithmic Function

Recall that logarithmic and exponential functions are inverses. Let

$$f(x) = \log_a x.$$

Then

$$\begin{aligned} y &= \log_a x, \\ x &= \log_a y, \\ a^x &= y. \end{aligned}$$

Therefore,

$$f^{-1}(x) = a^x.$$

Example. Determine the inverse of

$$f(x) = \ln(x+2),$$

and verify that the two functions are inverses. State the domain and range of $f^{-1}(x)$.

Solution. Begin with

$$y = \ln(x+2).$$

Interchange x and y:

$$x = \ln(y+2).$$

Rewrite in exponential form:

$$\begin{aligned} e^x &= y+2, \\ y &= e^x - 2. \end{aligned}$$

Therefore,

$$f^{-1}(x) = e^x - 2.$$

To verify, compute both compositions:

$$\begin{aligned} f\left(f^{-1}(x)\right) &= \ln\left((e^x - 2) + 2\right) \\ &= \ln(e^x) \\ &= x, \end{aligned}$$

and

$$\begin{aligned} f^{-1}(f(x)) &= e^{\ln(x+2)} - 2 \\ &= x + 2 - 2 \\ &= x. \end{aligned}$$

Thus, f and f^{-1} are inverses.
Since

$$f^{-1}(x) = e^x - 2,$$

its domain is

$$(-\infty, \infty),$$

and its range is

$$(-2, \infty).$$

Properties of the Logarithmic Function $f(x) = \log_a x$

1. The domain is the set of positive real numbers:

   $$x > 0,$$

   and the range is the set of all real numbers:

   $$(-\infty, \infty).$$

   To find the domain of a logarithmic function, always require its argument to be positive:

   $$\text{argument} > 0.$$

2. The x-intercept is 1, so the graph passes through the point

   $$(1, 0).$$

   There is no y-intercept.

3. The y-axis, $x = 0$, is the vertical asymptote. There is no horizontal asymptote.

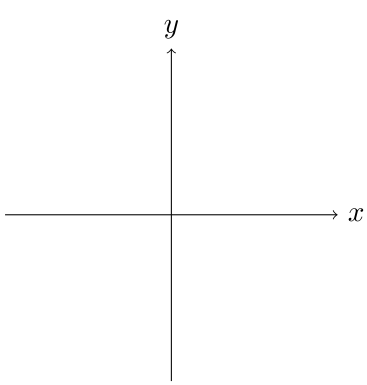
Example. Sketch the graph of

$$f(x) = -3 + \ln(x + 2)$$

using transformations. Determine the vertical asymptote, domain, and range.

Asymptote: ________________ **Domain:** ________________ **Range:** ________________

Basic function



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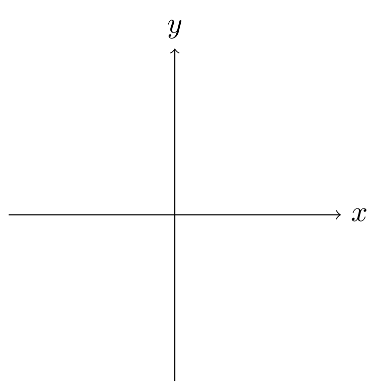

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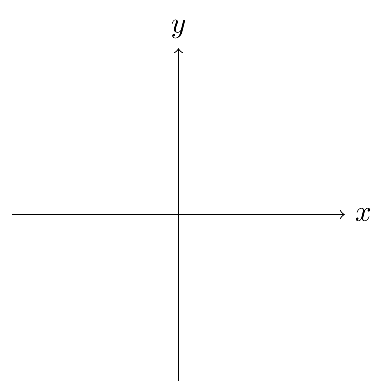


Solution. Begin with the basic logarithmic function

$$y = \ln x.$$

The transformation

$$\ln x \longrightarrow \ln(x+2)$$

shifts the graph 2 units to the left. Thus, the point $(1, 0)$ moves to

$$(-1, 0),$$

and the vertical asymptote moves from

$$x = 0$$

to

$$x = -2.$$

Next,

$$\ln(x+2) \longrightarrow -3 + \ln(x+2)$$

shifts the graph 3 units downward. Therefore,

$$(-1, 0) \longrightarrow (-1, -3).$$

The vertical shift does not change the vertical asymptote. Hence,

$$\text{Vertical asymptote: } x = -2.$$

The domain is

$$(-2, \infty),$$

and the range is

$$(-\infty, \infty).$$

Domain of the logarithmic function = Range of the exponential function = $(0, \infty)$
Range of the logarithmic function = Domain of the exponential function = $(-\infty, \infty)$

Figure 11: Domain and range of logarithmic and exponential functions

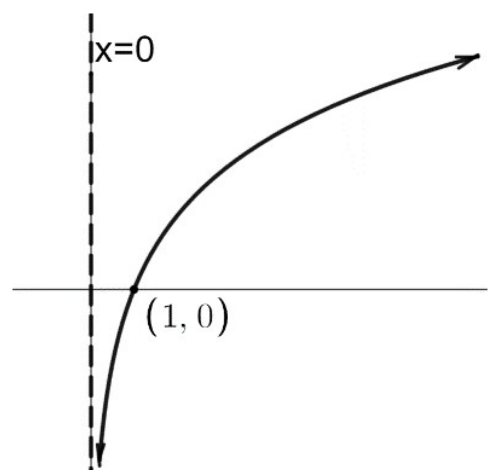


Figure 12: Basic logarithmic function

4.4: Properties of Logarithms and Exponential/Logarithmic Equations

For valid positive arguments,

$$\log_a(MN) = \log_a M + \log_a N,$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N, \qquad \log_a(M^p) = p\log_a M.$$

Also,

$$\log_a a = 1, \quad \log_a 1 = 0, \quad \log_a(a^x) = x, \quad a^{\log_a M} = M.$$

Solving Simple Logarithmic and Exponential Equations.

Solve the following equations by rewriting them in the other form. **Example.** Solve each equation.

1. $\ln(2x) = 3$
2. $\log_4(x-2) = 2$

Solution.

1. Rewrite the logarithmic equation in exponential form:

$$\begin{aligned} \ln(2x) &= 3, \\ 2x &= e^3, \\ x &= \frac{e^3}{2}. \end{aligned}$$

2. Rewrite the logarithmic equation in exponential form:

$$\begin{aligned} \log_4(x-2) &= 2, \\ x-2 &= 4^2, \\ x-2 &= 16, \\ x &= 18. \end{aligned}$$

Example. Solve each equation.

1. $e^x = 3$
2. $10^{2x} = 1.5$

Solution.

1. Rewrite the exponential equation in logarithmic form:

$$\begin{aligned} e^x &= 3, \\ x &= \ln 3. \end{aligned}$$

2. Rewrite the exponential equation in logarithmic form:

$$\begin{aligned} 10^{2x} &= 1.5, \\ 2x &= \log(1.5), \\ x &= \frac{\log(1.5)}{2}. \end{aligned}$$

Example. Solve each equation.

1. $e^{x+1} - 2 = 3$
2. $-3\log(x+2) + 6 = 3$

Solution.

1. First isolate the exponential expression:

$$\begin{aligned} e^{x+1} - 2 &= 3, \\ e^{x+1} &= 5. \end{aligned}$$

   Rewrite in logarithmic form:

$$\begin{aligned} x + 1 &= \ln 5, \\ x &= \ln 5 - 1. \end{aligned}$$

2. First isolate the logarithmic expression:

$$\begin{aligned} -3\log(x+2) + 6 &= 3, \\ -3\log(x+2) &= -3, \\ \log(x+2) &= 1. \end{aligned}$$

   Rewrite in exponential form:

$$\begin{aligned} x + 2 &= 10^1, \\ x + 2 &= 10, \\ x &= 8. \end{aligned}$$

Example. Expand $\ln\left(\dfrac{y\sqrt{x^2+1}}{(z-2)^3}\right)$.

Solution. $\ln y + \frac{1}{2}\ln(x^2+1) - 3\ln(z-2)$, assuming all displayed logarithmic arguments are positive.

Example. Solve $\log_3(x-5) + \log_3(x+3) = 2$.

Solution. Combine logs: $\log_3((x-5)(x+3)) = 2$. Hence $(x-5)(x+3) = 9$, so $x^2 - 2x - 24 = 0$, giving $x = 6, -4$. Domain requires $x > 5$, so $x = 6$ only.

Caution. Always check proposed solutions when logarithms, rational expressions, or even radicals impose domain restrictions.

Example. Solve exactly $3^{x+1} = 7$.

Solution. $(x+1)\ln 3 = \ln 7$, so $x = \dfrac{\ln 7}{\ln 3} - 1$.

Uniqueness Property of Logarithms

If M, N, and a are positive real numbers, with $a \neq 1$, then

$$M = N \quad \Longrightarrow \quad \log_a M = \log_a N,$$

and
$$\log_a M = \log_a N \quad \Longrightarrow \quad M = N.$$

This property can be used to solve exponential and logarithmic equations.
For example, consider
$$\log_a M = \log_a N.$$
By the uniqueness property of logarithms,
$$M = N.$$

Change-of-Base Formula
If $a \neq 1$, $b \neq 1$, and a, b, and M are positive real numbers, then
$$\log_a M = \frac{\log_b M}{\log_b a}$$

In particular, we may use either common logarithms or natural logarithms:
$$\log_a M = \frac{\log M}{\log a} = \frac{\ln M}{\ln a}.$$

Example. Use the change-of-base formula to find a decimal approximation of
$$\log_\pi(6.7).$$

Solution. Using the change-of-base formula,
$$\begin{aligned} \log_\pi(6.7) &= \frac{\ln(6.7)}{\ln(\pi)} \\ &\approx 1.6617. \end{aligned}$$

Solving Exponential and Logarithmic Equations

1. If an exponential equation can be rewritten using a common base, use the property
$$a^x = a^y \quad \Longrightarrow \quad x = y.$$

2. If the equation contains logarithms, first combine them into a **single logarithm**. If it contains an exponential expression, first isolate the exponential expression. Then rewrite the equation in exponential or logarithmic form and solve.
   Example. Solve
$$\log_3(x-5) + \log_3(x+3) = 2.$$

   Solution. Use the product property of logarithms:
$$\begin{aligned} \log_3(x-5) + \log_3(x+3) &= 2, \\ \log_3\big((x-5)(x+3)\big) &= 2. \end{aligned}$$

   Rewrite in exponential form:
$$\begin{aligned} (x-5)(x+3) &= 3^2, \\ (x-5)(x+3) &= 9. \end{aligned}$$

Expanding,

$$\begin{aligned} x^2 - 2x - 15 &= 9, \\ x^2 - 2x - 24 &= 0, \\ (x-6)(x+4) &= 0. \end{aligned}$$

Thus, the possible solutions are

$$x = 6 \qquad \text{or} \qquad x = -4.$$

However, the arguments of both logarithms must be positive. In particular,

$$x - 5 > 0,$$

so

$$x > 5.$$

Therefore, $x = -4$ is extraneous, and the solution is

$$\boxed{x = 6}.$$

3. If an equation has logarithms on both sides, use the properties of logarithms to rewrite the equation so that the logarithms have the same base. Then apply the uniqueness property:

$$\log_a M = \log_a N \quad \Longrightarrow \quad M = N.$$

Example. Solve

$$\log(3x-3) - \log 4 = \log(x+1).$$

Solution. Use the quotient property:

$$\begin{aligned} \log(3x-3) - \log 4 &= \log(x+1), \\ \log\left(\frac{3x-3}{4}\right) &= \log(x+1). \end{aligned}$$

By the uniqueness property,

$$\frac{3x-3}{4} = x + 1.$$

Therefore,

$$\begin{aligned} 3x - 3 &= 4x + 4, \\ -x &= 7, \\ x &= -7. \end{aligned}$$

However, the logarithmic arguments must be positive. In particular,

$$3x - 3 > 0 \quad \Longrightarrow \quad x > 1.$$

Thus, $x = -7$ is extraneous, so the equation has no real solution.

Example. Solve

$$\log_5(x+6) = \log_5 5 - \log_5(x+2).$$

Solution. Since

$$\log_5 5 = 1,$$

we may write

$$\log_5(x+6) = 1 - \log_5(x+2).$$

It is more useful to keep everything in logarithmic form:

$$1 = \log_5 5.$$

Thus,

$$\begin{aligned}\log_5(x+6) &= \log_5 5 - \log_5(x+2),\\ \log_5(x+6) &= \log_5\left(\frac{5}{x+2}\right).\end{aligned}$$

By the uniqueness property,

$$x+6 = \frac{5}{x+2}.$$

Multiplying by $x+2$,

$$\begin{aligned}(x+6)(x+2) &= 5,\\ x^2+8x+12 &= 5,\\ x^2+8x+7 &= 0,\\ (x+1)(x+7) &= 0.\end{aligned}$$

The possible solutions are

$$x = -1 \qquad \text{or} \qquad x = -7.$$

The domain restrictions require

$$x+6>0 \qquad \text{and} \qquad x+2>0,$$

so

$$x > -2.$$

Therefore, $x = -7$ is extraneous and $x = -1$.

Caution. Always check solutions when an equation contains expressions with domain restrictions, since algebraic manipulation may produce extraneous solutions.

Common sources of domain restrictions include:

- logarithmic expressions: the argument must be positive;
- rational expressions: the denominator cannot equal zero;
- even-root radical expressions: the radicand must be nonnegative.

4. If an exponential equation cannot be rewritten using a common base, logarithms can be used to solve the equation.

   If the equation contains only one exponential expression, isolate the exponential term and then rewrite the equation in logarithmic form.

Equivalently, use the property

$$M = N \implies \log M = \log N.$$

Example. Solve exactly

$$3^{x+1} = 7.$$

Solution. Take the natural logarithm of both sides:

$$\begin{aligned} \ln\left(3^{x+1}\right) &= \ln 7, \\ (x+1)\ln 3 &= \ln 7. \end{aligned}$$

Therefore,

$$\begin{aligned} x + 1 &= \frac{\ln 7}{\ln 3}, \\ x &= \frac{\ln 7}{\ln 3} - 1. \end{aligned}$$

Thus,

$$x = \frac{\ln 7}{\ln 3} - 1.$$

If an equation contains more than one exponential expression, simplify both sides as much as possible and then take the logarithm of both sides.

Example. Solve exactly

$$2^{x+2} = 6^{2x-5}.$$

Solution. Take the natural logarithm of both sides:

$$\begin{aligned} \ln\left(2^{x+2}\right) &= \ln\left(6^{2x-5}\right), \\ (x+2)\ln 2 &= (2x-5)\ln 6. \end{aligned}$$

Expand:

$$x\ln 2 + 2\ln 2 = 2x\ln 6 - 5\ln 6.$$

Collect the terms containing x:

$$\begin{aligned} 2\ln 2 + 5\ln 6 &= 2x\ln 6 - x\ln 2, \\ 2\ln 2 + 5\ln 6 &= x(2\ln 6 - \ln 2). \end{aligned}$$

Therefore,

$$x = \frac{2\ln 2 + 5\ln 6}{2\ln 6 - \ln 2}.$$

Example. Solve exactly

$$e^{2x-3} = 10^{3x+1}.$$

Solution. Take the natural logarithm of both sides:

$$\begin{aligned} \ln\left(e^{2x-3}\right) &= \ln\left(10^{3x+1}\right), \\ 2x - 3 &= (3x+1)\ln 10. \end{aligned}$$

Expand:
$$2x - 3 = 3x \ln 10 + \ln 10.$$

Collect the terms containing x:

$$2x - 3x \ln 10 = 3 + \ln 10,$$
$$x(2 - 3 \ln 10) = 3 + \ln 10.$$

Therefore,
$$x = \frac{3 + \ln 10}{2 - 3 \ln 10}.$$

Special Cases: Substitution

Sometimes an exponential equation can be rewritten as a quadratic equation by using substitution.

Example. Solve exactly
$$3e^{2x} + e^x = 10.$$

Solution. Let
$$u = e^x.$$

Then
$$e^{2x} = (e^x)^2 = u^2.$$

The equation becomes
$$3u^2 + u = 10,$$

or
$$3u^2 + u - 10 = 0.$$

Factor:
$$(3u - 5)(u + 2) = 0.$$

Thus,
$$u = \frac{5}{3} \qquad \text{or} \qquad u = -2.$$

Since
$$u = e^x > 0,$$

the solution $u = -2$ is impossible. Therefore,
$$e^x = \frac{5}{3}.$$

Taking the natural logarithm,
$$x = \ln\left(\frac{5}{3}\right).$$

5.1: Angle Measure, Special Triangles, and Special Angles

Some Definitions

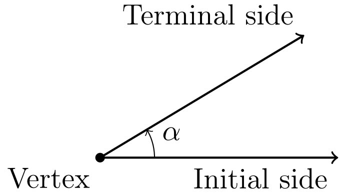
Ray: A portion of a line that begins at an endpoint and extends indefinitely in one direction.

Angle: The joining of two rays at a common endpoint, called the **vertex**.

Initial side: The ray at which the angle rotation begins. In standard position, the initial side lies along the positive x-axis.

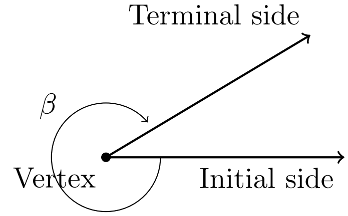
Terminal side: The ray where the angle rotation ends.

Counterclockwise Rotation



Counterclockwise rotation
Positive angle

Clockwise Rotation



Clockwise rotation
Negative angle

Common Greek letters used to represent angles include

$$\alpha \quad \text{(alpha)}, \qquad \beta \quad \text{(beta)}, \qquad \theta \quad \text{(theta)}.$$

Important:

$$\boxed{\begin{array}{c}\text{Counterclockwise rotation} \Longrightarrow \text{positive angle,} \\ \text{Clockwise rotation} \Longrightarrow \text{negative angle.}\end{array}}$$

An angle in standard position has vertex at the origin and initial side on the positive x-axis. Counterclockwise rotation is positive and clockwise rotation is negative.

Degrees.

The following diagrams illustrate one complete revolution, a right angle, and a straight angle.

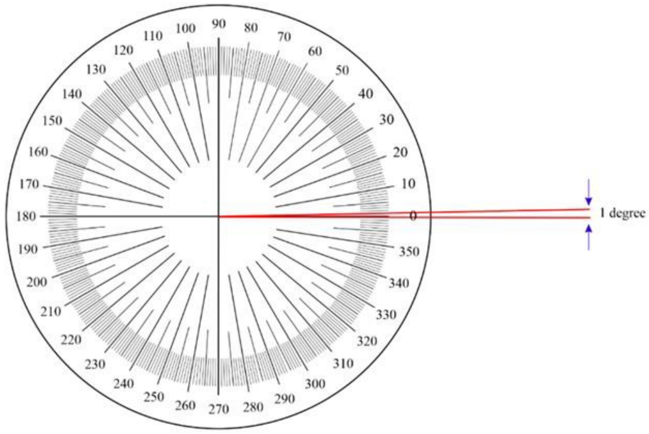
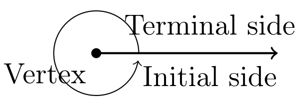


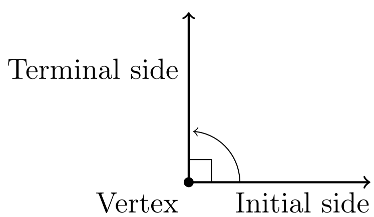
Figure 13: One degree as part of a full rotation



(a) One revolution

360°

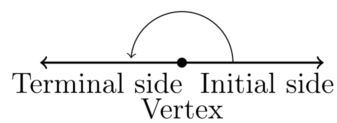
Counterclockwise rotation



(b) Right angle

$\frac{1}{4}$ revolution $= 90^\circ$

Counterclockwise rotation



(c) Straight angle

$\frac{1}{2}$ revolution $= 180^\circ$

Counterclockwise rotation

Types of Angles

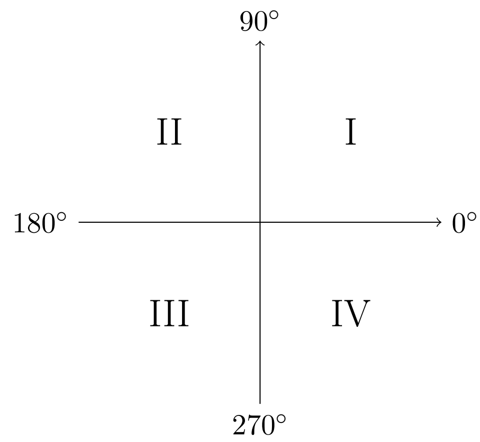
Acute angles: Angles whose measures are between 0° and 90°:

$$0^\circ < \theta < 90^\circ.$$

Obtuse angles: Angles whose measures are between 90° and 180°:

$$90^\circ < \theta < 180^\circ.$$

Quadrants:

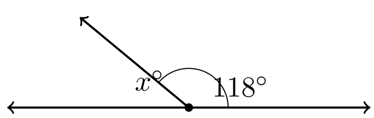


Supplementary and Complementary Angles

Supplementary angles:

Two angles are supplementary if the sum of their measures is

$$180°.$$



$$x + 118 = 180.$$

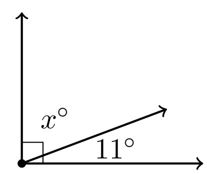
Solution.

$$x = 62°.$$

Complementary angles:

Two angles are complementary if the sum of their measures is

$$90°.$$



$$x + 11 = 90.$$

Solution.

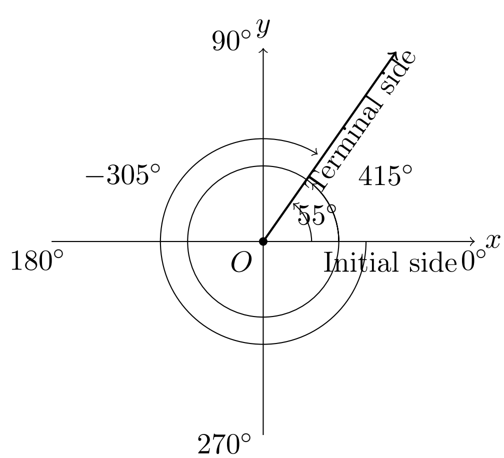
$$x = 79°.$$

Coterminal Angles

Two angles in standard position are **coterminal** if they have the same initial side and the same terminal side.

Coterminal angles differ by complete revolutions. Therefore,

$$\theta + 360°k, \qquad k \in \mathbb{Z}.$$



Example. Find two positive and two negative angles that are coterminal with

$$35^\circ.$$

Solution. Coterminal angles are obtained by adding or subtracting multiples of 360°.

Two positive coterminal angles are

$$35^\circ + 360^\circ = 395^\circ,$$
$$35^\circ + 720^\circ = 755^\circ.$$

Two negative coterminal angles are

$$35^\circ - 360^\circ = -325^\circ,$$
$$35^\circ - 720^\circ = -685^\circ.$$

Therefore, two positive and two negative coterminal angles are

$$395^\circ, \qquad 755^\circ, \qquad -325^\circ, \qquad -685^\circ.$$

Properties of Triangles

The following are important properties of triangles:

1. The sum of the interior angles of a triangle is
$$180^\circ.$$
2. The combined length of any two sides of a triangle exceeds the length of the third side. That is, if the side lengths are a, b, and c, then
$$a+b>c, \qquad a+c>b, \qquad b+c>a.$$
3. Larger angles are opposite larger sides. Similarly, larger sides are opposite larger angles.

Similar Triangles

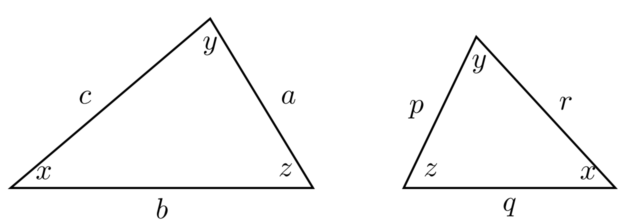
Similar triangles are triangles whose corresponding angles are equal and whose corresponding side lengths are proportional.

Suppose the corresponding sides of two similar triangles are a, b, c and p, q, r. Then

$$\frac{a}{p} = \frac{b}{q} = \frac{c}{r}.$$

Equivalent proportions may also be written, for example, as

$$\frac{a}{b} = \frac{p}{q}.$$



Example. The shadow of a flagpole has length 20 feet. At the same time, the shadow of a yardstick is 6 inches long. Find the height of the flagpole.

Solution. Since the flagpole and the yardstick cast shadows at the same time, the triangles formed by their heights and shadows are similar.

A yardstick is 3 feet tall, and its shadow is

$$6 \text{ inches} = \frac{1}{2} \text{ foot.}$$

Let h be the height of the flagpole. Using corresponding sides,

$$\frac{h}{20} = \frac{3}{1/2}.$$

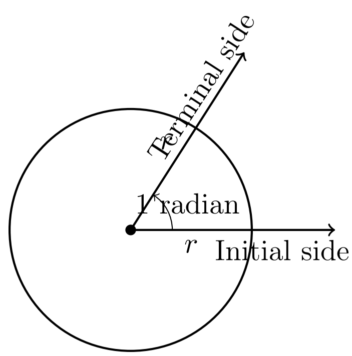
Therefore,

$$\frac{h}{20} = 6,$$
$$h = 120.$$

Thus, the flagpole is 120 feet tall.

Radian Measure

One radian is the measure of a central angle that intercepts an arc whose length is equal to the radius of the circle.



If an angle θ is measured in radians, then

$$\theta = \frac{s}{r},$$

where s is the intercepted arc length and r is the radius.

For one radian,

$$s = r,$$

so

$$\theta = \frac{r}{r} = 1.$$

Converting Between Degrees and Radians

Recall that one complete revolution is

$$360^\circ = 2\pi \text{ radians.}$$

Therefore,

$$180^\circ = \pi \text{ radians.}$$

To convert from degrees to radians, multiply by

$$\frac{\pi}{180^\circ}.$$

To convert from radians to degrees, multiply by

$$\frac{180^\circ}{\pi}.$$

Table 1: Common angle measures in degrees and their equivalent measures in radians.

Degrees	0°	30°	45°	60°	90°	120°	135°	150°	180°	210°	225°	240°	270°	300°	315°	330°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π

Example. Convert each angle as indicated.

1. 630° to exact radians.
2. 61° to decimal radians.
3. $\dfrac{3\pi}{2}$ radians to exact degrees.

Solution.

1. To convert degrees to radians, multiply by $\dfrac{\pi}{180^\circ}$:

$$\begin{aligned} 630^\circ \left(\frac{\pi}{180^\circ}\right) &= \frac{630\pi}{180} \\ &= \frac{7\pi}{2}. \end{aligned}$$

Therefore,

$$630^\circ = \frac{7\pi}{2} \text{ radians.}$$

2. To convert degrees to radians,

$$\begin{aligned} 61^\circ \left(\frac{\pi}{180^\circ}\right) &= \frac{61\pi}{180} \\ &\approx 1.0647. \end{aligned}$$

Therefore,

$$61^\circ \approx 1.0647 \text{ radians.}$$

3. To convert radians to degrees, multiply by $\dfrac{180^\circ}{\pi}$:

$$\begin{aligned} \frac{3\pi}{2}\left(\frac{180^\circ}{\pi}\right) &= \frac{3(180^\circ)}{2} \\ &= 270^\circ. \end{aligned}$$

Therefore,

$$\frac{3\pi}{2} \text{ radians} = 270^\circ.$$

Arc Length

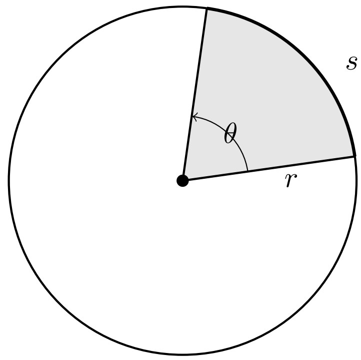


Figure 14: A central angle θ measured in radians intercepts an arc of length s in a circle of radius r. The relationship between arc length, radius, and central angle is $s = r\theta$.

For a circle of radius r, the circumference is

$$s = 2\pi r.$$

Since one complete revolution is 2π radians, an angle of θ radians intercepts an arc of length

$$s = r\theta.$$

Equivalently,

$$\theta = \frac{s}{r},$$

where θ must be measured in radians.

Example. Find θ if $s = 20$ miles and $r = 18$ miles. How many degrees is this?

Solution. Using the arc length formula,

$$s = r\theta.$$

Therefore,

$$\begin{aligned} 20 &= 18\theta, \\ \theta &= \frac{20}{18}, \\ \theta &= \frac{10}{9} \text{ radians.} \end{aligned}$$

To convert the angle to degrees,

$$\begin{aligned} \frac{10}{9}\left(\frac{180^\circ}{\pi}\right) &= \frac{200^\circ}{\pi} \\ &\approx 63.66^\circ. \end{aligned}$$

Thus,

$$\theta = \frac{10}{9} \text{ radians} \approx 63.66^\circ.$$

Area of a Sector

Recall that the area of a circle of radius r is

$$A = \pi r^2.$$

A complete circle corresponds to an angle of 2π radians. Therefore, a sector with central angle θ represents the fraction

$$\frac{\theta}{2\pi}$$

of the entire circle.

Thus, the area of a sector is

$$A = \frac{\theta}{2\pi}\left(\pi r^2\right),$$

so

$$A = \frac{1}{2}r^2\theta,$$

where θ is measured in **radians**.

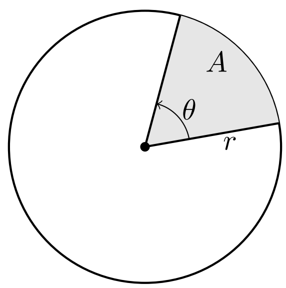


Figure 15: A sector of a circle with radius r and central angle θ. When θ is measured in radians, the area of the sector is $A = \frac{1}{2}r^2\theta$.

Example. Find the area of the sector of a circle of radius 50 inches and central angle 2 radians.

Solution. Using

$$A = \frac{1}{2}r^2\theta,$$

we have

$$\begin{aligned} A &= \frac{1}{2}(50)^2(2) \\ &= 2500. \end{aligned}$$

Therefore, the area of the sector is

$$2500 \text{ in}^2.$$

Example. A pizza has a diameter of 14 inches. A slice has a central angle of 50°, and the toppings extend to a radius of 6 inches. What is the area of the crust for one slice of pizza?

Solution. The pizza has radius

$$r = 7 \text{ inches}.$$

The toppings extend to a radius of 6 inches, so the crust is the portion between radii 6 and 7 inches.

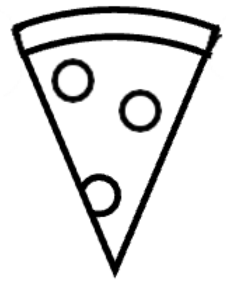


Figure 16: Pizza

First convert 50° to radians:

$$50^\circ\left(\frac{\pi}{180^\circ}\right)=\frac{5\pi}{18}.$$

The area of the entire sector of the pizza slice is

$$\begin{aligned}A_{\text{outer}}&=\frac{1}{2}(7)^2\left(\frac{5\pi}{18}\right)\\&=\frac{245\pi}{36}.\end{aligned}$$

The area of the sector containing the toppings is

$$\begin{aligned}A_{\text{inner}}&=\frac{1}{2}(6)^2\left(\frac{5\pi}{18}\right)\\&=5\pi.\end{aligned}$$

Therefore, the area of the crust is

$$\begin{aligned}A_{\text{crust}}&=A_{\text{outer}}-A_{\text{inner}}\\&=\frac{245\pi}{36}-5\pi\\&=\frac{65\pi}{36}.\end{aligned}$$

Thus,

$$A_{\text{crust}}=\frac{65\pi}{36}\text{ in}^2\approx 5.67\text{ in}^2.$$

Angular and Linear Speed

Suppose an object moves along a circle of radius r. If the object moves through an angle θ in time t, then its **angular speed** is

$$\omega=\frac{\theta}{t},$$

where θ is measured in radians.

Since the arc length is

$$s=r\theta,$$

the **linear speed** is

$$v = \frac{s}{t} = \frac{r\theta}{t}.$$

Therefore,

$$v = r\omega.$$

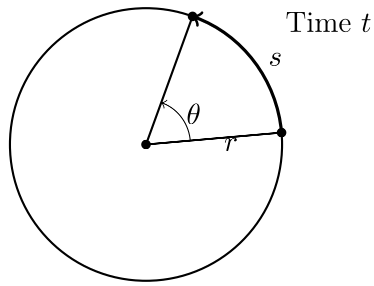


Figure 17: An object travels an arc of length s around a circle of radius r through a central angle θ during time t. Angular speed is $\omega = \theta/t$, linear speed is $v = s/t$, and $v = r\omega$.

Example. An object is traveling around a circle with a radius of 8 inches. If 3 revolutions are made in one minute, what is the angular speed in radians per second? What is the linear speed in feet per second?

Solution. Since one revolution is 2π radians,

$$3 \text{ revolutions} = 6\pi \text{ radians}.$$

Also,

$$1 \text{ minute} = 60 \text{ seconds}.$$

Thus, the angular speed is

$$\begin{aligned} \omega &= \frac{6\pi}{60} \\ &= \frac{\pi}{10} \text{ rad/sec}. \end{aligned}$$

For the linear speed, use

$$v = r\omega.$$

Since the radius is 8 inches,

$$\begin{aligned} v &= 8\left(\frac{\pi}{10}\right) \\ &= \frac{4\pi}{5} \text{ in/sec}. \end{aligned}$$

Convert inches to feet:

$$\begin{aligned} v &= \frac{4\pi}{5}\left(\frac{1 \text{ ft}}{12 \text{ in}}\right) \\ &= \frac{\pi}{15} \text{ ft/sec}. \end{aligned}$$

Therefore,

$$\omega = \frac{\pi}{10} \text{ rad/sec} \qquad \text{and} \qquad v = \frac{\pi}{15} \text{ ft/sec}.$$

Example. Pluto has a radius of 700 miles. As the planet rotates on its axis, a point on its equator travels at 267,558 miles per year.

Find the angular speed of a point on its equator in radians per year. Then find the number of rotations the planet makes per year.

Solution. We are given the linear speed

$$v = 267{,}558 \text{ miles/year}$$

and the radius

$$r = 700 \text{ miles.}$$

Using

$$v = r\omega,$$

we obtain

$$\begin{aligned}\omega &= \frac{v}{r}\\ &= \frac{267558}{700}\\ &\approx 382.226.\end{aligned}$$

Thus, the angular speed is approximately

$$382.226 \text{ rad/year.}$$

Since one complete rotation is 2π radians, the number of rotations per year is

$$\begin{aligned}\frac{\omega}{2\pi} &= \frac{382.226}{2\pi}\\ &\approx 60.833.\end{aligned}$$

Therefore, Pluto makes approximately

$$60.833 \text{ rotations per year.}$$

Example. The angular speed is 10 radians per minute. If this is on a wheel with a radius of 120 feet, what is the linear speed in miles per hour? Recall that 1 mile = 5280 feet.

Solution. Using

$$v = r\omega,$$

we obtain

$$\begin{aligned}v &= (120)(10)\\ &= 1200 \text{ ft/min.}\end{aligned}$$

Convert feet per minute to miles per hour:

$$\begin{aligned}1200\frac{\text{ft}}{\text{min}}\left(\frac{1 \text{ mile}}{5280 \text{ ft}}\right)\left(\frac{60 \text{ min}}{1 \text{ hour}}\right) &= \frac{72000}{5280} \text{ mph}\\ &\approx 13.64 \text{ mph.}\end{aligned}$$

Therefore, the linear speed is approximately 13.64 miles per hour.

5.2: Trigonometric Functions and the Unit Circle

Special Right Triangles

Two special right triangles are particularly useful when working with trigonometric functions.

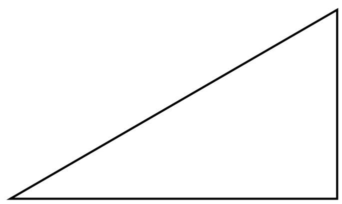
$30°$–$60°$–$90°$ **Triangle**

The side lengths of a $30°$–$60°$–$90°$ triangle are in the ratio

$$1 : \sqrt{3} : 2.$$

Thus, if the hypotenuse has length 1, then the side lengths are

$$\frac{1}{2}, \qquad \frac{\sqrt{3}}{2}, \qquad 1.$$



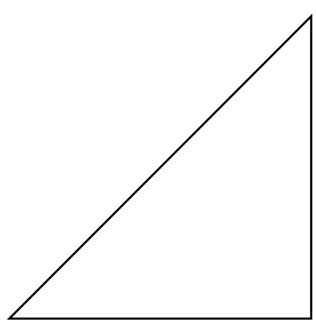
$45°$–$45°$–$90°$ **Triangle**

The side lengths of a $45°$–$45°$–$90°$ triangle are in the ratio

$$1 : 1 : \sqrt{2}.$$

If the hypotenuse has length 1, then the two legs each have length

$$\frac{\sqrt{2}}{2}.$$



The Unit Circle

The unit circle is the circle centered at the origin $(0, 0)$ with radius 1. Its equation is

$$x^2 + y^2 = 1.$$

Every point (x, y) on the unit circle therefore satisfies

$$x^2 + y^2 = 1.$$

Example. Find the points on the unit circle with

$$x = \frac{1}{2}.$$

Solution. Since

$$x^2 + y^2 = 1,$$

we have

$$\left(\frac{1}{2}\right)^2 + y^2 = 1.$$

Thus

$$\frac{1}{4} + y^2 = 1,$$

so

$$y^2 = \frac{3}{4}$$

and therefore

$$y = \pm\frac{\sqrt{3}}{2}.$$

Hence the two points are

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \qquad \text{and} \qquad \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

Symmetry of the Unit Circle

If (x, y) is a point on the unit circle, then symmetry gives the additional points

$$(-x, y), \qquad (-x, -y), \qquad (x, -y).$$

Example. Verify that

$$\left(\frac{3}{4}, \frac{\sqrt{7}}{4}\right)$$

is on the unit circle. Then use symmetry to find three other points on the unit circle.

Solution. We check that

$$\left(\frac{3}{4}\right)^2 + \left(\frac{\sqrt{7}}{4}\right)^2 = \frac{9}{16} + \frac{7}{16} = 1.$$

Thus the point lies on the unit circle.

By symmetry, three additional points are

$$\left(-\frac{3}{4}, \frac{\sqrt{7}}{4}\right), \qquad \left(-\frac{3}{4}, -\frac{\sqrt{7}}{4}\right), \qquad \left(\frac{3}{4}, -\frac{\sqrt{7}}{4}\right).$$

Reference Angles

The reference angle of an angle θ is the acute angle formed between the terminal side of θ and the x-axis. For an angle satisfying

$$0 \le \theta < 2\pi,$$

the reference angle θ_{ref} can be found according to the quadrant:

Quadrant	Reference Angle
I	θ
II	$\pi - \theta$
III	$\theta - \pi$
IV	$2\pi - \theta$

For angles measured in degrees,

Quadrant	Reference Angle
I	θ
II	$180^\circ - \theta$
III	$\theta - 180^\circ$
IV	$360^\circ - \theta$

For angles outside one complete revolution, first find a coterminal angle in

$$[0, 2\pi) \qquad \text{or} \qquad [0^\circ, 360^\circ).$$

Example. Find the reference angle for each angle:

$$225^\circ, \qquad \frac{5\pi}{6}, \qquad \frac{11\pi}{3}.$$

Solution. For 225°, the terminal side lies in Quadrant III. Thus

$$225^\circ - 180^\circ = 45^\circ.$$

For $\frac{5\pi}{6}$, the terminal side lies in Quadrant II. Thus

$$\pi - \frac{5\pi}{6} = \frac{\pi}{6}.$$

For $\frac{11\pi}{3}$, first find a coterminal angle:

$$\frac{11\pi}{3} - 2\pi = \frac{11\pi}{3} - \frac{6\pi}{3} = \frac{5\pi}{3}.$$

This angle lies in Quadrant IV, so

$$2\pi - \frac{5\pi}{3} = \frac{\pi}{3}.$$

Trigonometric Functions on the Unit Circle

Let θ be an angle in standard position, and let

$$P = (x, y)$$

be the point on the unit circle corresponding to θ.

The six trigonometric functions are defined by

$$\sin\theta = y, \qquad \cos\theta = x, \qquad \tan\theta = \frac{y}{x}, \quad x \neq 0,$$

and

$$\csc\theta = \frac{1}{y}, \quad y \neq 0, \qquad \sec\theta = \frac{1}{x}, \quad x \neq 0, \qquad \cot\theta = \frac{x}{y}, \quad y \neq 0.$$

Equivalently,

$$\csc\theta = \frac{1}{\sin\theta}, \qquad \sec\theta = \frac{1}{\cos\theta}, \qquad \cot\theta = \frac{1}{\tan\theta}.$$

Signs of the Trigonometric Functions

The signs of x and y depend on the quadrant:

Quadrant	x	y
I	$+$	$+$
II	$-$	$+$
III	$-$	$-$
IV	$+$	$-$

Therefore,

Quadrant	Positive Trigonometric Functions
I	$\sin, \cos, \tan, \csc, \sec, \cot$
II	$\sin, \csc$
III	$\tan, \cot$
IV	$\cos, \sec$

Example. Determine the quadrant in which θ lies if

$$\sin\theta < 0 \qquad \text{and} \qquad \cot\theta < 0.$$

Solution. Since $\sin\theta < 0$, θ must lie in Quadrant III or IV.

Since $\cot\theta < 0$, θ must lie in Quadrant II or IV.

Therefore,

$$\boxed{\text{Quadrant IV}}.$$

Angles on the Coordinate Axes

The four basic points on the unit circle are

θ	(x, y)
0	$(1, 0)$
$\frac{\pi}{2}$	$(0, 1)$
π	$(-1, 0)$
$\frac{3\pi}{2}$	$(0, -1)$

Consequently,

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$
$\sin\theta$	0	1	0	-1
$\cos\theta$	1	0	-1	0
$\tan\theta$	0	undefined	0	undefined

Example. Find the exact value of

$$\cot(-3\pi).$$

Solution. Since

$$-3\pi + 4\pi = \pi,$$

the angle -3π is coterminal with π. Therefore,

$$\sin\pi = 0, \qquad \cos\pi = -1.$$

Hence

$$\cot(-3\pi) = \frac{\cos(-3\pi)}{\sin(-3\pi)}$$

is undefined.

Example. Find the exact value of

$$\sec\left(\frac{5\pi}{2}\right).$$

Solution. Since

$$\frac{5\pi}{2} - 2\pi = \frac{\pi}{2},$$

we have

$$\cos\left(\frac{5\pi}{2}\right) = \cos\left(\frac{\pi}{2}\right) = 0.$$

Therefore,

$$\sec\left(\frac{5\pi}{2}\right)$$

is undefined.

Common Angles

For the first-quadrant common angles,

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
Degrees	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

A useful pattern is

$$\sin\theta = \frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$$

for

$$\theta = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2},$$

respectively.

The cosine values occur in reverse order:

$$\cos\theta = \frac{\sqrt{4}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{0}}{2}.$$

Using Reference Angles to Find Exact Values

To evaluate a trigonometric function at a common angle:

1. Find a coterminal angle in $[0, 2\pi)$ if necessary.
2. Determine the quadrant of the terminal side.
3. Find the reference angle.
4. Determine the exact value using the reference angle.
5. Apply the appropriate sign based on the quadrant.

Example. Find the exact value of
$$\csc\left(-\frac{3\pi}{4}\right).$$

Solution. A positive coterminal angle is
$$-\frac{3\pi}{4} + 2\pi = \frac{5\pi}{4}.$$
The terminal side lies in Quadrant III, and the reference angle is
$$\frac{\pi}{4}.$$
Since sine is negative in Quadrant III,
$$\sin\left(-\frac{3\pi}{4}\right) = -\frac{\sqrt{2}}{2}.$$
Therefore,
$$\csc\left(-\frac{3\pi}{4}\right) = -\sqrt{2}.$$

Example. Find the exact value of
$$\cot\left(\frac{17\pi}{6}\right).$$

Solution. First find a coterminal angle:
$$\frac{17\pi}{6} - 2\pi = \frac{5\pi}{6}.$$
The terminal side lies in Quadrant II, with reference angle
$$\frac{\pi}{6}.$$
Since tangent and cotangent are negative in Quadrant II,
$$\cot\left(\frac{17\pi}{6}\right) = -\sqrt{3}.$$

Exact Values from a Given Angle

Suppose
$$\theta = \frac{2\pi}{3}.$$
Then θ lies in Quadrant II and has reference angle $\pi/3$. Hence
$$\cos\theta = -\frac{1}{2}.$$

Find the exact values of the following:

$$\text{(a)} \quad \cos^2\theta, \qquad \text{(b)} \quad \cos(-\theta),$$

$$\text{(c)} \quad 2\cos\theta, \qquad \text{(d)} \quad \cos(2\theta).$$

Solution.
For (a),

$$\cos^2\theta = \left(-\frac{1}{2}\right)^2 = \frac{1}{4}.$$

For (b),

$$\cos(-\theta) = \cos\theta = -\frac{1}{2}.$$

For (c),

$$2\cos\theta = 2\left(-\frac{1}{2}\right) = -1.$$

For (d),

$$2\theta = \frac{4\pi}{3}.$$

Therefore,

$$\cos(2\theta) = \cos\left(\frac{4\pi}{3}\right) = -\frac{1}{2}.$$

Using a Calculator for Non-Common Angles

When evaluating trigonometric functions for non-common angles, first make sure the calculator is in the appropriate mode:

$$\text{degree mode} \qquad \text{or} \qquad \text{radian mode.}$$

Use a calculator to approximate:

$$\text{(a)} \quad \cot(3.2), \qquad \text{(b)} \quad \csc(150^\circ), \qquad \text{(c)} \quad \cos\left(\frac{5\pi}{7}\right).$$

For part (c), also determine the quadrant in which the angle terminates.

Application: Simple Harmonic Motion

A mass attached to a spring is moving with simple harmonic motion. After time t, the displacement h of the mass from its rest position is modeled by

$$h(t) = a\cos(\omega t - c),$$

where

$$a = \text{amplitude}, \qquad \omega = \text{angular frequency}, \qquad c = \text{phase constant.}$$

Example. Find the displacement of the mass when

$$a = 6.3\ \text{m}, \qquad \omega = 2.1\ \text{rad/s}, \qquad t = 3\ \text{s}, \qquad c = 1.4\ \text{rad}.$$

Solution. Substitute the given values:

$$h(3) = 6.3\cos\big((2.1)(3) - 1.4\big).$$

Thus

$$h(3) = 6.3\cos(4.9).$$

Using a calculator in radian mode,

$$h(3) \approx 1.17 \text{ m}.$$

Therefore, the mass is approximately

$$1.17 \text{ m}$$

above its rest position at $t = 3$ seconds.

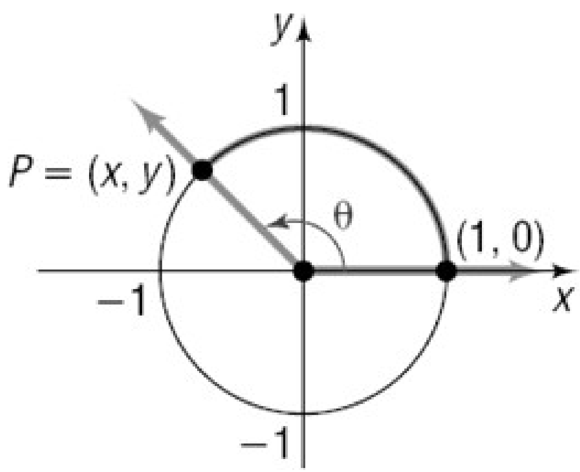


Figure 18: Point on the unit circle

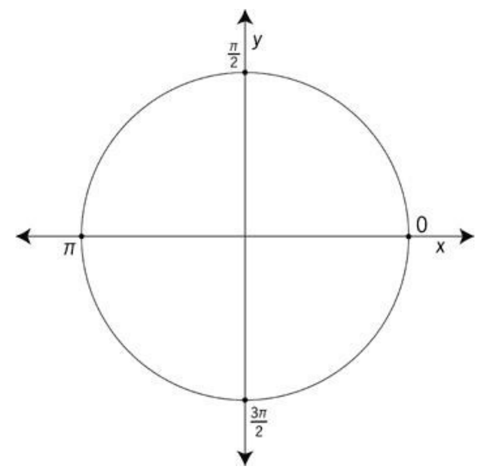


Figure 19: Unit circle with principal axes

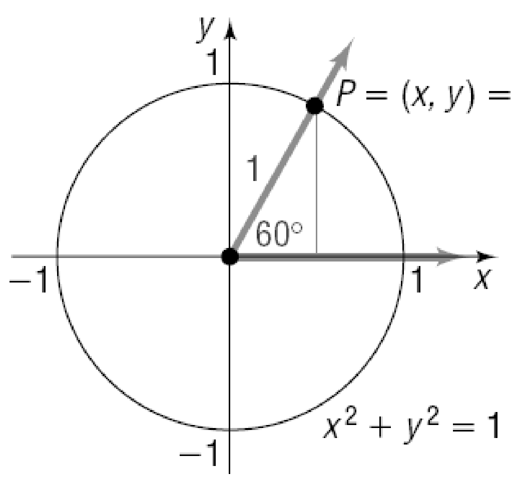


Figure 20: Unit-circle coordinates at 60 degrees

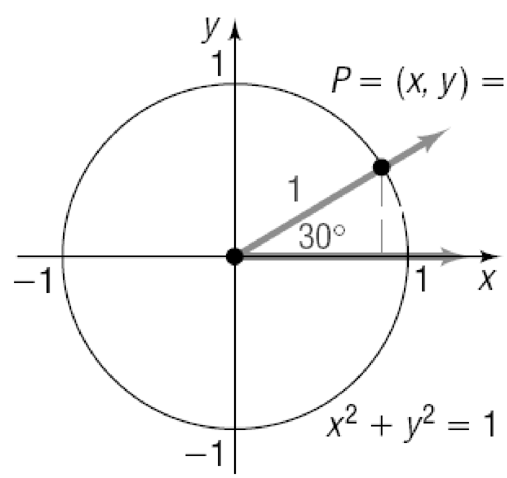


Figure 21: Unit-circle coordinates at 30 degrees

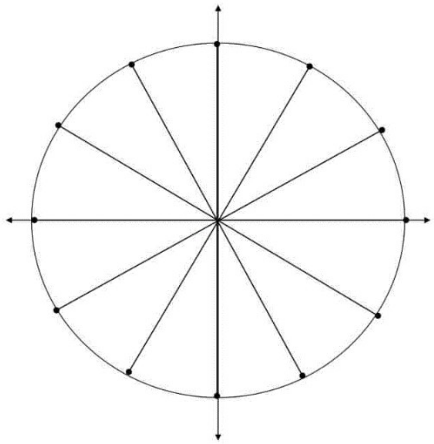


Figure 22: Blank unit-circle framework for special angles

Degrees	Radians	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
0°	0	0	1	0	–	1	–
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	2	$\frac{2\sqrt{3}}{3}$	$\sqrt{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	$\sqrt{2}$	$\sqrt{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2\sqrt{3}}{3}$	2	$\frac{\sqrt{3}}{3}$
90°	$\frac{\pi}{2}$	1	0	–	1	–	0

Figure 23: Exact trigonometric values for special angles

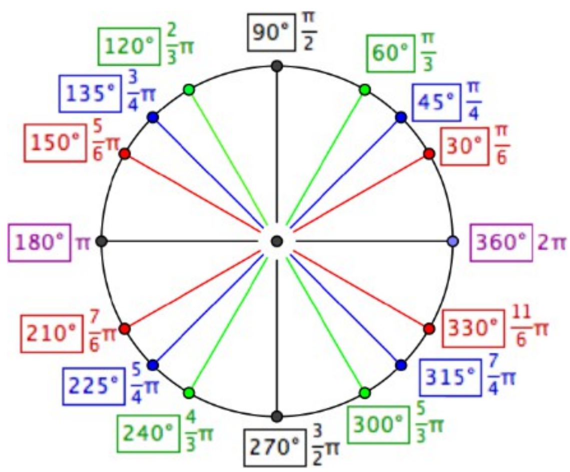


Figure 24: Common angles in degrees and radians

5.3: Graphs of Sine, Cosine, Secant, and Cosecant

A function is periodic if $f(x+P) = f(x)$ for all x in its domain; the smallest positive P is the period. For $y = A\sin(B(x-h)) + D$ or $y = A\cos(B(x-h)) + D$:

$$\text{amplitude} = |A|, \qquad \text{period} = \frac{2\pi}{|B|}, \qquad \text{midline } y = D.$$

The phase shift is h. Divide one period into four equal parts to locate the five key points.

Example. Graph $f(t) = -2\cos t$.

Solution. Amplitude 2, period 2π, midline 0. The negative sign reflects cosine across the horizontal axis. Key values at $0, \pi/2, \pi, 3\pi/2, 2\pi$ are $-2, 0, 2, 0, -2$.

Secant and cosecant are reciprocals of cosine and sine. Their vertical asymptotes occur where the corresponding denominator is zero.

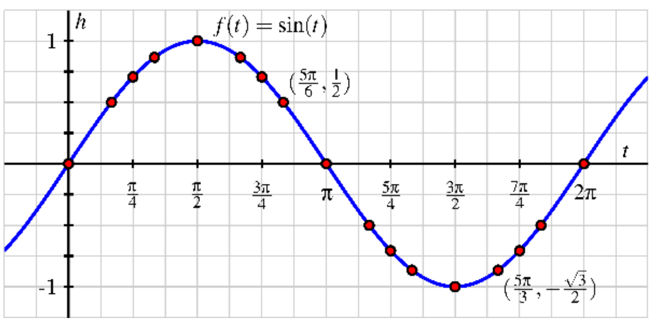


Figure 25: Graph of the sine function with key points

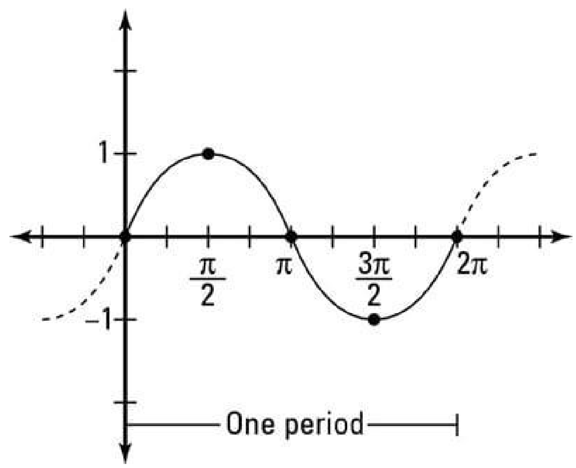


Figure 26: One period of the sine function

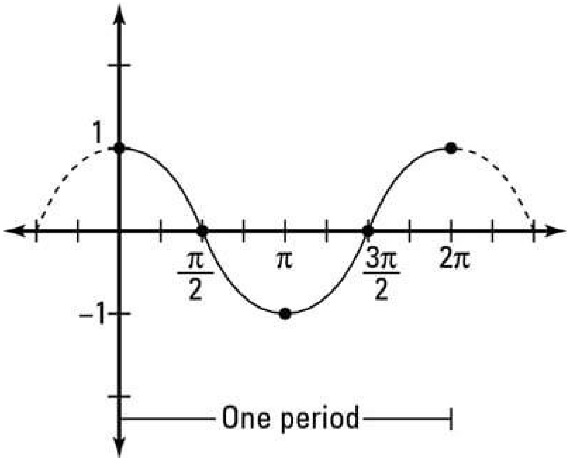


Figure 27: One period of the cosine function

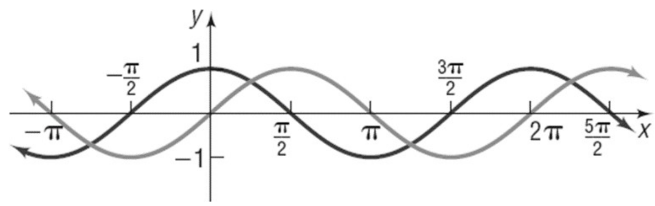


Figure 28: Sine and cosine graphs together

Graphing Transformations of Sine and Cosine

Consider functions of the form

$$y = A\sin(Bt) \qquad \text{and} \qquad y = A\cos(Bt).$$

The parameter A controls the **amplitude**, while the parameter B controls the **period**.

Amplitude

For

$$y = A\sin(Bt) \qquad \text{or} \qquad y = A\cos(Bt),$$

the amplitude is

$$|A|.$$

The amplitude gives the vertical distance from the midline to a maximum or minimum value.

If $A < 0$, the graph is also reflected across the horizontal axis.

Example. Graph

$$f(t) = -2\cos t.$$

Solution. The basic function is

$$y = \cos t.$$

Since

$$A = -2,$$

the amplitude is

$$|A| = 2.$$

The negative sign reflects the cosine graph across the horizontal axis. Thus, the range is

$$[-2, 2].$$

The period remains

$$2\pi.$$

Key points over one period are

t	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$-2\cos t$	-2	0	2	0	-2

Period

For the basic sine and cosine functions,

$$y = \sin t \qquad \text{and} \qquad y = \cos t,$$

the period is

$$2\pi.$$

For

$$y = A\sin(Bt) \qquad \text{or} \qquad y = A\cos(Bt),$$

the period is

$$P = \frac{2\pi}{|B|}.$$

Thus, increasing $|B|$ compresses the graph horizontally, while decreasing $|B|$ stretches the graph horizontally.

Example. Find the period of

$$f(t) = \sin(2t),$$

and then graph one period.

Solution. Here,

$$B = 2.$$

Therefore,

$$P = \frac{2\pi}{2} = \pi.$$

Hence one complete period occurs over an interval of length

$$\pi.$$

Using the rule of fourths,

$$\frac{P}{4} = \frac{\pi}{4}.$$

The five key points over one period are therefore

$$0, \qquad \frac{\pi}{4}, \qquad \frac{\pi}{2}, \qquad \frac{3\pi}{4}, \qquad \pi.$$

The corresponding function values are

t	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
$\sin(2t)$	0	1	0	-1	0

Example. Determine the amplitude and period of

$$y = -5\cos(-3t).$$

Solution. The amplitude is

$$|A| = |-5| = 5.$$

Since

$$B = -3,$$

the period is

$$P = \frac{2\pi}{|-3|} = \frac{2\pi}{3}.$$

Therefore,

$$\text{Amplitude} = 5$$

and

$$\text{Period} = \frac{2\pi}{3}.$$

Also, since cosine is an even function,

$$\cos(-3t) = \cos(3t),$$

so

$$y = -5\cos(-3t) = -5\cos(3t).$$

The Rule of Fourths

To graph one period of a sine or cosine function, divide the period into four equal intervals.
If

$$P = \frac{2\pi}{|B|},$$

then the quarter-period is

$$\frac{P}{4}.$$

The five key t-values over one period are

$$0, \qquad \frac{P}{4}, \qquad \frac{P}{2}, \qquad \frac{3P}{4}, \qquad P.$$

Example. Use the rule of fourths to find the maxima, minima, and t-intercepts and graph one period of

$$y = -2\sin\left(\frac{1}{4}t\right).$$

Solution. Here,

$$A = -2, \qquad B = \frac{1}{4}.$$

The amplitude is

$$|A| = 2.$$

The period is

$$P = \frac{2\pi}{1/4} = 8\pi.$$

Thus,

$$\frac{P}{4} = 2\pi.$$

The five key values over one period are

$$0, \qquad 2\pi, \qquad 4\pi, \qquad 6\pi, \qquad 8\pi.$$

Evaluate the function:

t	0	2π	4π	6π	8π
$-2\sin\left(\frac{1}{4}t\right)$	0	-2	0	2	0

Therefore, over one period:

$$t\text{-intercepts: } 0,\ 4\pi,\ 8\pi,$$

$$\text{minimum: } (2\pi, -2),$$

and

$$\text{maximum: } (6\pi, 2).$$

Summary of Transformations

For

$$y = A\sin(Bt) \qquad \text{or} \qquad y = A\cos(Bt),$$

remember:

$$\text{Amplitude} = |A|$$

and

$$\text{Period} = \frac{2\pi}{|B|}.$$

The transformations can be summarized as follows:

Parameter	Effect on the graph
$\lvert A\rvert > 1$	Vertical stretch
$0 < \lvert A\rvert < 1$	Vertical shrink
$A < 0$	Reflection across the horizontal axis
$\lvert B\rvert > 1$	Horizontal compression
$0 < \lvert B\rvert < 1$	Horizontal stretch

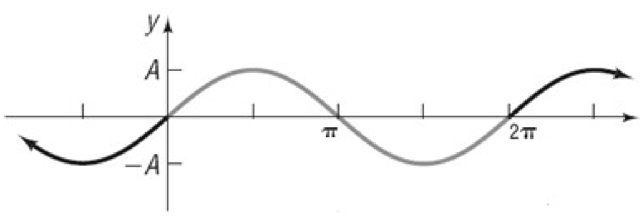


Figure 29: Sinusoidal graph with amplitude A

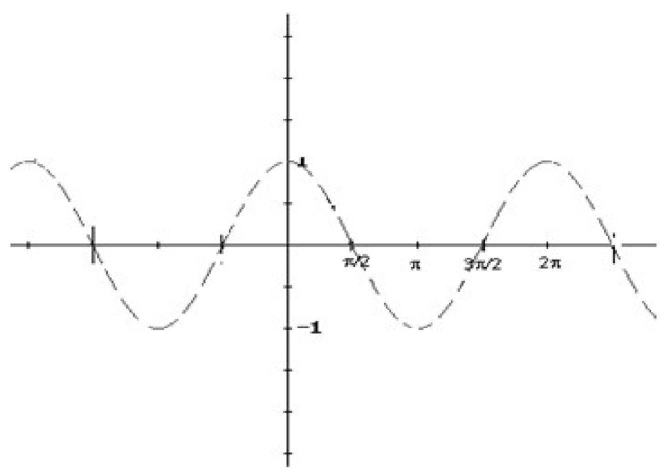


Figure 30: Cosine curve over multiple periods

Graphing with ALEKS

When graphing a transformed trigonometric function in ALEKS:

1. Select the appropriate parent function, such as sine or cosine.

Figure 31: Selection of functions

2. Apply any required translations.

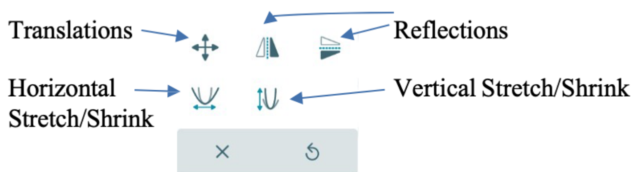


Figure 32: Transformation tools

3. Apply horizontal stretches or compressions.
4. Apply vertical stretches or compressions.
5. Apply reflections when necessary.

Before using the graphing tool, determine the amplitude and period algebraically.

Example. Find an equation for each graph shown.

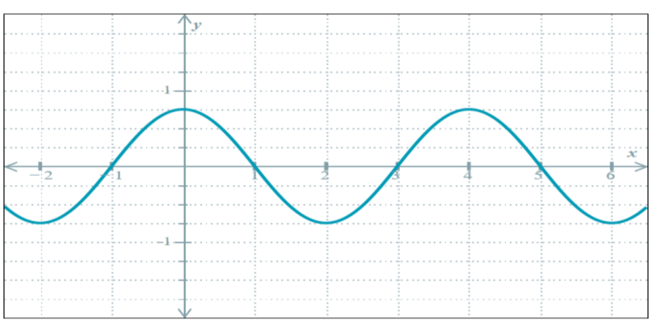


Figure 33: An oscallation curve.

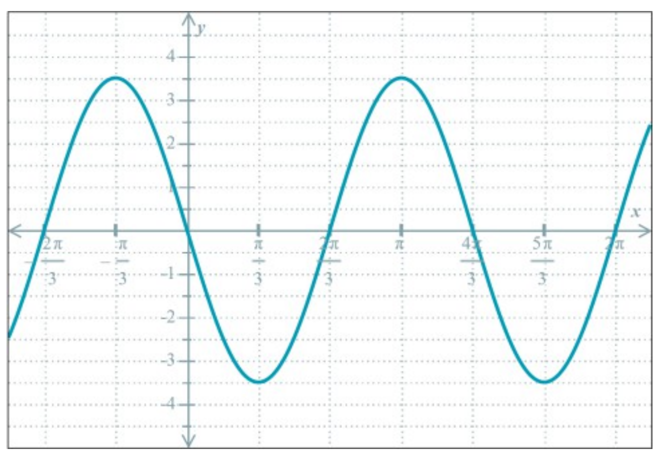


Figure 34: An oscallation curve.

Solution.

1. The first graph has maximum value $\frac{3}{4}$ and minimum value $-\frac{3}{4}$. Thus, the midline is

$$y = 0$$

and the amplitude is

$$A = \frac{3}{4}.$$

Consecutive maxima occur at $x = 0$ and $x = 4$, so the period is

$$P = 4.$$

Since

$$P = \frac{2\pi}{|B|},$$

we have

$$4 = \frac{2\pi}{|B|},$$

so

$$|B| = \frac{\pi}{2}.$$

Since the graph begins at a maximum when $x = 0$, a cosine function is convenient. Therefore, one equation for the graph is

$$y = \frac{3}{4}\cos\left(\frac{\pi}{2}x\right).$$

2. The second graph has maximum value $\frac{7}{2}$ and minimum value $-\frac{7}{2}$. Thus, the midline is

$$y = 0$$

and the amplitude is

$$A = \frac{7}{2}.$$

Consecutive maxima occur at

$$x = -\frac{\pi}{3} \qquad \text{and} \qquad x = \pi.$$

Therefore, the period is

$$P = \pi - \left(-\frac{\pi}{3}\right) = \frac{4\pi}{3}.$$

Using

$$P = \frac{2\pi}{|B|},$$

we obtain

$$\frac{4\pi}{3} = \frac{2\pi}{|B|},$$
$$|B| = \frac{3}{2}.$$

The graph passes through the origin and is decreasing there, so a negative sine function is convenient. Thus, one equation is

$$y = -\frac{7}{2}\sin\left(\frac{3}{2}x\right).$$

Equivalently, using cosine,

$$y = \frac{7}{2}\cos\left[\frac{3}{2}\left(x + \frac{\pi}{3}\right)\right].$$

Graphing Cosecant and Secant Functions

Recall the reciprocal relationships

$$\csc t = \frac{1}{\sin t}, \qquad \sec t = \frac{1}{\cos t}.$$

Therefore, the graphs of cosecant and secant are closely related to the graphs of sine and cosine, respectively.

	$\csc t$	$\sec t$
Domain	$t \neq k\pi, \quad k \in \mathbb{Z}$	$t \neq \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}$
Range	$(-\infty, -1] \cup [1, \infty)$	$(-\infty, -1] \cup [1, \infty)$
Period	2π	2π
Symmetry	Odd	Even
Vertical Asymptotes	$t = k\pi, \quad k \in \mathbb{Z}$	$t = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}$

How to Graph Secant and Cosecant

To graph a secant or cosecant function:

1. Graph the corresponding cosine or sine function as a guide.
2. Find where the guide function equals zero. These locations become vertical asymptotes.
3. Locate the maximum and minimum points of the guide function. These determine the vertices of the secant or cosecant branches.
4. Sketch the reciprocal branches approaching the vertical asymptotes.

Example. Sketch the graph of

$$y = -3\sec(2x).$$

Find three consecutive vertical asymptotes and two points on the graph. Label the asymptotes and the points.

Solution. Since

$$\sec(2x) = \frac{1}{\cos(2x)},$$

the vertical asymptotes occur where

$$\cos(2x) = 0.$$

Thus,

$$2x = \frac{\pi}{2} + k\pi, \qquad k \in \mathbb{Z},$$

and hence

$$x = \frac{\pi}{4} + \frac{k\pi}{2}.$$

Three consecutive vertical asymptotes are

$$x = -\frac{\pi}{4}, \qquad x = \frac{\pi}{4}, \qquad x = \frac{3\pi}{4}.$$

To find convenient points, use values where

$$\cos(2x) = \pm 1.$$

At $x = 0$,

$$y = -3\sec(0) = -3.$$

Therefore, one point is

$$(0, -3).$$

At $x = \frac{\pi}{2}$,

$$y = -3\sec(\pi) = 3.$$

Therefore, another point is

$$\left(\frac{\pi}{2}, 3\right).$$

Thus, three consecutive asymptotes are

$$x = -\frac{\pi}{4}, \qquad x = \frac{\pi}{4}, \qquad x = \frac{3\pi}{4},$$

and two convenient points are

$$(0, -3) \qquad \text{and} \qquad \left(\frac{\pi}{2}, 3\right).$$

The period of the function is

$$P = \frac{2\pi}{|2|} = \pi.$$

5.4: Graphs of Tangent and Cotangent; Inverse Trigonometric Functions

For $y = A\tan(B(x-h)) + D$ and $y = A\cot(B(x-h)) + D$,

$$\text{period} = \frac{\pi}{|B|}.$$

Tangent has vertical asymptotes where cosine is zero; cotangent has vertical asymptotes where sine is zero.

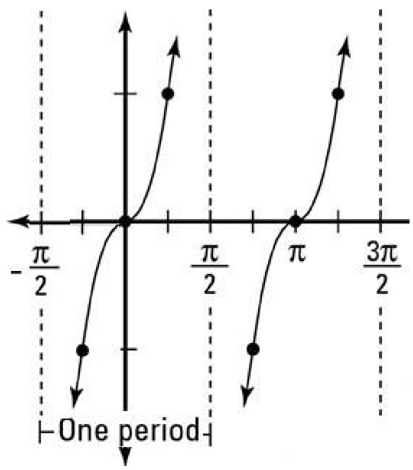


Figure 35: Graph of the tangent function

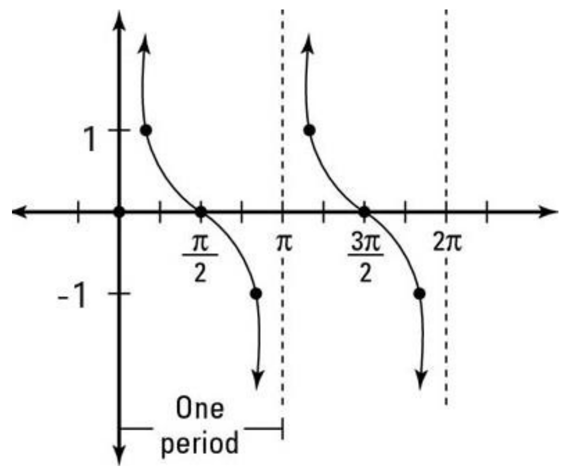


Figure 36: Graph of the cotangent function

	$\tan t$	$\cot t$
Domain	$t \neq \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}$	$t \neq k\pi, \quad k \in \mathbb{Z}$
Range	$(-\infty, \infty)$	$(-\infty, \infty)$
Period	π	π
Symmetry	Odd	Odd
Vertical Asymptotes	$t = \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}$	$t = k\pi, \quad k \in \mathbb{Z}$
Behavior	Increasing on each interval of its domain	Decreasing on each interval of its domain

The Tangent Function

For

$$y = \tan t,$$

one period can be graphed on

$$-\frac{\pi}{2} < t < \frac{\pi}{2}.$$

The vertical asymptotes are

$$t = -\frac{\pi}{2} \qquad \text{and} \qquad t = \frac{\pi}{2}.$$

Three useful points are

$$\left(-\frac{\pi}{4}, -1\right), \qquad (0,0), \qquad \left(\frac{\pi}{4}, 1\right).$$

The graph is increasing from $-\infty$ to ∞ between consecutive vertical asymptotes.

The Cotangent Function

For

$$y = \cot t,$$

one period can be graphed on

$$0 < t < \pi.$$

The vertical asymptotes are

$$t = 0 \qquad \text{and} \qquad t = \pi.$$

Three useful points are

$$\left(\frac{\pi}{4}, 1\right), \qquad \left(\frac{\pi}{2}, 0\right), \qquad \left(\frac{3\pi}{4}, -1\right).$$

The graph is decreasing from ∞ to $-\infty$ between consecutive vertical asymptotes.

Example. Sketch the graph of

$$y = 5\tan(\pi x).$$

Identify A, B, the period, and the vertical asymptotes.

Solution. Comparing

$$y = 5\tan(\pi x)$$

with

$$y = A\tan(Bx),$$

we have

$$A = 5 \qquad \text{and} \qquad B = \pi.$$

The period is

$$P = \frac{\pi}{|B|} = \frac{\pi}{\pi} = 1.$$

For tangent, vertical asymptotes occur when

$$Bx = \frac{\pi}{2} + k\pi, \qquad k \in \mathbb{Z}.$$

Thus,

$$\pi x = \frac{\pi}{2} + k\pi,$$
$$x = \frac{1}{2} + k.$$

Two consecutive vertical asymptotes surrounding the origin are

$$x = -\frac{1}{2} \qquad \text{and} \qquad x = \frac{1}{2}.$$

Useful points within this interval are obtained from the standard tangent points:

$$\left(-\frac{1}{4}, -5\right), \qquad (0, 0), \qquad \left(\frac{1}{4}, 5\right).$$

Therefore,

$$A = 5, \qquad B = \pi, \qquad P = 1$$

with vertical asymptotes

$$x = \frac{1}{2} + k, \qquad k \in \mathbb{Z}.$$

Example. Sketch the graph of

$$y = \frac{1}{3}\cot(2x).$$

Identify A, B, the period, and the vertical asymptotes.

Solution. Comparing

$$y = \frac{1}{3}\cot(2x)$$

with

$$y = A\cot(Bx),$$

we have

$$A = \frac{1}{3} \qquad \text{and} \qquad B = 2.$$

The period is

$$P = \frac{\pi}{|B|} = \frac{\pi}{2}.$$

For cotangent, vertical asymptotes occur when

$$Bx = k\pi, \qquad k \in \mathbb{Z}.$$

Thus,

$$\begin{aligned} 2x &= k\pi, \\ x &= \frac{k\pi}{2}. \end{aligned}$$

For one period, we may use the consecutive asymptotes

$$x = 0 \qquad \text{and} \qquad x = \frac{\pi}{2}.$$

Useful points between these asymptotes are

$$\left(\frac{\pi}{8}, \frac{1}{3}\right), \qquad \left(\frac{\pi}{4}, 0\right), \qquad \left(\frac{3\pi}{8}, -\frac{1}{3}\right).$$

Therefore,

$$A = \frac{1}{3}, \qquad B = 2, \qquad P = \frac{\pi}{2}$$

with vertical asymptotes

$$x = \frac{k\pi}{2}, \qquad k \in \mathbb{Z}.$$

Example. Find an equation for the graph shown.

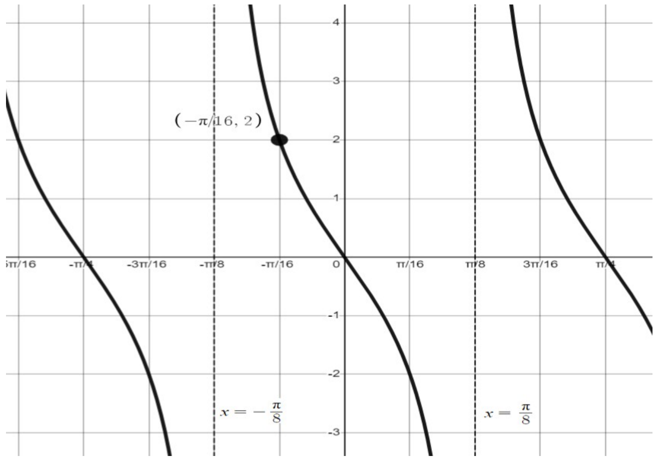


Figure 37: Transformed tangent graph

The graph has consecutive vertical asymptotes

$$x = -\frac{\pi}{8} \qquad \text{and} \qquad x = \frac{\pi}{8},$$

passes through the origin, is decreasing between consecutive asymptotes, and contains the point

$$\left(-\frac{\pi}{16}, 2\right).$$

Solution. Because the graph passes through the origin and has the shape of a tangent function, begin with

$$y = A\tan(Bx).$$

The distance between consecutive vertical asymptotes is the period:

$$P = \frac{\pi}{8} - \left(-\frac{\pi}{8}\right) = \frac{\pi}{4}.$$

Since

$$P = \frac{\pi}{|B|},$$

we have

$$\frac{\pi}{4} = \frac{\pi}{|B|},$$

so

$$|B| = 4.$$

We may therefore write

$$y = A\tan(4x).$$

Now use the point

$$\left(-\frac{\pi}{16}, 2\right).$$

Substituting gives

$$\begin{aligned} 2 &= A\tan\left(4\left(-\frac{\pi}{16}\right)\right) \\ &= A\tan\left(-\frac{\pi}{4}\right) \\ &= -A. \end{aligned}$$

Thus,

$$A = -2.$$

Therefore, an equation for the graph is

$$y = -2\tan(4x).$$

The negative coefficient also explains why the tangent branches are **decreasing** rather than increasing.

Asymptotes

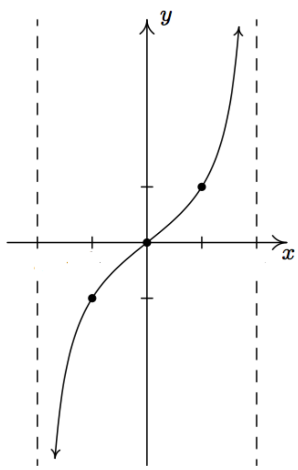


Figure 38: Graph of the secant function

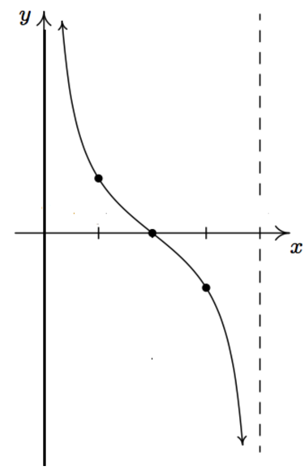


Figure 39: Graph of the cosecant function

5.5: Applications of Trigonometric Graphs

A general sinusoidal function can be written in the form

$$y = A\sin(B(t - C)) + D$$

or

$$y = A\cos(B(t - C)) + D.$$

The parameters determine the following features:

Parameter	Meaning
A	Amplitude $= \lvert A\rvert$
B	Determines the period $P = \dfrac{2\pi}{\lvert B\rvert}$
C	Horizontal shift (phase shift)
D	Vertical shift (midline or average value)

Thus, the midline of the graph is

$$y = D.$$

The maximum and minimum values are

$$y_{\text{max}} = D + |A|, \qquad y_{\text{min}} = D - |A|.$$

Vertical Shift

For

$$y = A\sin(Bt) + D \qquad \text{or} \qquad y = A\cos(Bt) + D,$$

the constant D shifts the graph vertically.
The horizontal line

$$y = D$$

is called the **midline**, and D is also the **average value** of the sinusoidal function.

Phase Shift

A **phase shift**, or horizontal shift, occurs when the input of the trigonometric function is translated horizontally.
For example, consider

$$y = 3\sin(2t + 4) - 1.$$

To identify the horizontal shift, first factor the coefficient of t inside the sine function:

$$y = 3\sin\big(2(t + 2)\big) - 1.$$

Therefore,

$$A = 3, \qquad B = 2, \qquad C = -2, \qquad D = -1.$$

Hence the graph has

$$\text{amplitude} = 3, \qquad P = \frac{2\pi}{2} = \pi,$$

a horizontal shift of 2 units to the left, and midline

$$y = -1.$$

Important: When the function is initially written as

$$y = A\sin(Bt + c) + D,$$

the phase shift is *not* simply c. Since

$$Bt + c = B\left(t + \frac{c}{B}\right),$$

the phase shift is

$$-\frac{c}{B}.$$

Primary Interval

The **primary interval** is an interval containing one complete period of the sinusoidal function. For a function written as

$$y = A\sin(Bt + c) + D$$

or

$$y = A\cos(Bt + c) + D,$$

one convenient primary interval is obtained by solving

$$0 \le Bt + c \le 2\pi.$$

Example. Find the primary interval for

$$y = -3\sin(\pi t + 4) + 2.$$

Solution. Set the angle between 0 and 2π:

$$0 \le \pi t + 4 \le 2\pi.$$

Subtracting 4 gives

$$-4 \le \pi t \le 2\pi - 4.$$

Dividing by π gives

$$-\frac{4}{\pi} \le t \le 2 - \frac{4}{\pi}.$$

Therefore, a primary interval is

$$\left[-\frac{4}{\pi}, 2 - \frac{4}{\pi}\right].$$

Example. Given

$$y = 3.1\sin\left(\frac{\pi}{3}t + \frac{2\pi}{3}\right) - 2.7,$$

identify the amplitude, period, phase shift, vertical shift, and endpoints of a primary interval.

Solution. First factor $\frac{\pi}{3}$ from the argument:

$$\frac{\pi}{3}t + \frac{2\pi}{3} = \frac{\pi}{3}(t+2).$$

Thus,

$$y = 3.1\sin\left(\frac{\pi}{3}(t+2)\right) - 2.7.$$

The amplitude is

$$|A| = 3.1.$$

The period is

$$P = \frac{2\pi}{|B|} = \frac{2\pi}{\pi/3} = 6.$$

The phase shift is

$$C = -2,$$

so the graph is shifted 2 units to the left.

The vertical shift and average value are

$$D = -2.7,$$

so the midline is

$$y = -2.7.$$

Since one period has length 6 and begins at $t = -2$, a convenient primary interval is

$$[-2, 4].$$

Therefore,

$$\begin{aligned} \text{Amplitude} &= 3.1, \\ \text{Period} &= 6, \\ \text{Phase shift} &= -2, \\ \text{Average value} &= -2.7, \\ \text{Primary interval} &= [-2, 4]. \end{aligned}$$

Example. Find the amplitude, period, average value, and phase shift of

$$y = -4\cos(2x + \pi) + 3.$$

Solution. Factor 2 from the argument:

$$2x + \pi = 2\left(x + \frac{\pi}{2}\right).$$

Thus,

$$y = -4\cos\left(2\left(x + \frac{\pi}{2}\right)\right) + 3.$$

The amplitude is

$$|A| = 4.$$

The period is

$$P = \frac{2\pi}{2} = \pi.$$

The average value is

$$D = 3,$$

so the midline is

$$y = 3.$$

The phase shift is

$$-\frac{\pi}{2},$$

which means the graph is shifted $\frac{\pi}{2}$ units to the left.
Therefore,

$$\begin{aligned} \text{Amplitude} &= 4, \\ \text{Period} &= \pi, \\ \text{Average value} &= 3, \\ \text{Phase shift} &= -\frac{\pi}{2}. \end{aligned}$$

Five Key Points: The Rule of Fourths

Once the period P is known, divide one complete period into four equal parts:

$$\frac{P}{4}.$$

If the phase shift is C, the five key x-values are

$$C, \qquad C + \frac{P}{4}, \qquad C + \frac{P}{2}, \qquad C + \frac{3P}{4}, \qquad C + P.$$

For

$$y = A\sin(B(x - C)) + D,$$

the corresponding y-values are

$$D, \qquad D + A, \qquad D, \qquad D - A, \qquad D.$$

For

$$y = A\cos(B(x - C)) + D,$$

the corresponding y-values are

$$D + A, \qquad D, \qquad D - A, \qquad D, \qquad D + A.$$

Position	Sine	Cosine
C	D	$D + A$
$C + \frac{P}{4}$	$D + A$	D
$C + \frac{P}{2}$	D	$D - A$
$C + \frac{3P}{4}$	$D - A$	D
$C + P$	D	$D + A$

Practice: Graphing Sinusoidal Functions

On Your Own. For each function, identify the amplitude, period, phase shift, and midline. Then use the rule of fourths to plot all maxima, minima, and midline points over one complete cycle.

1. $$y = -5\cos\left(\frac{2}{3}x\right) - 1$$

2. $$y = 2\sin\left(\frac{2}{5}x - \frac{4\pi}{5}\right) + 1$$

Finding an Equation from a Sinusoidal Graph

To determine a sinusoidal equation from a graph:

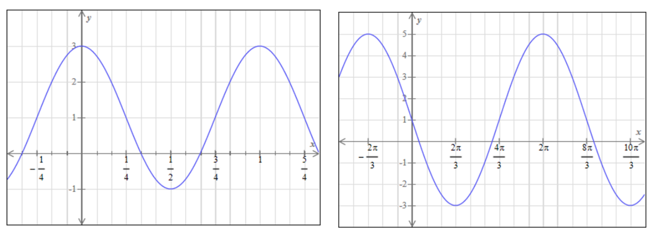


Figure 40: Trig graphs

1. Find the maximum and minimum values.
2. Determine the amplitude:
$$|A| = \frac{y_{\text{max}} - y_{\text{min}}}{2}.$$
3. Determine the midline:
$$D = \frac{y_{\text{max}} + y_{\text{min}}}{2}.$$
4. Determine the period P from the horizontal distance between consecutive corresponding points, such as two maxima.
5. Determine B using
$$|B| = \frac{2\pi}{P}.$$
6. Determine the phase shift by locating a convenient maximum, minimum, or midline crossing.

Cosine is often convenient when beginning at a maximum or minimum, while sine is often convenient when beginning at a midline crossing.

Application: Modeling a Population

The population sizes of many animal species rise and fall periodically over time. Suppose the population of a certain species is modeled by

$$p(t) = 2830 - 1130\cos(0.7t),$$

where $p(t)$ is the population size and t is measured in years.

Example. Find the amplitude, period, number of cycles per year, and maximum population.

Solution. The amplitude is

$$|A| = 1130.$$

The period is

$$P = \frac{2\pi}{0.7} = \frac{20\pi}{7} \approx 8.98 \text{ years}.$$

The number of cycles per year is the reciprocal of the period:

$$\frac{1}{P} = \frac{0.7}{2\pi} \approx 0.111 \text{ cycles per year}.$$

The average population is

$$D = 2830.$$

Therefore, the maximum population is

$$D + |A| = 2830 + 1130 = 3960.$$

Thus,

$$\begin{aligned} \text{Amplitude} &= 1130, \\ \text{Period} &\approx 8.98 \text{ years}, \\ \text{Cycles per year} &\approx 0.111, \\ \text{Maximum population} &= 3960. \end{aligned}$$

Finding a Sinusoidal Function from Data

Suppose periodic data have maximum value y_{max} and minimum value y_{min}.

Step 1: Find the amplitude.

$$|A| = \frac{y_{\text{max}} - y_{\text{min}}}{2}.$$

Step 2: Find the vertical shift or average value.

$$D = \frac{y_{\text{max}} + y_{\text{min}}}{2}.$$

Step 3: Find the period and determine B.

If the cycle repeats every P units, then

$$P = \frac{2\pi}{|B|},$$

so

$$|B| = \frac{2\pi}{P}.$$

Step 4: Determine the phase shift.

Choose a convenient point in the cycle:

- A maximum or minimum is often convenient when using cosine.
- A midline crossing is often convenient when using sine.

Use its horizontal location to determine the phase shift C.

Finally, write the model in one of the forms

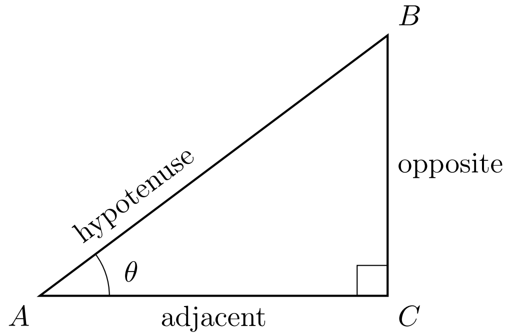
$$y = A\sin(B(t - C)) + D$$

or

$$y = A\cos(B(t - C)) + D.$$

5.6: Trigonometry of Right Triangles

Consider a right triangle with acute angle θ. Relative to the angle θ, the three sides are called the **opposite side**, **adjacent side**, and **hypotenuse**.



For a right triangle,

$$a^2 + b^2 = c^2,$$

where c is the length of the hypotenuse.

The two acute angles are complementary:

$$A + B = 90^\circ.$$

The Six Trigonometric Functions

The six trigonometric functions can be defined using the sides of a right triangle:

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

The reciprocal functions are

$$\csc\theta = \frac{\text{hypotenuse}}{\text{opposite}}, \qquad \sec\theta = \frac{\text{hypotenuse}}{\text{adjacent}}, \qquad \cot\theta = \frac{\text{adjacent}}{\text{opposite}}.$$

Finding Trigonometric Function Values Using Right Triangles

Example. Given

$$\tan A = \frac{2}{5},$$

find the exact values of the remaining five trigonometric functions.

Solution. Since

$$\tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{2}{5},$$

let the opposite side have length 2 and the adjacent side have length 5.

Using the Pythagorean Theorem, the hypotenuse is

$$c^2 = 2^2 + 5^2 = 29,$$

so

$$c = \sqrt{29}.$$

Therefore,

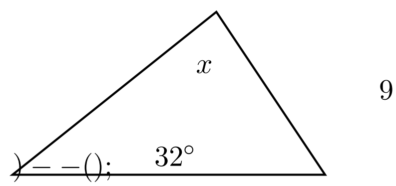
$$\sin A = \frac{2}{\sqrt{29}} = \frac{2\sqrt{29}}{29}, \qquad \cos A = \frac{5}{\sqrt{29}} = \frac{5\sqrt{29}}{29},$$

$$\csc A = \frac{\sqrt{29}}{2}, \qquad \sec A = \frac{\sqrt{29}}{5}, \qquad \cot A = \frac{5}{2}.$$

Finding a Missing Side

Example. Find the value of x in the right triangle shown below.

The side opposite the 32° angle has length 9, and the adjacent side has length x.



Solution. Relative to the 32° angle, the side of length 9 is opposite and x is adjacent. Therefore,

$$\tan 32^\circ = \frac{9}{x}.$$

Solving for x,

$$x \tan 32^\circ = 9,$$

so

$$x = \frac{9}{\tan 32^\circ}.$$

Thus,

$$x \approx 14.4.$$

Solving a Right Triangle

To **solve a triangle** means to determine all three side lengths and all three angle measures.

For a right triangle, one angle is already known:

$$90^\circ.$$

Since the two acute angles are complementary,

$$A + B = 90^\circ.$$

Example. Solve the right triangle if

$$A = 40^\circ \qquad \text{and} \qquad BC = 25 \text{ cm}.$$

Find all three angles and all three side lengths.

Solution. Since the triangle is a right triangle,

$$C = 90^\circ.$$

Therefore,

$$B = 90^\circ - 40^\circ = 50^\circ.$$

The remaining side lengths can then be determined using the appropriate trigonometric ratios.

Relative to angle A,

$$\sin 40^\circ = \frac{BC}{AB} = \frac{25}{AB}.$$

Thus,

$$AB = \frac{25}{\sin 40^\circ}.$$

Also,

$$\tan 40^\circ = \frac{BC}{AC} = \frac{25}{AC},$$

so

$$AC = \frac{25}{\tan 40^\circ}.$$

Therefore,

$$AB \approx 38.9 \text{ cm}, \qquad AC \approx 29.8 \text{ cm}.$$

Hence,

Angles	**Sides**
$A = 40^\circ$	$BC = 25$ cm
$B = 50^\circ$	$AC \approx 29.8$ cm
$C = 90^\circ$	$AB \approx 38.9$ cm

Applications of Right-Triangle Trigonometry

Many real-world problems involving heights and distances can be modeled using right triangles.

Example. A radio transmission tower is 200 feet high. A guy wire is attached to the tower 10 feet below the top and makes an angle of 69° with the ground. How long should the guy wire be?

Solution. Since the wire is attached 10 feet below the top, the attachment height is

$$200 - 10 = 190 \text{ ft}.$$

Let L denote the length of the wire. Relative to the 69° angle, the height 190 ft is the opposite side and L is the hypotenuse. Therefore,

$$\sin 69^\circ = \frac{190}{L}.$$

Solving for L gives

$$L = \frac{190}{\sin 69^\circ}.$$

Thus,

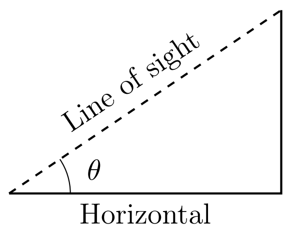
$$L \approx 203.5 \text{ ft}.$$

The guy wire should be approximately 204 feet long.

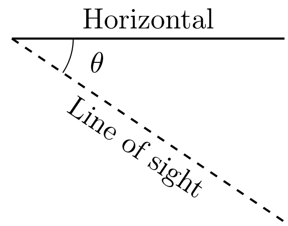
Angles of Elevation and Depression

An **angle of elevation** is the angle measured upward from a horizontal line of sight.

An **angle of depression** is the angle measured downward from a horizontal line of sight.



Angle of elevation



Angle of depression

Because horizontal lines are parallel, the angle of depression from an observer to an object is equal to the angle of elevation from the object to the observer.

Example. A person standing at the top of the Petronas Tower I in Kuala Lumpur, Malaysia, looks toward a residence. Suppose the tower has a height of 1483 ft and the angle of depression from the person's eyes to the residence is 5°.

How far is the residence from the base of the tower?

Solution. The angle of depression equals the corresponding angle of elevation, so the angle at the residence is

$$5^\circ.$$

Let d denote the horizontal distance from the residence to the base of the tower. Then

$$\tan 5^\circ = \frac{1483}{d}.$$

Solving for d gives

$$d = \frac{1483}{\tan 5^\circ}.$$

Therefore,

$$d \approx 16{,}951 \text{ ft}.$$

Thus, the residence is approximately 16,951 feet from the base of the tower.

Trigonometric Cofunctions

The two acute angles of a right triangle are complementary. Therefore, many trigonometric functions are related through complementary angles.

The **cofunction identities** are

$$\sin\theta = \cos\left(\frac{\pi}{2} - \theta\right), \qquad \cos\theta = \sin\left(\frac{\pi}{2} - \theta\right),$$

$$\tan\theta = \cot\left(\frac{\pi}{2} - \theta\right), \qquad \cot\theta = \tan\left(\frac{\pi}{2} - \theta\right),$$

$$\sec\theta = \csc\left(\frac{\pi}{2} - \theta\right), \qquad \csc\theta = \sec\left(\frac{\pi}{2} - \theta\right).$$

In degrees, the same identities can be written using

$$90^\circ - \theta.$$

For example,

$$\sin\theta = \cos(90^\circ - \theta).$$

Practice: Cofunction Identities

Find an equivalent trigonometric function for each expression.

1. $$\cot(25^\circ)$$
2. $$\sin\left(\frac{2\pi}{5}\right)$$
3. $$\csc\left(\frac{\pi}{7}\right)$$

Solution. Using the cofunction identities,

$$\cot(25^\circ) = \tan(90^\circ - 25^\circ) = \tan(65^\circ).$$

Also,

$$\sin\left(\frac{2\pi}{5}\right) = \cos\left(\frac{\pi}{2} - \frac{2\pi}{5}\right) = \cos\left(\frac{\pi}{10}\right).$$

Finally,

$$\csc\left(\frac{\pi}{7}\right) = \sec\left(\frac{\pi}{2} - \frac{\pi}{7}\right) = \sec\left(\frac{5\pi}{14}\right).$$

5.7: Trigonometric Values for Points Not on the Unit Circle

Let θ be an angle in standard position, and let

$$P = (x, y)$$

be a point on the terminal side of θ. Define

$$r = \sqrt{x^2 + y^2}.$$

Then

$$x^2 + y^2 = r^2.$$

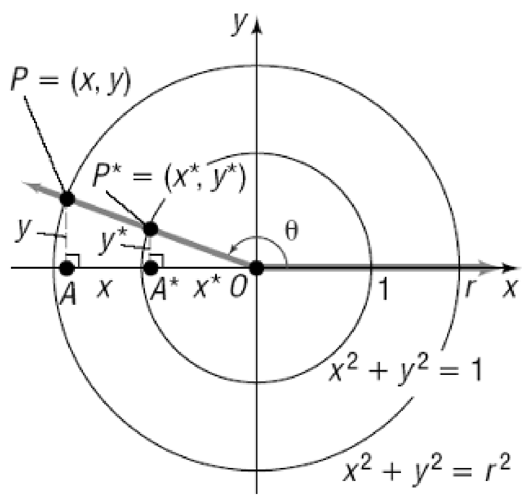


Figure 41: Circle with radius r

The six trigonometric functions are

$$\sin\theta = \frac{y}{r}, \qquad \cos\theta = \frac{x}{r}, \qquad \tan\theta = \frac{y}{x}, \quad x \neq 0,$$

and

$$\csc\theta = \frac{r}{y}, \quad y \neq 0, \qquad \sec\theta = \frac{r}{x}, \quad x \neq 0, \qquad \cot\theta = \frac{x}{y}, \quad y \neq 0.$$

These formulas generalize the unit-circle definitions. On the unit circle, $r = 1$, so they reduce to

$$\sin\theta = y, \qquad \cos\theta = x.$$

Example. The point $P = (-10, -2)$ lies on the terminal side of an angle θ. Find the exact values of the six trigonometric functions.

Solution. First find r:

$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ &= \sqrt{(-10)^2 + (-2)^2} \\ &= \sqrt{104} \\ &= 2\sqrt{26}. \end{aligned}$$

Since $x < 0$ and $y < 0$, the terminal side lies in Quadrant III.

Therefore,

$$\sin\theta = \frac{y}{r} = \frac{-2}{2\sqrt{26}} = -\frac{1}{\sqrt{26}} = -\frac{\sqrt{26}}{26},$$

$$\cos\theta = \frac{x}{r} = \frac{-10}{2\sqrt{26}} = -\frac{5}{\sqrt{26}} = -\frac{5\sqrt{26}}{26},$$

and

$$\tan\theta = \frac{y}{x} = \frac{-2}{-10} = \frac{1}{5}.$$

The reciprocal functions are

$$\csc\theta = -\sqrt{26},$$

$$\sec\theta = -\frac{\sqrt{26}}{5},$$

and

$$\cot\theta = 5.$$

Reference Angles and Exact Trigonometric Values

Example. Find the reference angle and then determine the exact value of the sine of each angle.

1. -315°
2. 420°

Solution.

1. Find a positive coterminal angle:

   $$-315^\circ + 360^\circ = 45^\circ.$$

   Thus, the reference angle is

   $$45^\circ.$$

   Therefore,

   $$\sin(-315^\circ) = \sin(45^\circ) = \frac{\sqrt{2}}{2}.$$

2. Find a coterminal angle between 0° and 360°:

   $$420^\circ - 360^\circ = 60^\circ.$$

   Thus, the reference angle is

   $$60^\circ.$$

   Therefore,

   $$\sin(420^\circ) = \sin(60^\circ) = \frac{\sqrt{3}}{2}.$$

Finding the Remaining Trigonometric Functions

Given

$$\cos\theta = \frac{5}{12} \qquad \text{and} \qquad \sin\theta < 0,$$

find the exact values of the other five trigonometric functions.

Solution. Since

$$\cos\theta > 0 \qquad \text{and} \qquad \sin\theta < 0,$$

the terminal side of θ lies in Quadrant IV.

Using

$$\cos\theta = \frac{x}{r} = \frac{5}{12},$$

let

$$x = 5, \qquad r = 12.$$

Use the Pythagorean relation:

$$x^2 + y^2 = r^2.$$

Thus,

$$\begin{aligned} 5^2 + y^2 &= 12^2, \\ 25 + y^2 &= 144, \\ y^2 &= 119. \end{aligned}$$

Since θ lies in Quadrant IV,

$$y = -\sqrt{119}.$$

Therefore,

$$\sin\theta = \frac{y}{r} = -\frac{\sqrt{119}}{12},$$

$$\tan\theta = \frac{y}{x} = -\frac{\sqrt{119}}{5},$$

$$\csc\theta = -\frac{12}{\sqrt{119}} = -\frac{12\sqrt{119}}{119},$$

$$\sec\theta = \frac{12}{5},$$

and

$$\cot\theta = -\frac{5}{\sqrt{119}} = -\frac{5\sqrt{119}}{119}.$$

Example. Given

$$\tan\theta = -\frac{5}{7} \qquad \text{and} \qquad \sin\theta > 0,$$

find the exact values of the other five trigonometric functions.

Solution. Since

$$\tan\theta < 0,$$

sine and cosine must have opposite signs.

Because
$$\sin\theta > 0,$$
we must have
$$\cos\theta < 0.$$
Therefore, θ lies in Quadrant II.
Using
$$\tan\theta = \frac{y}{x} = -\frac{5}{7},$$
take
$$y = 5, \qquad x = -7.$$
Then
$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ &= \sqrt{(-7)^2 + 5^2} \\ &= \sqrt{74}. \end{aligned}$$
Therefore,
$$\sin\theta = \frac{5}{\sqrt{74}} = \frac{5\sqrt{74}}{74},$$
$$\cos\theta = -\frac{7}{\sqrt{74}} = -\frac{7\sqrt{74}}{74},$$
$$\csc\theta = \frac{\sqrt{74}}{5},$$
$$\sec\theta = -\frac{\sqrt{74}}{7},$$
and
$$\cot\theta = -\frac{7}{5}.$$

6.1–6.2: Fundamental Trigonometric Identities and Verification

Two functions f and g are identical funcitons if $f(x) = g(x)$ for every value of x for which both functions are defined. Such equations are referred to as identities.

An equation that is not an identity is called a *conditional equation*.

The memorizatino and recall of these identities are essentialto your success for the rest of chapter 6 content as well a

Reciprocal identities

$$\csc\theta = \frac{1}{\sin\theta}, \quad \sec\theta = \frac{1}{\cos\theta}, \quad \cot\theta = \frac{1}{\tan\theta}.$$

Quotient identities

$$\tan\theta = \frac{\sin\theta}{\cos\theta}, \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}.$$

Pythagorean identities

$$\sin^2\theta + \cos^2\theta = 1, \quad 1 + \tan^2\theta = \sec^2\theta, \quad 1 + \cot^2\theta = \csc^2\theta.$$

Even-odd identities

$$\sin(-\theta) = -\sin\theta, \quad \cos(-\theta) = \cos\theta, \quad \tan(-\theta) = -\tan\theta.$$

Simplifying Trigonometric Expressions

Simplify the following expression using trigonometric identities:

$$\sin x \csc x + \cot^2 x.$$

Solution. Using the reciprocal identity

$$\csc x = \frac{1}{\sin x},$$

we have

$$\sin x \csc x = 1.$$

Therefore,

$$\sin x \csc x + \cot^2 x = 1 + \cot^2 x = \csc^2 x,$$

where we used the Pythagorean identity

$$1 + \cot^2 x = \csc^2 x.$$

Verifying Trigonometric Identities

When verifying trigonometric identities:

1. Work on only **one side** of the identity, usually the more complicated side, until it has been transformed into the other side. Do not assume that the two sides are equal while proving the identity.
2. Remember the fundamental trigonometric identities. In particular,
$$\sin^2 x + \cos^2 x = 1, \qquad 1 + \tan^2 x = \sec^2 x, \qquad 1 + \cot^2 x = \csc^2 x.$$
3. Rewriting expressions in terms of sine and cosine is often helpful:
$$\tan x = \frac{\sin x}{\cos x}, \qquad \cot x = \frac{\cos x}{\sin x}, \qquad \sec x = \frac{1}{\cos x}, \qquad \csc x = \frac{1}{\sin x}.$$
4. Algebra is essential. Useful techniques include factoring, recognizing special products, distributing, multiplying by conjugates, and combining fractions into a single quotient.

Example. Verify the identity
$$\tan x(\cot x + \tan x) = \sec^2 x.$$

Solution. Begin with the more complicated side:
$$\begin{aligned} \tan x(\cot x + \tan x) &= \tan x \cot x + \tan^2 x \\ &= 1 + \tan^2 x \\ &= \sec^2 x. \end{aligned}$$

Thus, the identity is verified.

Example. Verify the identity
$$\tan x \csc x = \sec x.$$

Solution. Rewrite the left-hand side in terms of sine and cosine:
$$\begin{aligned} \tan x \csc x &= \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} \\ &= \frac{1}{\cos x} \\ &= \sec x. \end{aligned}$$

Thus, the identity is verified.

Example. Verify the identity
$$\frac{\cot x - \tan x}{\cot^2 x - \tan^2 x} = \sin x \cos x.$$

Solution. Factor the denominator as a difference of squares:

$$\begin{aligned}\frac{\cot x - \tan x}{\cot^2 x - \tan^2 x} &= \frac{\cot x - \tan x}{(\cot x - \tan x)(\cot x + \tan x)} \\ &= \frac{1}{\cot x + \tan x} \\ &= \frac{1}{\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x}} \\ &= \frac{1}{\frac{\cos^2 x + \sin^2 x}{\sin x \cos x}} \\ &= \frac{1}{\frac{1}{\sin x \cos x}} \\ &= \sin x \cos x.\end{aligned}$$

Thus, the identity is verified.

Example. Verify the identity

$$\frac{\sin x}{1 + \sin x} - \frac{\sin x}{1 - \sin x} = -\frac{2}{\cot^2 x}.$$

Solution. Begin with the left-hand side and combine the fractions:

$$\begin{aligned}\frac{\sin x}{1 + \sin x} - \frac{\sin x}{1 - \sin x} &= \frac{\sin x(1 - \sin x) - \sin x(1 + \sin x)}{(1 + \sin x)(1 - \sin x)} \\ &= \frac{-2\sin^2 x}{1 - \sin^2 x} \\ &= \frac{-2\sin^2 x}{\cos^2 x} \\ &= -2\tan^2 x \\ &= -\frac{2}{\cot^2 x}.\end{aligned}$$

Thus, the identity is verified.

Example. Verify the identity

$$\tan^{11} x = (\sec^2 x - 1)^5 \tan x.$$

Solution. Using the Pythagorean identity

$$\sec^2 x - 1 = \tan^2 x,$$

we obtain

$$\begin{aligned}(\sec^2 x - 1)^5 \tan x &= (\tan^2 x)^5 \tan x \\ &= \tan^{10} x \tan x \\ &= \tan^{11} x.\end{aligned}$$

Thus, the identity is verified.

Example. Verify the identity

$$\frac{\cos x}{\sin x + 1} = -\frac{\sin x - 1}{\cos x}.$$

Solution. Begin with the left-hand side and multiply by the conjugate:

$$\begin{aligned}\frac{\cos x}{\sin x+1} &= \frac{\cos x}{\sin x+1}\cdot\frac{1-\sin x}{1-\sin x}\\ &= \frac{\cos x(1-\sin x)}{1-\sin^2 x}\\ &= \frac{\cos x(1-\sin x)}{\cos^2 x}\\ &= \frac{1-\sin x}{\cos x}\\ &= -\frac{\sin x-1}{\cos x}.\end{aligned}$$

Thus, the identity is verified.

Writing One Trigonometric Function in Terms of Another

Example. Write $\sin x$ in terms of $\cot x$.

Solution. Start with the Pythagorean identity

$$1+\cot^2 x=\csc^2 x.$$

Since

$$\csc x=\frac{1}{\sin x},$$

we obtain

$$1+\cot^2 x=\frac{1}{\sin^2 x}.$$

Therefore,

$$\sin^2 x=\frac{1}{1+\cot^2 x},$$

and hence

$$\sin x=\pm\frac{1}{\sqrt{1+\cot^2 x}}.$$

The sign depends on the quadrant containing x.

Determining Whether an Equation Is an Identity

Example. Show that

$$\cos(2x)=2\cos x$$

is **not** an identity.

Solution. To show that an equation is not an identity, it is enough to find one value of x for which the two sides are not equal.

Let

$$x=0.$$

Then

$$\cos(2x)=\cos(0)=1,$$

while

$$2\cos x = 2\cos(0) = 2.$$

Since

$$1 \neq 2,$$

the equation is not an identity.

Finding Trigonometric Values Using Identities

Example. Given

$$\tan x = \frac{15}{8}, \qquad x \text{ is in Quadrant III},$$

find the other five trigonometric function values.

Solution. Since

$$\tan x = \frac{15}{8},$$

we may take the opposite and adjacent sides to have magnitudes 15 and 8.

The hypotenuse is

$$r = \sqrt{15^2 + 8^2} = \sqrt{289} = 17.$$

Since x is in Quadrant III, both the x- and y-coordinates are negative. Thus, we may take

$$x = -8, \qquad y = -15, \qquad r = 17.$$

Therefore,

$$\sin x = -\frac{15}{17}, \qquad \cos x = -\frac{8}{17}, \qquad \csc x = -\frac{17}{15}, \qquad \sec x = -\frac{17}{8}, \qquad \cot x = \frac{8}{15}.$$

Example. Given

$$\cos x = -\frac{3}{5}, \qquad x \text{ is in Quadrant II},$$

find $\sin x$, $\sec x$, and $\cot x$.

Solution. Using

$$\sin^2 x + \cos^2 x = 1,$$

we have

$$\begin{aligned} \sin^2 x + \left(-\frac{3}{5}\right)^2 &= 1, \\ \sin^2 x + \frac{9}{25} &= 1, \\ \sin^2 x &= \frac{16}{25}. \end{aligned}$$

Since x is in Quadrant II, $\sin x > 0$. Therefore,

$$\sin x = \frac{4}{5}.$$

Next,

$$\sec x = \frac{1}{\cos x} = -\frac{5}{3},$$

and

$$\cot x = \frac{\cos x}{\sin x} = \frac{-3/5}{4/5} = -\frac{3}{4}.$$

6.3: Sum and Difference Formulas

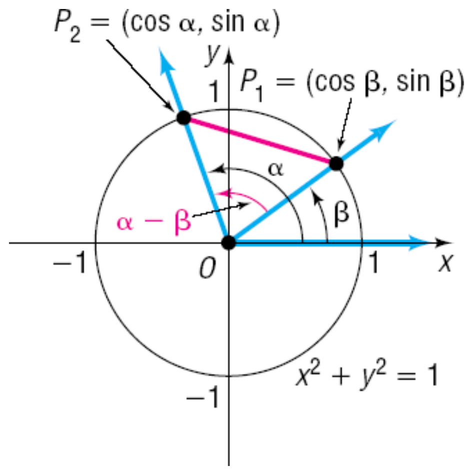


Figure 42: Unit-circle derivation for angle-difference formulas

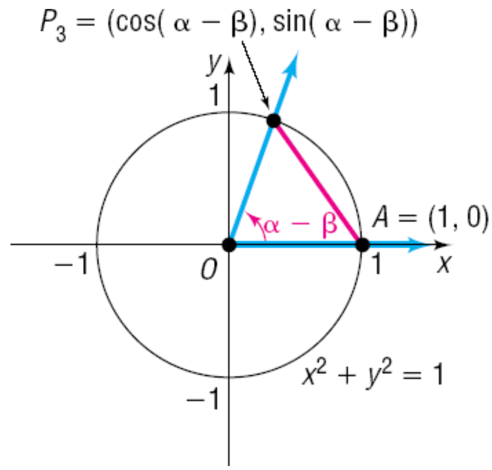


Figure 43: Rotated unit-circle configuration for angle differences

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta,$$
$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta,$$
$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}.$$

Example. Find $\sin(11\pi/12)$ exactly.

Solution. $11\pi/12 = 3\pi/4 + \pi/6$. Hence $\sin(11\pi/12) = \frac{\sqrt{2}}{2}\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\frac{1}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}$.

Sum and Difference Formulas

The sum and difference formulas allow us to find exact trigonometric values for angles that can be written as sums or differences of common angles.

Cosine Sum and Difference Formulas

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

and

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

A convenient way to remember the signs is

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta.$$

Example. Find the exact value of

$$\cos 105^\circ.$$

Solution. Write

$$105^\circ = 60^\circ + 45^\circ.$$

Using the cosine sum formula,

$$\begin{aligned}\cos 105^\circ &= \cos(60^\circ + 45^\circ)\\ &= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ\\ &= \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)\\ &= \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4}.\end{aligned}$$

Therefore,

$$\cos 105^\circ = \frac{\sqrt{2}-\sqrt{6}}{4}.$$

Example. Find the exact value of

$$\cos 5^\circ \cos 40^\circ - \sin 5^\circ \sin 40^\circ.$$

Solution. Recognize the cosine sum formula:

$$\cos\alpha\cos\beta - \sin\alpha\sin\beta = \cos(\alpha+\beta).$$

Therefore,

$$\begin{aligned}\cos 5^\circ \cos 40^\circ - \sin 5^\circ \sin 40^\circ &= \cos(5^\circ + 40^\circ)\\ &= \cos 45^\circ\\ &= \frac{\sqrt{2}}{2}.\end{aligned}$$

Example. Find the exact value of $\cos(\alpha+\beta)$ given that

$$\sin\alpha = \frac{3}{5},$$

with the terminal side of α in Quadrant I, and

$$\tan\beta = \frac{12}{5},$$

with the terminal side of β in Quadrant III.
Solution. Since

$$\sin\alpha = \frac{3}{5}$$

and α lies in Quadrant I, use a 3-4-5 triangle to obtain

$$\cos\alpha = \frac{4}{5}.$$

Since

$$\tan\beta = \frac{12}{5}$$

and β lies in Quadrant III, both sine and cosine are negative. Using a 5-12-13 triangle,

$$\sin\beta = -\frac{12}{13}, \qquad \cos\beta = -\frac{5}{13}.$$

Now apply the cosine sum formula:

$$\begin{aligned}\cos(\alpha+\beta) &= \cos\alpha\cos\beta - \sin\alpha\sin\beta \\ &= \left(\frac{4}{5}\right)\left(-\frac{5}{13}\right) - \left(\frac{3}{5}\right)\left(-\frac{12}{13}\right) \\ &= -\frac{20}{65} + \frac{36}{65} \\ &= \frac{16}{65}.\end{aligned}$$

Therefore,

$$\cos(\alpha+\beta) = \frac{16}{65}.$$

Sine Sum and Difference Formulas

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$

and

$$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta.$$

Equivalently,

$$\sin(\alpha\pm\beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta.$$

Example. Find the exact value of

$$\sin\left(\frac{11\pi}{12}\right).$$

Solution. Write

$$\frac{11\pi}{12} = \frac{2\pi}{3} + \frac{\pi}{4}.$$

Using the sine sum formula,

$$\begin{aligned}\sin\left(\frac{11\pi}{12}\right) &= \sin\left(\frac{2\pi}{3} + \frac{\pi}{4}\right) \\ &= \sin\left(\frac{2\pi}{3}\right)\cos\left(\frac{\pi}{4}\right) + \cos\left(\frac{2\pi}{3}\right)\sin\left(\frac{\pi}{4}\right) \\ &= \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(-\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) \\ &= \frac{\sqrt{6}-\sqrt{2}}{4}.\end{aligned}$$

Therefore,

$$\sin\left(\frac{11\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}.$$

Tangent Sum and Difference Formulas

The tangent formulas can be derived using

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

together with the sine and cosine sum and difference formulas.
The resulting identities are

$$\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$$

and

$$\tan(\alpha-\beta) = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}.$$

Equivalently,

$$\tan(\alpha\pm\beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}.$$

Example. Find the exact value of

$$\tan\left(\frac{\pi}{12}\right).$$

Solution. Write

$$\frac{\pi}{12} = \frac{\pi}{4} - \frac{\pi}{6}.$$

Using the tangent difference formula,

$$\begin{aligned}
\tan\left(\frac{\pi}{12}\right) &= \frac{\tan\left(\frac{\pi}{4}\right) - \tan\left(\frac{\pi}{6}\right)}{1 + \tan\left(\frac{\pi}{4}\right)\tan\left(\frac{\pi}{6}\right)} \\
&= \frac{1 - \frac{\sqrt{3}}{3}}{1 + \frac{\sqrt{3}}{3}}.
\end{aligned}$$

Multiplying the numerator and denominator by 3 gives

$$\frac{3-\sqrt{3}}{3+\sqrt{3}}.$$

Rationalizing,

$$\begin{aligned}
\frac{3-\sqrt{3}}{3+\sqrt{3}} &= \frac{(3-\sqrt{3})^2}{9-3} \\
&= \frac{12 - 6\sqrt{3}}{6} \\
&= 2 - \sqrt{3}.
\end{aligned}$$

Therefore,

$$\tan\left(\frac{\pi}{12}\right) = 2 - \sqrt{3}.$$

6.4: Double-Angle and Half-Angle Identities

$$\sin 2\theta = 2\sin\theta\cos\theta,$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta,$$
$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}.$$

Double-Angle Formulas

The sum formulas can be used to derive formulas for twice an angle.
For sine,

$$\begin{aligned}\sin(2\theta) &= \sin(\theta+\theta)\\ &= \sin\theta\cos\theta + \cos\theta\sin\theta\\ &= 2\sin\theta\cos\theta.\end{aligned}$$

Thus,

$$\sin(2\theta) = 2\sin\theta\cos\theta.$$

For cosine,

$$\begin{aligned}\cos(2\theta) &= \cos(\theta+\theta)\\ &= \cos^2\theta - \sin^2\theta.\end{aligned}$$

Therefore,

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta.$$

Using $\sin^2\theta + \cos^2\theta = 1$, we obtain two additional forms:

$$\cos(2\theta) = 2\cos^2\theta - 1$$

and

$$\cos(2\theta) = 1 - 2\sin^2\theta.$$

For tangent,

$$\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}.$$

Example. If

$$\sin\theta = \frac{1}{6}, \qquad 0 < \theta < \frac{\pi}{2},$$

find the exact value of each of the following:

$$\text{(a) } \sin(2\theta), \qquad \text{(b) } \cos(2\theta), \qquad \text{(c) } \tan(2\theta).$$

Solution. Since θ is in Quadrant I, $\cos\theta > 0$. Using the Pythagorean identity,

$$\begin{aligned}\cos\theta &= \sqrt{1-\sin^2\theta}\\ &= \sqrt{1-\frac{1}{36}}\\ &= \frac{\sqrt{35}}{6}.\end{aligned}$$

For part (a),

$$\begin{aligned}\sin(2\theta) &= 2\sin\theta\cos\theta\\ &= 2\left(\frac{1}{6}\right)\left(\frac{\sqrt{35}}{6}\right)\\ &= \frac{\sqrt{35}}{18}.\end{aligned}$$

For part (b),

$$\begin{aligned}\cos(2\theta) &= 1-2\sin^2\theta\\ &= 1-2\left(\frac{1}{6}\right)^2\\ &= 1-\frac{1}{18}\\ &= \frac{17}{18}.\end{aligned}$$

For part (c), first note that

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1}{\sqrt{35}}.$$

Hence,

$$\begin{aligned}\tan(2\theta) &= \frac{2\tan\theta}{1-\tan^2\theta}\\ &= \frac{\frac{2}{\sqrt{35}}}{1-\frac{1}{35}}\\ &= \frac{\sqrt{35}}{17}.\end{aligned}$$

Power-Reducing Identities

The double-angle identities for cosine can be rearranged to reduce even powers of sine and cosine. From

$$\cos(2\theta) = 1-2\sin^2\theta,$$

we obtain

$$\sin^2\theta = \frac{1-\cos(2\theta)}{2}.$$

Similarly, from

$$\cos(2\theta) = 2\cos^2\theta - 1,$$

we obtain

$$\cos^2\theta = \frac{1+\cos(2\theta)}{2}.$$

Dividing these identities gives

$$\tan^2\theta = \frac{1-\cos(2\theta)}{1+\cos(2\theta)}.$$

Example. Write $\sin^4 x$ in terms of cosines to the first power.

Solution. Since

$$\sin^4 x = (\sin^2 x)^2,$$

apply the power-reducing identity:

$$\begin{aligned}\sin^4 x &= \left(\frac{1-\cos(2x)}{2}\right)^2\\ &= \frac{1}{4}\left(1-2\cos(2x)+\cos^2(2x)\right).\end{aligned}$$

The expression still contains $\cos^2(2x)$, so apply the power-reducing identity again:

$$\cos^2(2x) = \frac{1+\cos(4x)}{2}.$$

Therefore,

$$\begin{aligned}\sin^4 x &= \frac{1}{4}\left(1-2\cos(2x)+\frac{1+\cos(4x)}{2}\right)\\ &= \frac{3}{8}-\frac{1}{2}\cos(2x)+\frac{1}{8}\cos(4x).\end{aligned}$$

Half-Angle Identities

Sum-to-product formulas	Product-to-sum formulas
$\sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2}$	$\sin u\cos v = \frac{1}{2}[\sin(u+v)+\sin(u-v)]$
$\sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$	$\cos u\sin v = \frac{1}{2}[\sin(u+v)-\sin(u-v)]$
$\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2}$	$\cos u\cos v = \frac{1}{2}[\cos(u-v)+\cos(u+v)]$
$\cos x - \cos y = -2\sin\frac{x+y}{2}\sin\frac{x-y}{2}$	$\sin u\sin v = \frac{1}{2}[\cos(u-v)-\cos(u+v)]$

Figure 44: Sum-to-product and product-to-sum identities

The power-reducing identities also lead directly to the half-angle identities. Replacing θ by $u/2$ gives

$$\sin\left(\frac{u}{2}\right) = \pm\sqrt{\frac{1-\cos u}{2}}$$

and

$$\cos\left(\frac{u}{2}\right) = \pm\sqrt{\frac{1+\cos u}{2}}.$$

For tangent,

$$\tan\left(\frac{u}{2}\right) = \pm\sqrt{\frac{1-\cos u}{1+\cos u}}$$

or, equivalently,

$$\tan\left(\frac{u}{2}\right) = \frac{1-\cos u}{\sin u} = \frac{\sin u}{1+\cos u}.$$

Important. The sign in a half-angle formula is determined by the quadrant containing the angle $u/2$.

Example. Can $\cos 22.5^\circ$ be found exactly?

Solution. Since

$$22.5^\circ = \frac{45^\circ}{2},$$

we use the cosine half-angle identity. Because 22.5° is in Quadrant I, the cosine is positive:

$$\begin{aligned}\cos 22.5^\circ &= \sqrt{\frac{1+\cos 45^\circ}{2}}\\ &= \sqrt{\frac{1+\frac{\sqrt{2}}{2}}{2}}\\ &= \frac{\sqrt{2+\sqrt{2}}}{2}.\end{aligned}$$

Example. If

$$\sin\theta = \frac{3}{5}, \qquad 0 < \theta < \frac{\pi}{2},$$

find the exact values of

$$\text{(a)}\ \sin\left(\frac{\theta}{2}\right), \qquad \text{(b)}\ \cos\left(\frac{\theta}{2}\right).$$

Solution. Since θ is in Quadrant I,

$$\cos\theta = \frac{4}{5}.$$

Also,

$$0 < \frac{\theta}{2} < \frac{\pi}{4},$$

so both sine and cosine of $\theta/2$ are positive.

For part (a),

$$\begin{aligned}\sin\left(\frac{\theta}{2}\right) &= \sqrt{\frac{1-\cos\theta}{2}}\\ &= \sqrt{\frac{1-\frac{4}{5}}{2}}\\ &= \sqrt{\frac{1}{10}}\\ &= \frac{\sqrt{10}}{10}.\end{aligned}$$

For part (b),

$$\begin{aligned}\cos\left(\frac{\theta}{2}\right) &= \sqrt{\frac{1+\cos\theta}{2}} \\ &= \sqrt{\frac{1+\frac{4}{5}}{2}} \\ &= \sqrt{\frac{9}{10}} \\ &= \frac{3\sqrt{10}}{10}.\end{aligned}$$

Example. Use a half-angle formula to find the exact value of

$$\sin 105^\circ.$$

Solution. Since

$$105^\circ = \frac{210^\circ}{2},$$

use the sine half-angle formula. Since 105° lies in Quadrant II, sine is positive:

$$\begin{aligned}\sin 105^\circ &= \sqrt{\frac{1-\cos 210^\circ}{2}} \\ &= \sqrt{\frac{1+\frac{\sqrt{3}}{2}}{2}} \\ &= \frac{\sqrt{2+\sqrt{3}}}{2}.\end{aligned}$$

Example. Use a half-angle formula to find the exact value of

$$\tan\left(\frac{5\pi}{12}\right).$$

Solution. Since

$$\frac{5\pi}{12} = \frac{1}{2}\left(\frac{5\pi}{6}\right),$$

use

$$\tan\left(\frac{u}{2}\right) = \frac{\sin u}{1+\cos u}.$$

Thus,

$$\begin{aligned}\tan\left(\frac{5\pi}{12}\right) &= \frac{\sin\left(\frac{5\pi}{6}\right)}{1+\cos\left(\frac{5\pi}{6}\right)} \\ &= \frac{\frac{1}{2}}{1-\frac{\sqrt{3}}{2}} \\ &= \frac{1}{2-\sqrt{3}} \\ &= 2+\sqrt{3}.\end{aligned}$$

Product-to-Sum and Sum-to-Product Formulas

The sum and difference identities can also be rearranged to convert products of trigonometric functions into sums.

Some useful formulas are

$$\sin\alpha\sin\beta = \frac{1}{2}\left[\cos(\alpha-\beta)-\cos(\alpha+\beta)\right]$$

and

$$\cos\alpha+\cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right).$$

Example. Find the exact value of the product, without using a calculator:

$$\sin 67.5^\circ \sin 22.5^\circ.$$

Solution. Using the product-to-sum identity,

$$\begin{aligned}\sin 67.5^\circ \sin 22.5^\circ &= \frac{1}{2}\left[\cos(67.5^\circ-22.5^\circ)-\cos(67.5^\circ+22.5^\circ)\right]\\ &= \frac{1}{2}\left[\cos 45^\circ-\cos 90^\circ\right]\\ &= \frac{1}{2}\left(\frac{\sqrt{2}}{2}\right)\\ &= \frac{\sqrt{2}}{4}.\end{aligned}$$

Example. Find the exact value of

$$\cos\left(\frac{5\pi}{12}\right)+\cos\left(\frac{\pi}{12}\right).$$

Solution. Using the sum-to-product identity,

$$\begin{aligned}\cos\left(\frac{5\pi}{12}\right)+\cos\left(\frac{\pi}{12}\right) &= 2\cos\left(\frac{\frac{5\pi}{12}+\frac{\pi}{12}}{2}\right)\cos\left(\frac{\frac{5\pi}{12}-\frac{\pi}{12}}{2}\right)\\ &= 2\cos\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{6}\right)\\ &= 2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)\\ &= \frac{\sqrt{6}}{2}.\end{aligned}$$

6.5: The Inverse Sine, Cosine, and Tangent Functions

Review of Functions and Their Inverses

Recall the following properties of a function f and its inverse f^{-1}.

1. $f^{-1}(f(x)) = x$ for every x in the domain of f, and $f(f^{-1}(x)) = x$ for every x in the domain of f^{-1}.
2. The domain of f is the range of f^{-1}, and the range of f is the domain of f^{-1}.
3. The graphs of f and f^{-1} are symmetric with respect to the line $y = x$.
4. If $y = f(x)$ has an inverse function, then interchanging x and y gives $x = f(y)$. Solving this equation for y gives
$$y = f^{-1}(x).$$

The Inverse Sine Function

The sine function
$$y = \sin x$$
is not one-to-one on $(-\infty, \infty)$. Therefore, it does not have an inverse on its entire domain.

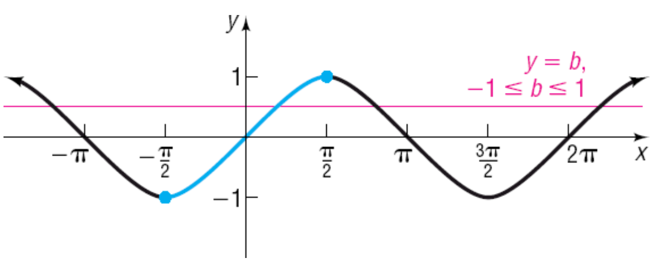


Figure 45: Restricted sine domain and range

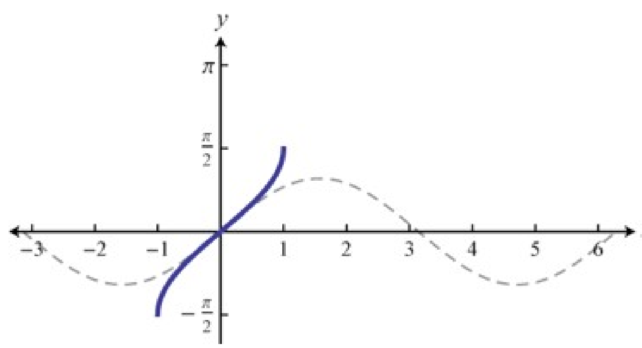


Figure 46: Restricted sine function for defining arcsine

To define an inverse, we restrict the domain of sine to
$$-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}.$$
On this interval, sine is one-to-one and has range $[-1, 1]$. Thus,
$$\sin : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \longrightarrow [-1, 1].$$
Its inverse is called the **inverse sine function** and is written
$$y = \sin^{-1} x$$
or
$$y = \arcsin x.$$

Therefore,

$$\boxed{y = \sin^{-1} x \quad \Longleftrightarrow \quad x = \sin y, \qquad -1 \le x \le 1, \quad -\frac{\pi}{2} \le y \le \frac{\pi}{2}.}$$

In particular,

$$\mathrm{Dom}(\sin^{-1}) = [-1, 1], \qquad \mathrm{Range}(\sin^{-1}) = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right].$$

Example. Find the exact value of

$$\sin^{-1}\left(\frac{1}{2}\right).$$

Solution. Let

$$y = \sin^{-1}\left(\frac{1}{2}\right).$$

Then

$$\sin y = \frac{1}{2}, \qquad -\frac{\pi}{2} \le y \le \frac{\pi}{2}.$$

Since

$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2},$$

we obtain

$$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}.$$

Example. Find the exact value of

$$\sin^{-1}\left(-\frac{\sqrt{2}}{2}\right).$$

Solution. We seek an angle y satisfying

$$\sin y = -\frac{\sqrt{2}}{2}$$

with

$$-\frac{\pi}{2} \le y \le \frac{\pi}{2}.$$

Therefore,

$$y = -\frac{\pi}{4},$$

and hence

$$\sin^{-1}\left(-\frac{\sqrt{2}}{2}\right) = -\frac{\pi}{4}.$$

Using a Calculator

If the value is not a common unit-circle value, use a calculator.

Be sure to check whether the calculator is in **degree mode** or **radian mode**.

For example, find

$$\sin^{-1}\left(\frac{3}{5}\right).$$

Solution. In radian mode,

$$\sin^{-1}\left(\frac{3}{5}\right) \approx 0.644.$$

Properties of Sine and Inverse Sine

The inverse-function identities are

$$\sin^{-1}(\sin x) = x, \qquad -\frac{\pi}{2} \le x \le \frac{\pi}{2},$$

and

$$\sin(\sin^{-1} x) = x, \qquad -1 \le x \le 1.$$

Important. The identity

$$\sin^{-1}(\sin x) = x$$

is only guaranteed when

$$-\frac{\pi}{2} \le x \le \frac{\pi}{2}.$$

Example. Find the exact value of

$$\sin^{-1}\left(\sin\frac{4\pi}{3}\right).$$

Solution. Since

$$\sin\frac{4\pi}{3} = -\frac{\sqrt{3}}{2},$$

we have

$$\sin^{-1}\left(\sin\frac{4\pi}{3}\right) = \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right).$$

The inverse sine must return a value in

$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right].$$

Therefore,

$$\sin^{-1}\left(\sin\frac{4\pi}{3}\right) = -\frac{\pi}{3}.$$

Example. Find the exact value of

$$\sin^{-1}\left(\sin\frac{7\pi}{4}\right).$$

Solution. Since

$$\sin\frac{7\pi}{4} = -\frac{\sqrt{2}}{2},$$

we obtain

$$\sin^{-1}\left(\sin\frac{7\pi}{4}\right) = -\frac{\pi}{4}.$$

Example. Find the exact value of

$$\sin\left(\sin^{-1}\frac{1}{2}\right).$$

Solution. Since $\frac{1}{2} \in [-1, 1]$, we can use the inverse-function identity:

$$\sin\left(\sin^{-1}\frac{1}{2}\right) = \frac{1}{2}.$$

The Inverse Cosine Function

The cosine function is not one-to-one on its entire domain. We therefore restrict cosine to

$$0 \leq x \leq \pi.$$

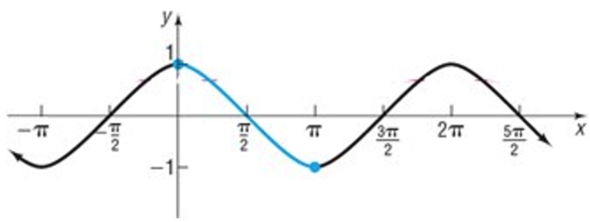


Figure 47: Restricted cosine domain and range

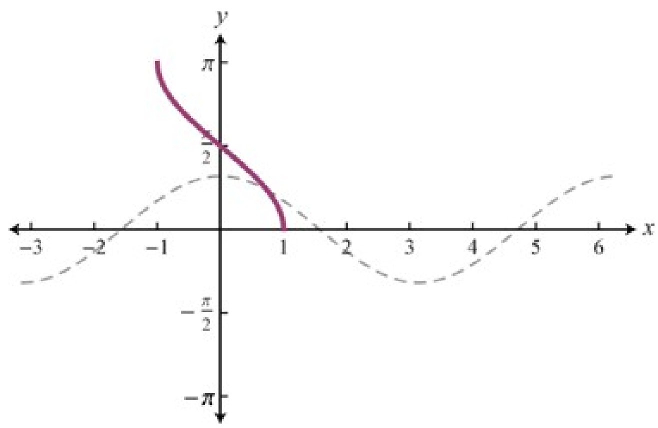


Figure 48: Restricted cosine function for defining arccosine

On this interval,

$$\cos : [0, \pi] \longrightarrow [-1, 1]$$

is one-to-one.

Its inverse is the **inverse cosine function**,

$$y = \cos^{-1} x$$

or

$$y = \arccos x.$$

Thus,

$$\boxed{y = \cos^{-1} x \quad \Longleftrightarrow \quad x = \cos y, \qquad -1 \leq x \leq 1, \quad 0 \leq y \leq \pi.}$$

Therefore,

$$\operatorname{Dom}(\cos^{-1}) = [-1, 1], \qquad \operatorname{Range}(\cos^{-1}) = [0, \pi].$$

Example. Find

$$\cos^{-1}\left(\frac{1}{2}\right).$$

Solution. We seek $y \in [0, \pi]$ such that

$$\cos y = \frac{1}{2}.$$

Since

$$\cos \frac{\pi}{3} = \frac{1}{2},$$

we obtain

$$\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}.$$

Example. Find

$$\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right).$$

Solution. We seek $y \in [0, \pi]$ such that

$$\cos y = -\frac{\sqrt{3}}{2}.$$

Therefore,

$$y = \frac{5\pi}{6},$$

so

$$\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{5\pi}{6}.$$

Properties of Cosine and Inverse Cosine

We have

$$\cos^{-1}(\cos x) = x, \qquad 0 \le x \le \pi,$$

and

$$\cos(\cos^{-1} x) = x, \qquad -1 \le x \le 1.$$

Example. Find the exact value of

$$\cos^{-1}\left(\cos\frac{5\pi}{3}\right).$$

Solution. Since

$$\cos\frac{5\pi}{3} = \frac{1}{2},$$

we obtain

$$\cos^{-1}\left(\cos\frac{5\pi}{3}\right) = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}.$$

Example. Find the exact value of

$$\cos^{-1}\left(\cos\frac{9\pi}{8}\right).$$

Solution. Since

$$\pi < \frac{9\pi}{8} < 2\pi,$$

the angle $\frac{9\pi}{8}$ is outside the range $[0, \pi]$ of $\cos^{-1}$.
Using

$$\cos(2\pi - \theta) = \cos\theta,$$

we have

$$\cos\frac{9\pi}{8} = \cos\frac{7\pi}{8}.$$

Because

$$\frac{7\pi}{8} \in [0, \pi],$$

it follows that

$$\cos^{-1}\left(\cos\frac{9\pi}{8}\right) = \frac{7\pi}{8}.$$

Example. Find

$$\cos\left(\cos^{-1}\frac{3}{4}\right).$$

Solution. Since $\frac{3}{4} \in [-1, 1]$,

$$\cos\left(\cos^{-1}\frac{3}{4}\right) = \frac{3}{4}.$$

The Inverse Tangent Function

The tangent function is not one-to-one on its entire domain. We restrict tangent to

$$-\frac{\pi}{2} < x < \frac{\pi}{2}.$$

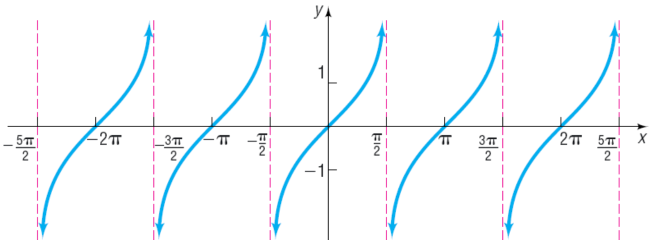


Figure 49: Graph of tangent with vertical asymptotes

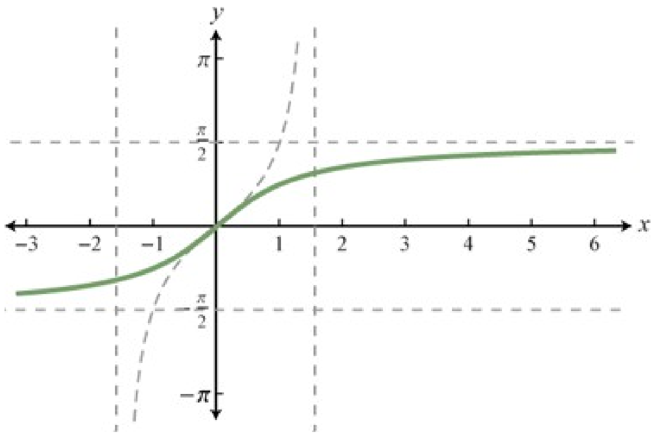


Figure 50: Graph of inverse tangent

On this interval,

$$\tan : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \longrightarrow (-\infty, \infty)$$

is one-to-one.

Its inverse is the **inverse tangent function**,

$$y = \tan^{-1} x$$

or

$$y = \arctan x.$$

Thus,

$$\boxed{y = \tan^{-1} x \quad \Longleftrightarrow \quad x = \tan y, \qquad -\infty < x < \infty, \quad -\frac{\pi}{2} < y < \frac{\pi}{2}.}$$

Therefore,

$$\mathrm{Dom}(\tan^{-1}) = (-\infty, \infty),$$

and

$$\mathrm{Range}(\tan^{-1}) = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Properties of Tangent and Inverse Tangent

We have
$$\tan^{-1}(\tan x) = x, \qquad -\frac{\pi}{2} < x < \frac{\pi}{2},$$
and
$$\tan(\tan^{-1} x) = x, \qquad -\infty < x < \infty.$$

Example. Find the exact value of
$$\tan^{-1}\left(-\frac{\sqrt{3}}{3}\right).$$

Solution. We seek an angle y satisfying
$$\tan y = -\frac{\sqrt{3}}{3}, \qquad -\frac{\pi}{2} < y < \frac{\pi}{2}.$$
Since
$$\tan\left(-\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{3},$$
we obtain
$$\tan^{-1}\left(-\frac{\sqrt{3}}{3}\right) = -\frac{\pi}{6}.$$

Example. Use a calculator to approximate
$$\tan^{-1}(8).$$

Solution. In radian mode,
$$\tan^{-1}(8) \approx 1.446.$$

Example. Find the exact value of
$$\tan^{-1}\left(\tan\left(-\frac{5\pi}{3}\right)\right).$$

Solution. The angle $-\frac{5\pi}{3}$ is not in the principal range
$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$
Since tangent has period π,
$$\tan\left(-\frac{5\pi}{3}\right) = \tan\left(-\frac{5\pi}{3} + 2\pi\right) = \tan\frac{\pi}{3}.$$
Therefore,
$$\tan^{-1}\left(\tan\left(-\frac{5\pi}{3}\right)\right) = \frac{\pi}{3}.$$

Compositions Involving Inverse Trigonometric Functions

Example. Find
$$\cos\left(\sin^{-1}\left(-\frac{1}{2}\right)\right).$$

Solution. Let
$$\theta = \sin^{-1}\left(-\frac{1}{2}\right).$$

Then

$$\theta = -\frac{\pi}{6}.$$

Therefore,

$$\cos\left(\sin^{-1}\left(-\frac{1}{2}\right)\right) = \cos\left(-\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}.$$

Example. Find

$$\tan\left(\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)\right).$$

Solution. Let

$$\theta = \cos^{-1}\left(-\frac{\sqrt{2}}{2}\right).$$

Since $\theta \in [0, \pi]$,

$$\theta = \frac{3\pi}{4}.$$

Therefore,

$$\tan\left(\cos^{-1}\left(-\frac{\sqrt{2}}{2}\right)\right) = \tan\frac{3\pi}{4} = -1.$$

Example. Find

$$\sec\left(\cos^{-1}\frac{\sqrt{3}}{2}\right).$$

Solution. Since

$$\cos\left(\cos^{-1}\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{2},$$

we have

$$\sec\left(\cos^{-1}\frac{\sqrt{3}}{2}\right) = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}.$$

Example. Find

$$\cos^{-1}\left(\sin\frac{5\pi}{3}\right).$$

Solution. Since

$$\sin\frac{5\pi}{3} = -\frac{\sqrt{3}}{2},$$

we have

$$\cos^{-1}\left(\sin\frac{5\pi}{3}\right) = \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{5\pi}{6}.$$

Example. Find

$$\sin^{-1}\left(\cos\frac{\pi}{6}\right).$$

Solution. Since

$$\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2},$$

we obtain

$$\sin^{-1}\left(\cos\frac{\pi}{6}\right) = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}.$$

Inverse Trigonometric Expressions Using Triangles

Inverse trigonometric functions can also be used to evaluate expressions that are not standard unit-circle values.

Example. Find
$$\sec\left(\tan^{-1}\left(-\frac{1}{7}\right)\right).$$

Solution. Let
$$\theta = \tan^{-1}\left(-\frac{1}{7}\right).$$

Then
$$\tan\theta = -\frac{1}{7}.$$

Since
$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

and $\tan\theta < 0$, the angle θ lies in Quadrant IV.

Construct a right triangle with
$$\text{opposite} = -1, \qquad \text{adjacent} = 7.$$

The hypotenuse is
$$\sqrt{1^2 + 7^2} = \sqrt{50} = 5\sqrt{2}.$$

Thus
$$\cos\theta = \frac{7}{5\sqrt{2}},$$

and therefore
$$\sec\theta = \frac{5\sqrt{2}}{7}.$$

Hence
$$\sec\left(\tan^{-1}\left(-\frac{1}{7}\right)\right) = \frac{5\sqrt{2}}{7}.$$

Example. Find
$$\cot\left(\sin^{-1}\frac{1}{3}\right).$$

Solution. Let
$$\theta = \sin^{-1}\frac{1}{3}.$$

Then
$$\sin\theta = \frac{1}{3}.$$

Since $\theta \in [-\pi/2, \pi/2]$ and $\sin\theta > 0$, θ lies in Quadrant I.

Take
$$\text{opposite} = 1, \qquad \text{hypotenuse} = 3.$$

Then
$$\text{adjacent} = \sqrt{3^2 - 1^2} = \sqrt{8} = 2\sqrt{2}.$$

Therefore,
$$\cot\theta = \frac{\text{adjacent}}{\text{opposite}} = 2\sqrt{2}.$$

Thus

$$\cot\left(\sin^{-1}\frac{1}{3}\right) = 2\sqrt{2}.$$

Example. Simplify

$$\sin\left(\cos^{-1}\frac{x}{\sqrt{x^2+9}}\right).$$

Solution. Let

$$\theta = \cos^{-1}\left(\frac{x}{\sqrt{x^2+9}}\right).$$

Then

$$\cos\theta = \frac{x}{\sqrt{x^2+9}}.$$

Using

$$\sin^2\theta + \cos^2\theta = 1,$$

we obtain

$$\sin^2\theta = 1 - \frac{x^2}{x^2+9} = \frac{9}{x^2+9}.$$

Because

$$0 \le \theta \le \pi,$$

we know $\sin\theta \ge 0$. Hence

$$\sin\theta = \frac{3}{\sqrt{x^2+9}}.$$

Therefore,

$$\sin\left(\cos^{-1}\frac{x}{\sqrt{x^2+9}}\right) = \frac{3}{\sqrt{x^2+9}}.$$

Calculator Practice

Approximate each expression correct to two decimal places:

$$\sin^{-1}(0.521), \qquad \csc^{-1}(-2.5), \qquad \sec^{-1}(3), \qquad \cot^{-1}(-2).$$

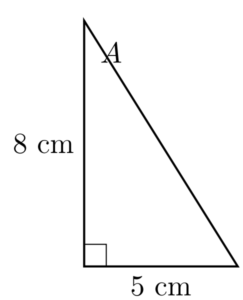
When using a calculator, remember that most calculators have direct buttons only for

$$\sin^{-1}, \qquad \cos^{-1}, \qquad \tan^{-1}.$$

Reciprocal identities may be used for the other inverse trigonometric functions, subject to the appropriate principal-value conventions.

Solving a Right Triangle

Example. Solve the right triangle whose legs have lengths 8 cm and 5 cm.



Solution. First find the hypotenuse c using the Pythagorean Theorem:

$$c^2 = 8^2 + 5^2 = 89.$$

Thus

$$c = \sqrt{89} \approx 9.43 \text{ cm}.$$

For angle A,

$$\tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{5}{8}.$$

Therefore,

$$A = \tan^{-1}\left(\frac{5}{8}\right) \approx 32.01^\circ.$$

Let the remaining acute angle be B. Since

$$A + B = 90^\circ,$$

we obtain

$$B = 90^\circ - A \approx 57.99^\circ.$$

Thus the triangle has

Angles	**Sides**
90°	$\sqrt{89}$ cm ≈ 9.43 cm
32.01°	5 cm
57.99°	8 cm

6.6: Trigonometric Equations

Principal Roots: The solution to a trigonometric equation found by using inverse trigonometric functions.

Solutions over the interval $[0, 2\pi)$**:** Use the principal root and trigonometric behavior to find all of the other solutions in the interval.

Solutions from the set of real numbers: Use the principal root and periodic behavior to state all possible solutions.

Example. Solve exactly

$$2\cos x + 2 = 1.$$

Find the principal root, all solutions over the interval $[0, 2\pi)$, and all real solutions.

Solution. First isolate the trigonometric function:

$$2\cos x = -1, \qquad \cos x = -\frac{1}{2}.$$

The principal root is

$$x = \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}.$$

Since cosine is negative in Quadrants II and III, the solutions over $[0, 2\pi)$ are

$$x = \frac{2\pi}{3}, \frac{4\pi}{3}.$$

Using the periodicity of cosine, all real solutions are

$$x = \frac{2\pi}{3} + 2\pi k \qquad \text{or} \qquad x = \frac{4\pi}{3} + 2\pi k, \qquad k \in \mathbb{Z}.$$

Example. Solve each equation exactly for the principal root, all solutions over the interval $0 \le \theta < 2\pi$, and all real solutions.

$$\text{(a)} \quad 2\sin\theta + 6 = 5 \qquad\qquad \text{(b)} \quad 8\tan\theta + 7\sqrt{3} = -\sqrt{3}.$$

Solution. (a) First isolate the sine function:

$$2\sin\theta + 6 = 5 \quad \Longrightarrow \quad 2\sin\theta = -1 \quad \Longrightarrow \quad \sin\theta = -\frac{1}{2}.$$

The principal root is

$$\theta = \sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}.$$

Since sine is negative in Quadrants III and IV, the solutions over $[0, 2\pi)$ are

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}.$$

Thus, all real solutions are

$$\theta = \frac{7\pi}{6} + 2\pi k \qquad \text{or} \qquad \theta = \frac{11\pi}{6} + 2\pi k, \qquad k \in \mathbb{Z}.$$

(b) First isolate the tangent function:

$$8\tan\theta + 7\sqrt{3} = -\sqrt{3} \quad\Longrightarrow\quad 8\tan\theta = -8\sqrt{3} \quad\Longrightarrow\quad \tan\theta = -\sqrt{3}.$$

The principal root is

$$\theta = \tan^{-1}(-\sqrt{3}) = -\frac{\pi}{3}.$$

Since tangent is negative in Quadrants II and IV, the solutions over $[0, 2\pi)$ are

$$\theta = \frac{2\pi}{3}, \frac{5\pi}{3}.$$

Since tangent has period π, all real solutions can be written as

$$\theta = -\frac{\pi}{3} + k\pi, \qquad k \in \mathbb{Z}.$$

Example. Solve the following for all solutions in $[0, 2\pi)$:

$$(\cot x + 1)(2\sin x - \sqrt{3}) = 0.$$

Solution. By the zero-product property,

$$\cot x + 1 = 0 \qquad \text{or} \qquad 2\sin x - \sqrt{3} = 0.$$

For the first equation,

$$\cot x = -1.$$

Since $\cot x = -1$ in Quadrants II and IV,

$$x = \frac{3\pi}{4}, \frac{7\pi}{4}.$$

For the second equation,

$$2\sin x - \sqrt{3} = 0 \quad\Longrightarrow\quad \sin x = \frac{\sqrt{3}}{2}.$$

Since sine is positive in Quadrants I and II,

$$x = \frac{\pi}{3}, \frac{2\pi}{3}.$$

Therefore, all solutions in $[0, 2\pi)$ are

$$x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{7\pi}{4}.$$

Example. Solve for all solutions in $[0, 2\pi)$:

$$2\cos^2\theta + \sqrt{3}\cos\theta = 0.$$

Solution. Factor the equation:

$$\cos\theta(2\cos\theta + \sqrt{3}) = 0.$$

By the zero-product property,

$$\cos\theta = 0 \qquad \text{or} \qquad 2\cos\theta + \sqrt{3} = 0.$$

If

$$\cos\theta = 0,$$

then
$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}.$$

For the second equation,
$$2\cos\theta + \sqrt{3} = 0 \quad \Longrightarrow \quad \cos\theta = -\frac{\sqrt{3}}{2}.$$

Thus,
$$\theta = \frac{5\pi}{6}, \frac{7\pi}{6}.$$

Therefore,
$$\theta = \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}.$$

Example. Solve for all solutions in $[0, 2\pi)$:
$$6\sin^2\theta - 13\sin\theta = -7.$$

Solution. Move all terms to one side:
$$6\sin^2\theta - 13\sin\theta + 7 = 0.$$

Factor:
$$(3\sin\theta - ?)(2\sin\theta - ?) = 0.$$

Since this expression does not factor conveniently over the integers, let
$$u = \sin\theta.$$

Then
$$6u^2 - 13u + 7 = 0.$$

Using the quadratic formula,
$$u = \frac{13 \pm \sqrt{(-13)^2 - 4(6)(7)}}{12} = \frac{13 \pm 1}{12}.$$

Therefore,
$$u = 1 \qquad \text{or} \qquad u = \frac{7}{6}.$$

Since
$$-1 \le \sin\theta \le 1,$$

the equation
$$\sin\theta = \frac{7}{6}$$

has no real solution.
Thus,
$$\sin\theta = 1,$$

which gives
$$\theta = \frac{\pi}{2}.$$

Therefore, the only solution in $[0, 2\pi)$ is

$$\theta = \frac{\pi}{2}.$$

Example. Solve for all solutions in $[0, 2\pi)$:

$$-2\sin(4x) = \sqrt{3}.$$

Solution. First isolate the sine function:

$$\sin(4x) = -\frac{\sqrt{3}}{2}.$$

Because

$$0 \le x < 2\pi,$$

multiplying the interval by 4 gives

$$0 \le 4x < 8\pi.$$

The reference angle is

$$\frac{\pi}{3}.$$

Since sine is negative in Quadrants III and IV,

$$4x = \frac{4\pi}{3} + 2\pi k \qquad \text{or} \qquad 4x = \frac{5\pi}{3} + 2\pi k.$$

Dividing by 4 gives

$$x = \frac{\pi}{3} + \frac{k\pi}{2} \qquad \text{or} \qquad x = \frac{5\pi}{12} + \frac{k\pi}{2}.$$

Restricting these solutions to $[0, 2\pi)$ gives

$$x = \frac{\pi}{3}, \frac{5\pi}{12}, \frac{5\pi}{6}, \frac{11\pi}{12}, \frac{4\pi}{3}, \frac{17\pi}{12}, \frac{11\pi}{6}, \frac{23\pi}{12}.$$

Example. Solve for all solutions in $[0, 2\pi)$:

$$3\sec^2 x = 4.$$

Solution. First isolate $\sec^2 x$:

$$\sec^2 x = \frac{4}{3}.$$

Using

$$\sec^2 x = \frac{1}{\cos^2 x},$$

we obtain

$$\cos^2 x = \frac{3}{4}.$$

Taking square roots,

$$\cos x = \pm\frac{\sqrt{3}}{2}.$$

Therefore,

$$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}.$$

What would all the solutions be?

For all real x, the solutions can be written compactly as

$$x = k\pi \pm \frac{\pi}{6}, \qquad k \in \mathbb{Z}.$$

Example. Use a calculator to solve on the interval

$$0 \leq \theta < 2\pi : \qquad \cos\theta = \tan(3\theta).$$

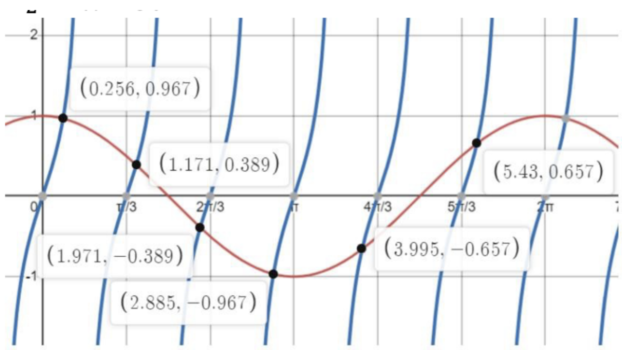


Figure 51: Graph of $Y_1 = \cos\theta$ and $Y_2 = \tan 3\theta$

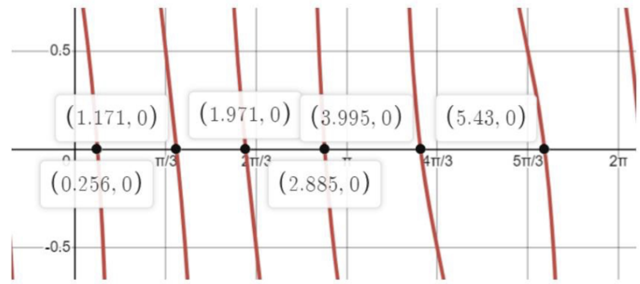


Figure 52: Graph of $Y_1 = \cos\theta - \tan 3\theta$

Solution. This equation cannot be conveniently solved using elementary algebraic methods. We can solve it graphically with a calculator.

One approach is to graph

$$Y_1 = \cos\theta \qquad \text{and} \qquad Y_2 = \tan(3\theta).$$

The solutions occur at the intersection points of the two graphs.

Alternatively, move all terms to one side and graph

$$Y_1 = \cos\theta - \tan(3\theta).$$

The solutions are the zeros of this function.

Using a calculator, the solutions on $[0, 2\pi)$ are approximately

$$\theta \approx 0.256,\ 1.171,\ 1.971,\ 2.885,\ 3.995,\ 5.430.$$

6.7: Trigonometric Equations and Applications

Example. Solve on the interval $0 \leq x < 2\pi$ using trigonometric identities:

$$2 \sin x \sin(2x) + \sin(2x) = 0.$$

Solution. Factor out the common term $\sin(2x)$:

$$\sin(2x)(2 \sin x + 1) = 0.$$

By the zero-product property,

$$\sin(2x) = 0 \qquad \text{or} \qquad 2 \sin x + 1 = 0.$$

For

$$\sin(2x) = 0,$$

we have

$$2x = k\pi.$$

Thus,

$$x = \frac{k\pi}{2}.$$

On the interval $0 \leq x < 2\pi$,

$$x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}.$$

For

$$2 \sin x + 1 = 0,$$

we obtain

$$\sin x = -\frac{1}{2}.$$

Thus,

$$x = \frac{7\pi}{6}, \frac{11\pi}{6}.$$

Therefore,

$$x = 0, \frac{\pi}{2}, \pi, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}.$$

Example. Solve each equation over the interval $0 \leq \theta < 2\pi$:

$$\text{(a)} \quad \cos(2\theta) + 14 \sin^2 \theta = 10, \qquad \text{(b)} \quad \sin^2 \theta = 3(\cos \theta + 1).$$

Solution. (a) Use the double-angle identity

$$\cos(2\theta) = 1 - 2 \sin^2 \theta.$$

Then

$$1 - 2 \sin^2 \theta + 14 \sin^2 \theta = 10,$$

so

$$12 \sin^2 \theta = 9.$$

Therefore,

$$\sin^2 \theta = \frac{3}{4},$$

and hence

$$\sin\theta = \pm\frac{\sqrt{3}}{2}.$$

Thus,

$$\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}.$$

(b) Use the Pythagorean identity

$$\sin^2\theta = 1 - \cos^2\theta.$$

Then

$$1 - \cos^2\theta = 3(\cos\theta + 1).$$

Expanding and moving all terms to one side gives

$$1 - \cos^2\theta = 3\cos\theta + 3,$$

$$\cos^2\theta + 3\cos\theta + 2 = 0.$$

Factor:

$$(\cos\theta + 1)(\cos\theta + 2) = 0.$$

Thus,

$$\cos\theta = -1 \qquad \text{or} \qquad \cos\theta = -2.$$

Since $-1 \le \cos\theta \le 1$, the equation

$$\cos\theta = -2$$

has no real solution. Therefore,

$$\cos\theta = -1,$$

which gives

$$\theta = \pi.$$

Example. Solve the equation over the interval $0 \le x < 2\pi$:

$$\sec x = -\sqrt{1 + \tan x}.$$

Solution. Square both sides:

$$\sec^2 x = 1 + \tan x.$$

Using the identity

$$\sec^2 x = 1 + \tan^2 x,$$

we obtain

$$1 + \tan^2 x = 1 + \tan x.$$

Thus,

$$\tan^2 x - \tan x = 0.$$

Factor:

$$\tan x(\tan x - 1) = 0.$$

Therefore,

$$\tan x = 0 \qquad \text{or} \qquad \tan x = 1.$$

On $[0, 2\pi)$, these give the candidates
$$x = 0, \pi, \frac{\pi}{4}, \frac{5\pi}{4}.$$

Because we squared the original equation, each candidate must be checked.
For $x = 0$,
$$\sec 0 = 1 \neq -\sqrt{1 + \tan 0} = -1.$$

For $x = \pi$,
$$\sec \pi = -1 = -\sqrt{1 + \tan \pi}.$$

For $x = \frac{\pi}{4}$,
$$\sec \frac{\pi}{4} = \sqrt{2} \neq -\sqrt{2}.$$

For $x = \frac{5\pi}{4}$,
$$\sec \frac{5\pi}{4} = -\sqrt{2} = -\sqrt{1 + \tan \frac{5\pi}{4}}.$$

Therefore,
$$x = \pi, \frac{5\pi}{4}.$$

Example. Solve the equation over the interval $0 \leq x < 2\pi$:
$$\csc x + 1 = -\cot x.$$

Solution. Rewrite the equation in terms of sine and cosine:
$$\frac{1}{\sin x} + 1 = -\frac{\cos x}{\sin x}.$$

Since $\sin x \neq 0$, multiply both sides by $\sin x$:
$$1 + \sin x = -\cos x.$$

Thus,
$$\sin x + \cos x = -1.$$

Using
$$\sin x + \cos x = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right),$$

we obtain
$$\sqrt{2} \sin\left(x + \frac{\pi}{4}\right) = -1,$$

so
$$\sin\left(x + \frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}.$$

The corresponding candidates are
$$x = \pi, \frac{3\pi}{2}.$$

However, $x = \pi$ is not in the domain of the original equation because
$$\csc \pi$$

is undefined.

Therefore, the only solution is

$$x = \frac{3\pi}{2}.$$

Example. Solve on the interval $0 \le x < 2\pi$:

$$2\cos x \cos 3x - 2\sin x \sin 3x = \sqrt{2}.$$

Solution. Use the sum identity

$$\cos(A+B) = \cos A \cos B - \sin A \sin B.$$

Thus,

$$2\big(\cos x \cos 3x - \sin x \sin 3x\big) = \sqrt{2},$$

which gives

$$2\cos(4x) = \sqrt{2}.$$

Therefore,

$$\cos(4x) = \frac{\sqrt{2}}{2}.$$

The general solutions for $4x$ are

$$4x = \frac{\pi}{4} + 2\pi k \qquad \text{or} \qquad 4x = -\frac{\pi}{4} + 2\pi k.$$

Dividing by 4,

$$x = \frac{\pi}{16} + \frac{k\pi}{2} \qquad \text{or} \qquad x = -\frac{\pi}{16} + \frac{k\pi}{2}.$$

Restricting to $0 \le x < 2\pi$ gives

$$x = \frac{\pi}{16}, \frac{7\pi}{16}, \frac{9\pi}{16}, \frac{15\pi}{16}, \frac{17\pi}{16}, \frac{23\pi}{16}, \frac{25\pi}{16}, \frac{31\pi}{16}.$$

Applications

Example. Suppose a projectile is fired from a cannon with initial velocity v_0 and angle of elevation θ. The horizontal distance $R(\theta)$, in feet, is given by

$$R(\theta) = \frac{v_0^2 \sin(2\theta)}{32}.$$

If $v_0 = 80$ ft/s, what angle of elevation should be used to hit a target 133 ft in front of the cannon?

Solution. Substitute

$$v_0 = 80 \qquad \text{and} \qquad R(\theta) = 133.$$

Then

$$133 = \frac{80^2 \sin(2\theta)}{32}.$$

Since

$$\frac{80^2}{32} = 200,$$

we have

$$133 = 200\sin(2\theta).$$

Thus,

$$\sin(2\theta) = \frac{133}{200} = 0.665.$$

Using inverse sine,

$$2\theta = \sin^{-1}(0.665) \approx 41.68^\circ.$$

Since sine is positive in Quadrants I and II, the second possibility is

$$2\theta = 180^\circ - 41.68^\circ = 138.32^\circ.$$

Therefore,

$$\theta \approx 20.84^\circ$$

or

$$\theta \approx 69.16^\circ.$$

Thus, the target can be reached using either angle of elevation

$$\theta \approx 20.84^\circ \qquad \text{or} \qquad \theta \approx 69.16^\circ.$$

Example. The population sizes of many animal species rise and fall over time. Suppose that the population size of a certain species is modeled by

$$p(t) = 4000 - 1200\cos\left(\frac{2\pi}{9}t\right),$$

where $p(t)$ represents the total population size and t is the time, in years.

Suppose we start at $t = 0$. During the first 9 years, when will the population size be 4500?

Solution. Set

$$p(t) = 4500.$$

Then

$$4500 = 4000 - 1200\cos\left(\frac{2\pi}{9}t\right).$$

Subtract 4000:

$$500 = -1200\cos\left(\frac{2\pi}{9}t\right).$$

Thus,

$$\cos\left(\frac{2\pi}{9}t\right) = -\frac{5}{12}.$$

Let

$$u = \frac{2\pi}{9}t.$$

Since $0 \le t \le 9$, we have

$$0 \le u \le 2\pi.$$

The principal solution is

$$u = \cos^{-1}\left(-\frac{5}{12}\right) \approx 2.001.$$

Since cosine is negative in Quadrants II and III, the second solution is

$$u = 2\pi - 2.001 \approx 4.283.$$

Using

$$t = \frac{9u}{2\pi},$$

we obtain

$$t \approx \frac{9(2.001)}{2\pi} \approx 2.87$$

and

$$t \approx \frac{9(4.283)}{2\pi} \approx 6.13.$$

Therefore, during the first 9 years, the population will be 4500 at approximately

$$t \approx 2.87 \text{ years} \qquad \text{and} \qquad t \approx 6.13 \text{ years}.$$

7.1: Oblique Triangles and the Law of Sines

Oblique Triangles

An **oblique triangle** is a triangle that does not contain a right angle.

Deriving the Law of Sines

Consider a triangle with sides a, b, and c opposite angles A, B, and C, respectively. Draw an altitude of length h from vertex B.

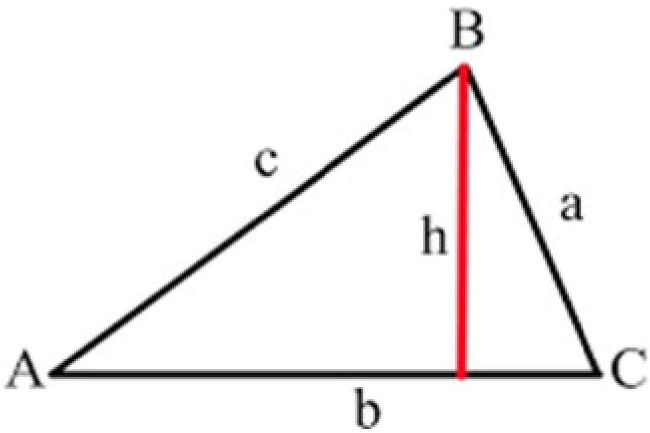


Figure 53: Altitude used in the Law of Sines

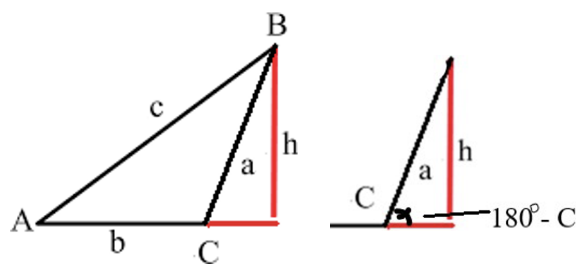


Figure 54: A partition of oblique triangle

From the resulting right triangles,

$$\sin A = \frac{h}{c} \qquad \text{and} \qquad \sin C = \frac{h}{a}.$$

Thus,

$$c \sin A = h \qquad \text{and} \qquad a \sin C = h.$$

Therefore,

$$c \sin A = a \sin C.$$

Dividing by ac gives

$$\frac{\sin A}{a} = \frac{\sin C}{c}.$$

The same relationship holds when the altitude falls outside the triangle. For an obtuse angle C,

$$\sin(180^\circ - C) = \sin C,$$

so the same derivation applies.

Drawing an altitude from another vertex similarly gives

$$\frac{\sin B}{b} = \frac{\sin C}{c}.$$

Hence we obtain the **Law of Sines**.

$$\boxed{\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}}$$

Equivalently,

$$\boxed{\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}}$$

where the side a is opposite angle A, the side b is opposite angle B, and the side c is opposite angle C.

When Can We Use the Law of Sines?

The Law of Sines is especially useful when we know an angle and its opposite side.

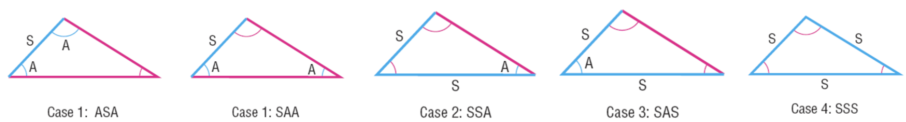


Figure 55: Four cases where Law of Sines work the best

Case	Known Information
ASA	Two angles and the included side
SAA	Two angles and a nonincluded side
SSA	Two sides and an angle opposite one of them
SAS	Two sides and the included angle
SSS	Three sides

The Law of Sines can be used directly for **ASA**, **SAA**, and **SSA** triangles. The SSA case requires special care because there may be zero, one, or two possible triangles.

Solving ASA and SAA Triangles

Example. Solve $\triangle ABC$ if

$$b = 3 \text{ cm}, \qquad A = 85^\circ, \qquad C = 50^\circ.$$

Solution. First find the remaining angle:

$$A + B + C = 180^\circ.$$

Thus,

$$B = 180^\circ - 85^\circ - 50^\circ = 45^\circ.$$

Now use the Law of Sines:

$$\frac{a}{\sin 85^\circ} = \frac{3}{\sin 45^\circ}.$$

Therefore,

$$a = \frac{3\sin 85^\circ}{\sin 45^\circ} \approx 4.23 \text{ cm}.$$

Similarly,

$$\frac{c}{\sin 50^\circ} = \frac{3}{\sin 45^\circ},$$

so

$$c = \frac{3\sin 50^\circ}{\sin 45^\circ} \approx 3.25 \text{ cm}.$$

Therefore,

$$A = 85^\circ, \qquad B = 45^\circ, \qquad C = 50^\circ,$$

and

$$a \approx 4.23 \text{ cm}, \qquad b = 3 \text{ cm}, \qquad c \approx 3.25 \text{ cm}.$$

The Ambiguous Case: SSA

Suppose angle A and sides a and b are known. The altitude corresponding to side b is

$$h = b \sin A.$$

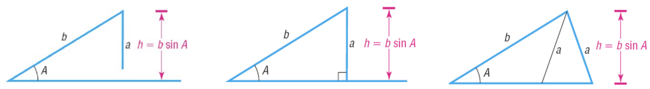


Figure 56: Illustrative grpahs for SSA triangle

When A is acute, the number of possible triangles depends on the relationship among a, b, and h.

Condition	**Number of Triangles**
$a < h$	0
$a = h$	1 right triangle
$h < a < b$	2
$a \geq b$	1

This is called the **ambiguous case** of the Law of Sines.

Example. Given $\triangle ABC$ with

$$A = 30^\circ \qquad \text{and} \qquad b = 10 \text{ in},$$

what length of side a will produce a right triangle?

Solution. For the right triangle,

$$\sin 30^\circ = \frac{a}{10}.$$

Thus,

$$a = 10 \sin 30^\circ = 5.$$

Therefore, the altitude is

$$h = 5 \text{ in}.$$

Hence:

$$a < 5 \quad \Longrightarrow \quad \text{no triangle},$$

$$a = 5 \quad \Longrightarrow \quad \text{one right triangle},$$

$$5 < a < 10 \quad \Longrightarrow \quad \text{two triangles},$$

and

$$a \geq 10 \quad \Longrightarrow \quad \text{one triangle}.$$

Example. Given $\triangle ABC$ with

$$A = 30^\circ, \qquad b = 10 \text{ in}, \qquad a = 3 \text{ in},$$

solve the triangle.

Solution. Compute the altitude:

$$h = b \sin A = 10 \sin 30^\circ = 5.$$

Since

$$a = 3 < 5 = h,$$

side a is too short to form a triangle.

Therefore, no triangle exists.

Example. Given $\triangle ABC$ with

$$A = 30^\circ, \qquad b = 10 \text{ in}, \qquad a = 6 \text{ in},$$

solve the triangle.

Solution. Since

$$h = b \sin A = 10 \sin 30^\circ = 5$$

and

$$5 < a = 6 < 10 = b,$$

two triangles are possible.

Using the Law of Sines,

$$\frac{\sin B}{10} = \frac{\sin 30^\circ}{6}.$$

Thus,

$$\sin B = \frac{10 \sin 30^\circ}{6} = \frac{5}{6}.$$

The first possible angle is

$$B_1 = \sin^{-1}\left(\frac{5}{6}\right) \approx 56.44^\circ.$$

The second possible angle is

$$B_2 = 180^\circ - 56.44^\circ \approx 123.56^\circ.$$

For the first triangle,

$$C_1 = 180^\circ - 30^\circ - 56.44^\circ \approx 93.56^\circ.$$

Using the Law of Sines,

$$\frac{c_1}{\sin 93.56^\circ} = \frac{6}{\sin 30^\circ},$$

so

$$c_1 = \frac{6 \sin 93.56^\circ}{\sin 30^\circ} \approx 11.98 \text{ in}.$$

For the second triangle,

$$C_2 = 180^\circ - 30^\circ - 123.56^\circ \approx 26.44^\circ.$$

Thus,

$$\frac{c_2}{\sin 26.44^\circ} = \frac{6}{\sin 30^\circ},$$

which gives

$$c_2 = \frac{6 \sin 26.44^\circ}{\sin 30^\circ} \approx 5.34 \text{ in}.$$

Therefore, there are two possible triangles:

	A	B	C
Triangle 1	$30°$	$56.44°$	$93.56°$
Triangle 2	$30°$	$123.56°$	$26.44°$

with corresponding sides

$$(a, b, c) \approx (6, 10, 11.98)$$

or

$$(a, b, c) \approx (6, 10, 5.34).$$

Example. Given $\triangle ABC$ with

$$A = 30^\circ, \qquad b = 10 \text{ in}, \qquad a = 14 \text{ in},$$

solve the triangle.

Solution. Using the Law of Sines,

$$\frac{\sin B}{10} = \frac{\sin 30^\circ}{14}.$$

Thus,

$$\sin B = \frac{10 \sin 30^\circ}{14} = \frac{5}{14}.$$

Therefore,

$$B = \sin^{-1}\left(\frac{5}{14}\right) \approx 20.92^\circ.$$

The supplementary angle

$$180^\circ - 20.92^\circ = 159.08^\circ$$

cannot be used because

$$30^\circ + 159.08^\circ > 180^\circ.$$

Thus, only one triangle exists.

The third angle is

$$C = 180^\circ - 30^\circ - 20.92^\circ \approx 129.08^\circ.$$

Using the Law of Sines,

$$\frac{c}{\sin 129.08^\circ} = \frac{14}{\sin 30^\circ}.$$

Therefore,

$$c = \frac{14 \sin 129.08^\circ}{\sin 30^\circ} \approx 21.74 \text{ in}.$$

Hence,

$$A = 30^\circ, \qquad B \approx 20.92^\circ, \qquad C \approx 129.08^\circ,$$

and

$$a = 14 \text{ in}, \qquad b = 10 \text{ in}, \qquad c \approx 21.74 \text{ in}.$$

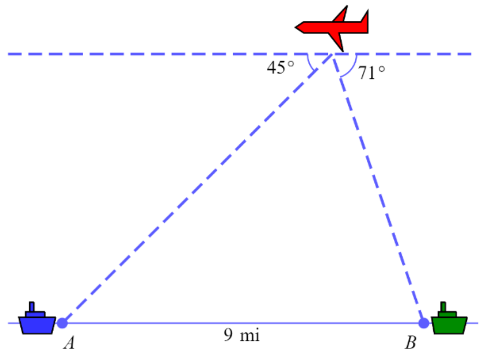


Figure 57: Illustrative digram

Application of the Law of Sines

Example. A spy plane is flying over the ocean. Two aircraft carriers are 9 miles apart. The angle of depression to the carrier that the plane has just flown over is 71°, and the angle of depression to the carrier ahead of the plane is 45°. The range of a missile on the plane is 8 miles. Can either carrier be reached?

Solution. Let the two carriers be A and B, and let the plane be P.

Because the horizontal line through the plane is parallel to the ocean surface, the angles of depression are equal to the corresponding angles of elevation. Therefore,

$$A = 45^\circ \qquad \text{and} \qquad B = 71^\circ.$$

The angle at the plane is

$$P = 180^\circ - 45^\circ - 71^\circ = 64^\circ.$$

The distance between the carriers is

$$AB = 9 \text{ mi}.$$

Using the Law of Sines,

$$\frac{PA}{\sin 71^\circ} = \frac{9}{\sin 64^\circ}.$$

Thus,

$$PA = \frac{9 \sin 71^\circ}{\sin 64^\circ} \approx 9.47 \text{ mi}.$$

For the other carrier,

$$\frac{PB}{\sin 45^\circ} = \frac{9}{\sin 64^\circ},$$

so

$$PB = \frac{9 \sin 45^\circ}{\sin 64^\circ} \approx 7.08 \text{ mi}.$$

The missile has a range of 8 miles. Since

$$PA \approx 9.47 > 8,$$

carrier A is out of range.

However,

$$PB \approx 7.08 < 8,$$

so carrier B is within range.

Therefore, only the carrier approximately 7.08 miles from the plane can be reached.

7.2: Law of Cosines

Law of Cosines

The Law of Sines cannot be used directly to solve **SAS** or **SSS** triangles. For these cases, we use the **Law of Cosines**.

Derivation of the Law of Cosines

Consider a triangle with sides a, b, and c, where side a is opposite angle A. Place the triangle in the coordinate plane so that one vertex is at the origin and side b lies along the x-axis.

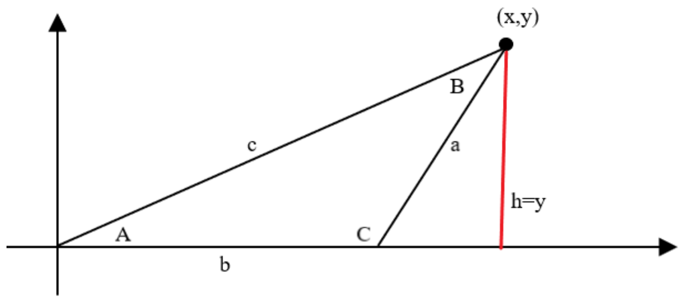


Figure 58: Drive the Law of Cosines

If the third vertex has coordinates (x, y), then

$$x = c\cos A \qquad \text{and} \qquad y = c\sin A.$$

Also, by the distance formula,

$$(x-b)^2 + y^2 = a^2.$$

Expanding,

$$x^2 - 2bx + b^2 + y^2 = a^2.$$

Substitute

$$x = c\cos A, \qquad y = c\sin A.$$

Then

$$(c\cos A)^2 - 2bc\cos A + b^2 + (c\sin A)^2 = a^2.$$

Thus,

$$c^2\cos^2 A - 2bc\cos A + b^2 + c^2\sin^2 A = a^2.$$

Factor the c^2 terms:

$$b^2 + c^2(\cos^2 A + \sin^2 A) - 2bc\cos A = a^2.$$

Using

$$\cos^2 A + \sin^2 A = 1,$$

we obtain

$$a^2 = b^2 + c^2 - 2bc\cos A.$$

Law of Cosines

For any triangle with sides a, b, and c opposite angles A, B, and C, respectively,

$$\boxed{a^2 = b^2 + c^2 - 2bc\cos A}$$

$$\boxed{b^2 = a^2 + c^2 - 2ac\cos B}$$

$$\boxed{c^2 = a^2 + b^2 - 2ab\cos C}$$

In words:

> The square of one side of a triangle equals the sum of the squares of the other two sides minus twice their product times the cosine of their included angle.

The Law of Cosines is especially useful for

SAS and SSS triangles.

SAS Triangles

Example. Solve the triangle if

$$a = 7, \qquad b = 9, \qquad C = 60^\circ.$$

Find side c and angles A and B.

Solution. Since two sides and the included angle are known, use the Law of Cosines to find c:

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

Substitute the given values:

$$c^2 = 7^2 + 9^2 - 2(7)(9)\cos 60^\circ.$$

Thus,

$$c^2 = 49 + 81 - 126\left(\frac{1}{2}\right) = 67.$$

Therefore,

$$c = \sqrt{67} \approx 8.19.$$

Now use the Law of Cosines to find angle A:

$$a^2 = b^2 + c^2 - 2bc\cos A.$$

Solving for $\cos A$,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}.$$

Hence,

$$\cos A = \frac{9^2 + (\sqrt{67})^2 - 7^2}{2(9)(\sqrt{67})}.$$

Therefore,

$$A = \cos^{-1}\left(\frac{81 + 67 - 49}{18\sqrt{67}}\right) \approx 47.78^\circ.$$

Finally,

$$B = 180^\circ - A - C,$$

so

$$B = 180^\circ - 47.78^\circ - 60^\circ \approx 72.22^\circ.$$

Thus,

$$c \approx 8.19, \qquad A \approx 47.78^\circ, \qquad B \approx 72.22^\circ.$$

Example. Solve the triangle if

$$a = 9, \qquad b = 3, \qquad C = 127^\circ.$$

Find side c and angles A and B.

Solution. Use the Law of Cosines:

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

Thus,

$$c^2 = 9^2 + 3^2 - 2(9)(3)\cos 127^\circ.$$

Therefore,

$$c = \sqrt{90 - 54\cos 127^\circ} \approx 11.07.$$

To find A, use

$$a^2 = b^2 + c^2 - 2bc\cos A.$$

Solving for $\cos A$,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}.$$

Thus,

$$A = \cos^{-1}\left(\frac{3^2 + (11.07)^2 - 9^2}{2(3)(11.07)}\right) \approx 40.50^\circ.$$

The remaining angle is

$$B = 180^\circ - 127^\circ - 40.50^\circ \approx 12.50^\circ.$$

Therefore,

$$c \approx 11.07, \qquad A \approx 40.50^\circ, \qquad B \approx 12.50^\circ.$$

SSS Triangles

When all three sides are known, the Law of Cosines can be rearranged to find an angle:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc},$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac},$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$

Example. Solve the triangle if

$$a = 3 \text{ m}, \qquad b = 5 \text{ m}, \qquad c = 6 \text{ m}.$$

Solution. Use the Law of Cosines to find angle A:

$$a^2 = b^2 + c^2 - 2bc\cos A.$$

Thus,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{5^2 + 6^2 - 3^2}{2(5)(6)}.$$

Therefore,

$$\cos A = \frac{52}{60} = \frac{13}{15},$$

so

$$A = \cos^{-1}\left(\frac{13}{15}\right) \approx 29.93^\circ.$$

Next, find B:

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}.$$

Thus,

$$\cos B = \frac{3^2 + 6^2 - 5^2}{2(3)(6)} = \frac{20}{36} = \frac{5}{9}.$$

Hence,

$$B = \cos^{-1}\left(\frac{5}{9}\right) \approx 56.25^\circ.$$

Finally,

$$C = 180^\circ - A - B,$$

so

$$C = 180^\circ - 29.93^\circ - 56.25^\circ \approx 93.82^\circ.$$

Therefore,

$$A \approx 29.93^\circ, \qquad B \approx 56.25^\circ, \qquad C \approx 93.82^\circ.$$

Application of the Law of Cosines

Example. Two airplanes leave an airport, and the angle between their flight paths is 40°. After one hour, one plane has traveled 300 miles while the other has traveled 200 miles. How far apart are the planes at this time?

Solution. The flight paths form two sides of a triangle with lengths

$$300 \text{ mi} \qquad \text{and} \qquad 200 \text{ mi},$$

and the included angle is

$$40^\circ.$$

Let d represent the distance between the two planes. By the Law of Cosines,

$$d^2 = 300^2 + 200^2 - 2(300)(200)\cos 40^\circ.$$

Thus,

$$d = \sqrt{300^2 + 200^2 - 2(300)(200)\cos 40^\circ}.$$

Using a calculator,

$$d \approx 195.13.$$

Therefore, the airplanes are approximately 195.13 miles apart.

8.1: Systems of Two Linear Equations

System of linear equations

A **system of linear equations** consists of two or more linear equations involving the same variables. A solution to a system is an ordered pair that satisfies *every* equation in the system.

For a system of two linear equations in two variables, there are three possibilities:

- the lines intersect at exactly one point;
- the lines are parallel and never intersect;
- the equations represent the same line.

Consider the following three systems:

$$\begin{cases} x + 2y = 7, \\ x - y = 4 \end{cases} \quad \begin{cases} 2x - y = -7, \\ 2x - y = -3 \end{cases} \quad \begin{cases} 3x - y = -6, \\ -6x + 2y = 12 \end{cases}$$

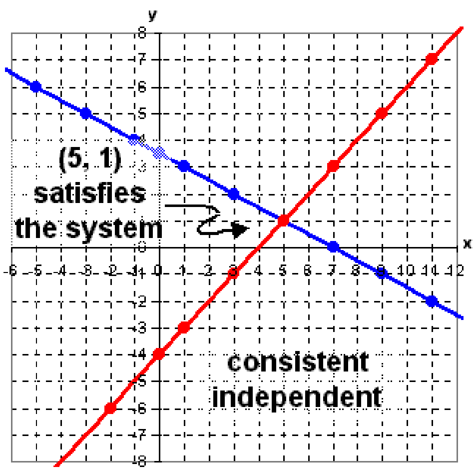


Figure 59: Consistent independent linear system

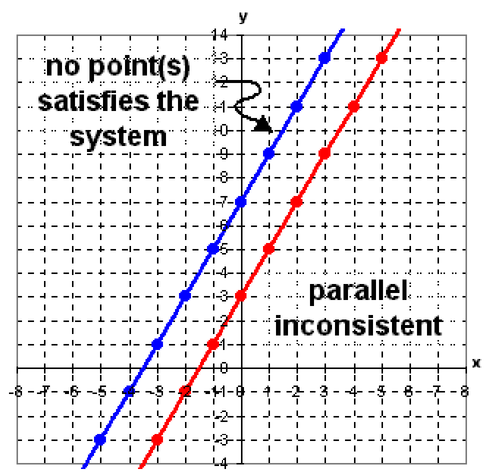


Figure 60: Inconsistent linear system

Classification of Linear Systems

Consistent System. A system is **consistent** if it has at least one solution.

Inconsistent System. A system is **inconsistent** if it has no solution. Geometrically, the two lines are parallel and distinct.

Independent System. A system is **independent** if it has exactly one solution. Geometrically, the two lines intersect at exactly one point.

Dependent System (coincident lines). A system is **dependent** if it has infinitely many solutions. Geometrically, the two equations represent the same line.

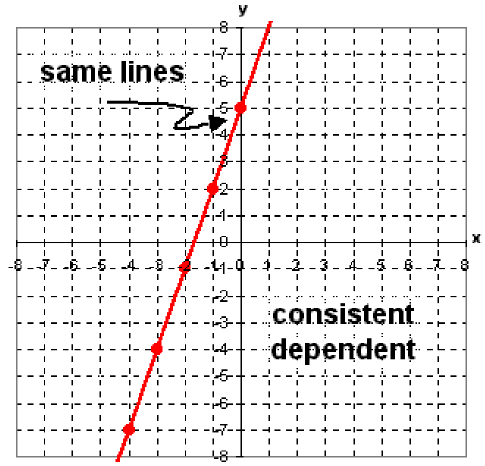


Figure 61: Consistent dependent linear system

Recall Algebraic Methods

Two common algebraic methods for solving systems are the **substitution method** and the **elimination method**.

Substitution Method

1. Choose one equation and isolate one variable.
2. Substitute the resulting expression into the other equation.
3. Solve the resulting equation for one variable.
4. Substitute this value into either original equation to find the remaining variable.
5. Check the solution in both original equations.

Elimination Method

1. Write the equations with like terms aligned.
2. If necessary, multiply one or both equations by constants so that one variable has opposite or equal coefficients.
3. Add or subtract the equations to eliminate one variable.
4. Solve for the remaining variable.
5. Substitute the value into either original equation to find the other variable.
6. Check the solution in both original equations.

Verifying a Solution

Example. Verify that $(4, 5)$ is a solution of

$$\begin{cases} 2x - y = 3, \\ 5x + 4y = 40. \end{cases}$$

Solution. Substitute

$$x = 4, \qquad y = 5$$

into both equations.

For the first equation,

$$2(4) - 5 = 8 - 5 = 3.$$

For the second equation,

$$5(4) + 4(5) = 20 + 20 = 40.$$

Since $(4, 5)$ satisfies both equations,

$$(4, 5)$$

is a solution of the system.

Recognizing Systems with No Solution

Example. Explain why the system

$$\begin{cases} 3x + y = 4, \\ 9x + 3y = 13 \end{cases}$$

has no solution.

Solution. Multiply the first equation by 3:

$$3(3x + y) = 3(4),$$

which gives

$$9x + 3y = 12.$$

However, the second equation is

$$9x + 3y = 13.$$

The same expression cannot equal both 12 and 13. Equivalently, subtracting the equations gives

$$0 = 1,$$

which is a contradiction.

Therefore, the system is inconsistent and has

$$\text{no solution.}$$

Systems with Infinitely Many Solutions

Example. The system

$$\begin{cases} x + 2y = 4, \\ 7x + 14y = 28 \end{cases}$$

has infinitely many solutions. State the general solution and a parameterized solution.

Solution. The second equation is seven times the first:

$$7(x + 2y) = 7(4),$$

so both equations represent the same line.

Thus, the system has infinitely many solutions.

Solving the first equation for x gives

$$x = 4 - 2y.$$

Hence, the general solution can be written as

$$(x, y) = (4 - 2y, y).$$

To write the solution parametrically, let
$$y = t, \qquad t \in \mathbb{R}.$$
Then
$$x = 4 - 2t.$$
Therefore,
$$(x, y) = (4 - 2t, t), \qquad t \in \mathbb{R}.$$

Solving a System

Example. Solve the system
$$\begin{cases} \frac{3}{4}x + \frac{1}{3}y = -2, \\ \frac{1}{2}x + \frac{1}{5}y = 3. \end{cases}$$

Solution. First eliminate the fractions.
Multiply the first equation by 12:
$$12\left(\frac{3}{4}x + \frac{1}{3}y\right) = 12(-2),$$
so
$$9x + 4y = -24.$$
Multiply the second equation by 10:
$$10\left(\frac{1}{2}x + \frac{1}{5}y\right) = 10(3),$$
so
$$5x + 2y = 30.$$
Thus, the equivalent system is
$$\begin{cases} 9x + 4y = -24, \\ 5x + 2y = 30. \end{cases}$$
Multiply the second equation by -2:
$$-10x - 4y = -60.$$
Add this to the first equation:
$$\begin{array}{r} 9x + 4y = -24, \\ \underline{-10x - 4y = -60,} \\ -x = -84. \end{array}$$
Therefore,
$$x = 84.$$
Substitute into
$$5x + 2y = 30 :$$
$$5(84) + 2y = 30.$$
Thus,
$$420 + 2y = 30,$$

so

$$2y = -390$$

and

$$y = -195.$$

Therefore, the solution is

$$(84, -195).$$

8.2: Systems of Three or More Linear Equations

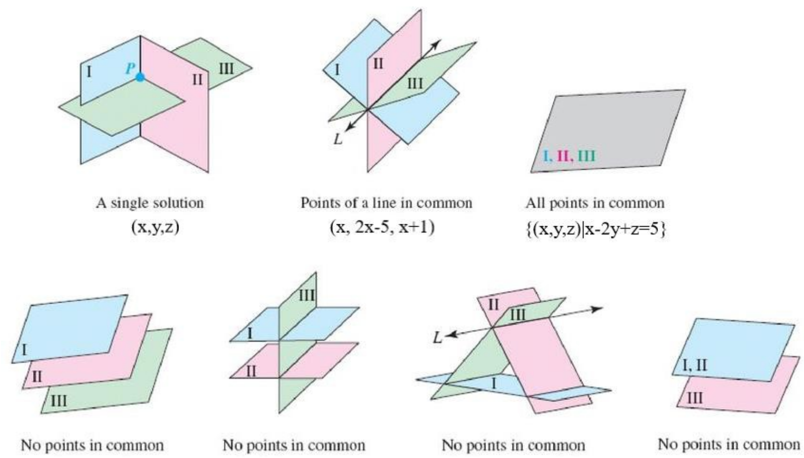


Figure 62: Geometric interpretations of three-plane systems

Elimination Process for Systems of Three Equations and Three Unknowns

1. Write each equation in standard form:

$$Ax + By + Cz = D.$$

2. Use any of the following operations to reduce the system:
   (a) Change the order of the equations.
   (b) Replace any equation by a nonzero constant multiple of that equation.
   (c) Replace an equation by the sum of two equations from the system.

3. Continue eliminating variables until the system has been reduced to one equation in one unknown. Then use **back substitution** to solve for the remaining unknowns.

Example. Solve the following system using elimination:

$$\begin{cases} -19 - 3y + 4z = -x, \\ 2x + y + z = 3, \\ -2x + 3y - 3z = -17. \end{cases}$$

Solution. First, write the first equation in standard form:

$$-19 - 3y + 4z = -x \quad \Longrightarrow \quad x - 3y + 4z = 19.$$

Thus, the system becomes

$$\begin{cases} x - 3y + 4z = 19 & (1), \\ 2x + y + z = 3 & (2), \\ -2x + 3y - 3z = -17 & (3). \end{cases}$$

We first eliminate x using equations (2) and (3). Adding the two equations gives

$$\begin{array}{r} 2x + y + z = 3, \\ -2x + 3y - 3z = -17, \\ \hline 4y - 2z = -14. \end{array}$$

Therefore,

$$4y - 2z = -14. \tag{4}$$

Next, eliminate x using equations (1) and (3). Multiply equation (1) by 2:

$$2x - 6y + 8z = 38.$$

Add this equation to equation (3):

$$\begin{array}{r} 2x - 6y + 8z = 38, \\ -2x + 3y - 3z = -17, \\ \hline -3y + 5z = 21. \end{array}$$

Therefore,

$$-3y + 5z = 21. \tag{5}$$

We have reduced the original system to a system of two equations in two unknowns:

$$\begin{cases} 4y - 2z = -14, \\ -3y + 5z = 21. \end{cases}$$

Simplify equation (4) by dividing by 2:

$$2y - z = -7.$$

Thus,

$$z = 2y + 7.$$

Substitute this into equation (5):

$$-3y + 5(2y + 7) = 21.$$

Simplifying,

$$-3y + 10y + 35 = 21,$$

so

$$7y = -14,$$

and therefore

$$y = -2.$$

Now find z:

$$z = 2(-2) + 7 = 3.$$

Finally, substitute $y = -2$ and $z = 3$ into equation (1):

$$x - 3(-2) + 4(3) = 19.$$

Thus,

$$x + 6 + 12 = 19,$$

so

$$x = 1.$$

Therefore, the solution to the system is

$$(x, y, z) = (1, -2, 3).$$

8.5: Using Matrices to Solve Systems

Matrices and Augmented Matrices

A **matrix** is a rectangular array of numbers arranged in rows and columns.

For example,

$$A = \begin{bmatrix} 2 & -3 & 5 \\ 1 & 4 & -2 \end{bmatrix}$$

is a 2×3 matrix because it has 2 rows and 3 columns.

An **augmented matrix** is a matrix that represents a system of linear equations. The vertical bar separates the coefficients of the variables from the constants.

For example, the system

$$\begin{cases} a_{11}x + a_{12}y = b_1, \\ a_{21}x + a_{22}y = b_2 \end{cases}$$

has augmented matrix

$$\left[\begin{array}{cc|c} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \end{array}\right].$$

Example. Write the augmented matrix for the system

$$\begin{cases} 7x - 2y = 7, \\ -7x - 5y = -2. \end{cases}$$

Solution. The coefficients of x and y form the first two columns, and the constants form the augmented column. Thus,

$$\left[\begin{array}{cc|c} 7 & -2 & 7 \\ -7 & -5 & -2 \end{array}\right].$$

Example. Write the system of equations represented by

$$\left[\begin{array}{cc|c} 1 & 6 & -3 \\ -6 & 4 & 6 \end{array}\right].$$

Solution. Each row represents one equation. Therefore,

$$\begin{cases} x + 6y = -3, \\ -6x + 4y = 6. \end{cases}$$

Elementary Row Operations

The following operations may be performed on a matrix without changing the solution set of the corresponding system.

1. **Interchange two rows:**

$$R_i \leftrightarrow R_j.$$

2. **Multiply a row by a nonzero constant:**

$$R_i \to kR_i, \qquad k \neq 0.$$

3. **Replace a row by the sum of that row and a multiple of another row:**

$$R_i \to R_i + kR_j.$$

Example. Perform row operations on

$$\left[\begin{array}{ccc|c} 3 & 1 & 7 & 4 \\ 1 & 1 & -1 & -2 \\ -8 & 2 & 5 & 9 \end{array}\right]$$

to obtain row echelon form.

Solution. Begin by interchanging the first two rows:

$$R_1 \leftrightarrow R_2.$$

Thus,

$$\left[\begin{array}{ccc|c} 3 & 1 & 7 & 4 \\ 1 & 1 & -1 & -2 \\ -8 & 2 & 5 & 9 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 3 & 1 & 7 & 4 \\ -8 & 2 & 5 & 9 \end{array}\right].$$

Eliminate the entries below the first pivot:

$$R_2 \to R_2 - 3R_1, \qquad R_3 \to R_3 + 8R_1.$$

This gives

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & -2 & 10 & 10 \\ 0 & 10 & -3 & -7 \end{array}\right].$$

Make the second pivot equal to 1:

$$R_2 \to -\frac{1}{2}R_2.$$

Then

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & -5 & -5 \\ 0 & 10 & -3 & -7 \end{array}\right].$$

Eliminate the entry below the second pivot:

$$R_3 \to R_3 - 10R_2.$$

Therefore,

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & -5 & -5 \\ 0 & 0 & 47 & 43 \end{array}\right].$$

Finally,

$$R_3 \to \frac{1}{47} R_3,$$

giving

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & -5 & -5 \\ 0 & 0 & 1 & \frac{43}{47} \end{array}\right].$$

This matrix is in row echelon form.

Row Echelon Form

A matrix is in **row echelon form (REF)** if:

1. The first nonzero entry in each nonzero row is 1.
2. Each leading 1 is to the right of the leading 1 in the row above it.
3. All entries below each leading 1 are 0.
4. Any rows consisting entirely of zeros are at the bottom.

A typical augmented matrix in REF has the form

$$\left[\begin{array}{ccc|c} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \end{array}\right].$$

Example. Given the system

$$\begin{cases} 3x + y = 5, \\ x - 2y = -3, \end{cases}$$

write the augmented matrix, put it in row echelon form using row operations, and state the solution.

Solution. The augmented matrix is

$$\left[\begin{array}{cc|c} 3 & 1 & 5 \\ 1 & -2 & -3 \end{array}\right].$$

Interchange the rows:

$$R_1 \leftrightarrow R_2.$$

Then

$$\left[\begin{array}{cc|c} 1 & -2 & -3 \\ 3 & 1 & 5 \end{array}\right].$$

Eliminate the 3 below the first pivot:

$$R_2 \to R_2 - 3R_1.$$

Thus,

$$\left[\begin{array}{cc|c} 1 & -2 & -3 \\ 0 & 7 & 14 \end{array}\right].$$

Divide the second row by 7:

$$R_2 \to \frac{1}{7} R_2.$$

Therefore,

$$\left[\begin{array}{cc|c} 1 & -2 & -3 \\ 0 & 1 & 2 \end{array}\right].$$

The second row gives

$$y = 2.$$

Substitute into the first equation:

$$x - 2(2) = -3,$$

so

$$x = 1.$$

Therefore,

$$(x, y) = (1, 2).$$

Reduced Row Echelon Form

A matrix is in **reduced row echelon form (RREF)** if:

1. It is in row echelon form.
2. Each leading entry is 1.
3. Every leading 1 is the only nonzero entry in its column.
4. Any rows consisting entirely of zeros are at the bottom.

A typical augmented matrix in RREF has the form

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array}\right].$$

From REF to RREF

Continuing the previous 3×4 matrix,

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & -5 & -5 \\ 0 & 0 & 1 & \frac{43}{47} \end{array}\right],$$

we can eliminate the entries above the pivots.
First,

$$R_2 \to R_2 + 5R_3,$$

giving

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & 0 & -\frac{20}{47} \\ 0 & 0 & 1 & \frac{43}{47} \end{array}\right].$$

Next,

$$R_1 \to R_1 + R_3,$$

so

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & -\frac{51}{47} \\ 0 & 1 & 0 & -\frac{20}{47} \\ 0 & 0 & 1 & \frac{43}{47} \end{array}\right].$$

Finally,

$$R_1 \to R_1 - R_2.$$

Thus,

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{31}{47} \\ 0 & 1 & 0 & -\frac{20}{47} \\ 0 & 0 & 1 & \frac{43}{47} \end{array}\right]$$

is the reduced row echelon form.

Interpreting an Augmented Matrix

Example. For each matrix, write the corresponding system and determine whether the system is consistent or inconsistent. If the system is consistent, state the solution.

(a)

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 7 \end{array}\right].$$

Solution. The corresponding system is

$$x = -2, \qquad y = 0, \qquad z = 7.$$

Thus, the system is consistent with the unique solution

$$(x, y, z) = (-2, 0, 7).$$

(b)

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 0 & 7 \end{array}\right].$$

Solution. The last row represents

$$0 = 7,$$

which is impossible.

Therefore, the system is **inconsistent** and has no solution.

(c)

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 9 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Solution. The system is

$$\begin{cases} x + 2z = -1, \\ y - z = 9. \end{cases}$$

There is no pivot in the z-column, so z is a free variable.

Let

$$z = t.$$

Then

$$x = -1 - 2t, \qquad y = 9 + t.$$

Therefore, the system is consistent with infinitely many solutions:

$$(x, y, z) = (-1 - 2t,\ 9 + t,\ t), \qquad t \in \mathbb{R}.$$

(d)

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Solution. The system is

$$x = 0, \qquad y = 0, \qquad z = 0.$$

Thus, the unique solution is

$$(x, y, z) = (0, 0, 0).$$

(e)

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -5 \\ 0 & 1 & 5 & 7 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Solution. The corresponding system is

$$\begin{cases} x + 2y = -5, \\ y + 5z = 7. \end{cases}$$

Let

$$z = t.$$

Then

$$y = 7 - 5t.$$

Substituting into the first equation,

$$x + 2(7 - 5t) = -5,$$

so

$$x = -19 + 10t.$$

Therefore,

$$(x, y, z) = (-19 + 10t,\ 7 - 5t,\ t), \qquad t \in \mathbb{R}.$$

(f)

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Solution. The matrix represents the single equation

$$x + y + 2z = -1.$$

Both y and z are free variables. Let

$$y = s, \qquad z = t.$$

Then

$$x = -1 - s - 2t.$$

Therefore,

$$(x, y, z) = (-1 - s - 2t,\ s,\ t), \qquad s, t \in \mathbb{R}.$$

Using Technology to Obtain RREF

For a TI-83/84 calculator:

1. Press

   `2nd MATRIX`

   to open the matrix menu.
2. Move to `EDIT`, choose a matrix such as `[A]`, and press `ENTER`.
3. Enter the dimensions of the augmented matrix and then enter all entries row by row.
4. Return to the home screen.
5. Press

   `2nd MATRIX`,

   move to `MATH`, and select

   `rref(`.
6. Return to the matrix menu, select the matrix you entered, and close the parenthesis:

   `rref([A])`.
7. Press `ENTER`. The calculator displays the reduced row echelon form.

Remark. Obtaining RREF is also known as carrying out **Gauss–Jordan elimination**.

Solving Systems Using RREF

Example. Use a calculator to solve

$$\begin{cases} x - y - z = 1, \\ -x + 2y - 3z = -3, \\ 4x - 3y - 8z = 2. \end{cases}$$

Solution. The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 1 \\ -1 & 2 & -3 & -3 \\ 4 & -3 & -8 & 2 \end{array}\right].$$

Using RREF,

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 1 \\ -1 & 2 & -3 & -3 \\ 4 & -3 & -8 & 2 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & -5 & -1 \\ 0 & 1 & -4 & -2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Thus,

$$x - 5z = -1, \qquad y - 4z = -2.$$

Let

$$z = t.$$

Then

$$x = 5t - 1, \qquad y = 4t - 2.$$

Therefore,

$$(x, y, z) = (5t - 1,\ 4t - 2,\ t), \qquad t \in \mathbb{R}.$$

Example. Use a calculator to solve

$$\begin{cases} x - y = 5, \\ 7x - 3z = 49, \\ 7y + 2z = 14. \end{cases}$$

Solution. Notice that the missing variables have coefficient 0. Thus, the augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 5 \\ 7 & 0 & -3 & 49 \\ 0 & 7 & 2 & 14 \end{array}\right].$$

Using RREF,

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 5 \\ 7 & 0 & -3 & 49 \\ 0 & 7 & 2 & 14 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Therefore,

$$x = 7, \qquad y = 2, \qquad z = 0.$$

Hence,

$$(x, y, z) = (7, 2, 0).$$

Applications of Systems of Equations

Example. Mixture Problem

Bryan saves quarters and dimes in his piggy bank. He has 225 coins with a total value of \$45.00. How many quarters and how many dimes does he have?

Solution. Let

$$x = \text{number of quarters}, \qquad y = \text{number of dimes}.$$

The total number of coins gives

$$x + y = 225.$$

The total value gives

$$0.25x + 0.10y = 45.$$

Thus, the system is

$$\begin{cases} x + y = 225, \\ 0.25x + 0.10y = 45. \end{cases}$$

The augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 1 & 225 \\ 0.25 & 0.10 & 45 \end{array}\right].$$

Using RREF,

$$\left[\begin{array}{cc|c} 1 & 1 & 225 \\ 0.25 & 0.10 & 45 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & 0 & 150 \\ 0 & 1 & 75 \end{array}\right].$$

Therefore, Bryan has

150 quarters

and

75 dimes.

Example. Motion Problem

A cruise ship travels 1435 miles from Caracas to the coast of Belize. The Caribbean Current flows in the direction of the trip to Belize. The trip to Belize takes 70 hours, while the return trip against the current takes 82 hours. Find the speed of the current and the cruising speed of the ship in still water.

Solution. Let

$$x = \text{speed of the current}, \qquad y = \text{speed of the ship in still water.}$$

Recall that

$$\text{Distance} = \text{Rate} \times \text{Time}.$$

Traveling *with* the current, the effective speed is

$$x + y.$$

Thus,

$$1435 = 70(x + y).$$

Traveling *against* the current, the effective speed is

$$y - x.$$

Thus,

$$1435 = 82(y - x).$$

Therefore,

$$\begin{cases} 70x + 70y = 1435, \\ -82x + 82y = 1435. \end{cases}$$

The corresponding augmented matrix is

$$\left[\begin{array}{cc|c} 70 & 70 & 1435 \\ -82 & 82 & 1435 \end{array}\right].$$

Its RREF is

$$\left[\begin{array}{cc|c} 1 & 0 & 1.5 \\ 0 & 1 & 19 \end{array}\right].$$

Therefore,

$$x = 1.5 \qquad \text{and} \qquad y = 19.$$

Hence, the speed of the current is

$$1.5 \text{ mph},$$

and the cruising speed of the ship is

$$19 \text{ mph}.$$

Example. Investment and Finance

A trustee invests \$2.5 million in three different investments: a savings fund paying 4% annual interest, a money market fund paying 7%, and government bonds paying 8%. The total interest earned after one year is \$0.178 million. Suppose the amount invested in government bonds is \$0.3 million more than *twice* the amount invested in the money market fund. Find the amount invested in each.

Solution. Let

$$x = \text{amount invested in savings},$$

$$y = \text{amount invested in the money market fund},$$

and

$$z = \text{amount invested in government bonds},$$

where all amounts are measured in millions of dollars.

The total investment gives

$$x + y + z = 2.5.$$

The interest equation is

$$0.04x + 0.07y + 0.08z = 0.178.$$

The relationship between the government bonds and money market investment is

$$z = 2y + 0.3,$$

or

$$-2y + z = 0.3.$$

Thus,

$$\begin{cases} x + y + z = 2.5, \\ 0.04x + 0.07y + 0.08z = 0.178, \\ -2y + z = 0.3. \end{cases}$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2.5 \\ 0.04 & 0.07 & 0.08 & 0.178 \\ 0 & -2 & 1 & 0.3 \end{array}\right].$$

Using RREF,

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2.5 \\ 0.04 & 0.07 & 0.08 & 0.178 \\ 0 & -2 & 1 & 0.3 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0.4 \\ 0 & 1 & 0 & 0.6 \\ 0 & 0 & 1 & 1.5 \end{array}\right].$$

Therefore,

$$x = 0.4, \qquad y = 0.6, \qquad z = 1.5.$$

Thus,

$$\$0.4 \text{ million}$$

is invested in savings,

$$\$0.6 \text{ million}$$

is invested in the money market fund, and

$$\$1.5 \text{ million}$$

is invested in government bonds.

8.7–8.8: Determinants and Cramer's Rule

Determinants play an important role in solving systems of linear equations.

Determinants of 2×2 Matrices

For real numbers a, b, c, d, the determinant of

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is defined by

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Example. Find the value of the determinant

$$\begin{vmatrix} 2 & -3 \\ 8 & 3 \end{vmatrix}.$$

Solution. Using

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$

we have

$$\begin{vmatrix} 2 & -3 \\ 8 & 3 \end{vmatrix} = (2)(3) - (-3)(8).$$

Thus,

$$6 + 24 = 30.$$

Therefore,

$$\begin{vmatrix} 2 & -3 \\ 8 & 3 \end{vmatrix} = 30.$$

The Theory Behind Cramer's Rule

Consider the system

$$\begin{cases} ax + by = s, \\ cx + dy = t. \end{cases}$$

To eliminate y, multiply the first equation by d and the second equation by b:

$$adx + bdy = sd,$$

$$bcx + bdy = bt.$$

Subtract the second equation from the first:

$$(ad - bc)x = sd - bt.$$

If

$$ad - bc \neq 0,$$

then

$$x = \frac{sd - bt}{ad - bc}.$$

Notice that

$$ad - bc = \begin{vmatrix} a & b \\ c & d \end{vmatrix}.$$

Also,

$$sd - bt = \begin{vmatrix} s & b \\ t & d \end{vmatrix}.$$

Therefore,

$$x = \frac{\begin{vmatrix} s & b \\ t & d \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}.$$

Similarly,

$$y = \frac{\begin{vmatrix} a & s \\ c & t \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}.$$

Cramer's Rule

For the system

$$\begin{cases} ax + by = s, \\ cx + dy = t, \end{cases}$$

define

$$D = \begin{vmatrix} a & b \\ c & d \end{vmatrix},$$

$$D_x = \begin{vmatrix} s & b \\ t & d \end{vmatrix}, \qquad D_y = \begin{vmatrix} a & s \\ c & t \end{vmatrix}.$$

If

$$D \neq 0,$$

then

$$\boxed{x = \frac{D_x}{D}, \qquad y = \frac{D_y}{D}.}$$

If

$$D = 0,$$

then Cramer's Rule cannot be used to obtain a unique solution. The system may have no solution or infinitely many solutions.

Example. Solve the system using Cramer's Rule:

$$\begin{cases} x + y = 8, \\ x - y = 2. \end{cases}$$

Solution. The coefficient determinant is

$$D = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = (1)(-1) - (1)(1) = -2.$$

Since

$$D \neq 0,$$

the system has a unique solution.

For x, replace the first column by the constants:

$$D_x = \begin{vmatrix} 8 & 1 \\ 2 & -1 \end{vmatrix} = 8(-1) - (1)(2) = -10.$$

Thus,

$$x = \frac{D_x}{D} = \frac{-10}{-2} = 5.$$

For y, replace the second column by the constants:

$$D_y = \begin{vmatrix} 1 & 8 \\ 1 & 2 \end{vmatrix} = (1)(2) - (8)(1) = -6.$$

Therefore,

$$y = \frac{D_y}{D} = \frac{-6}{-2} = 3.$$

Hence,

$$(x, y) = (5, 3).$$

Determinants of 3×3 Matrices

For a 3×3 determinant, the signs of the cofactors follow the pattern

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

For

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix},$$

we may expand along any row or column.

Expanding along the first row gives

$$\boxed{\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix}.}$$

Each 2×2 determinant is called a **minor**.

Important. You may expand along any row or column, but the sign associated with each position must be respected.

Example. Find the value of

$$\begin{vmatrix} 4 & 3 & -5 \\ 1 & -4 & 1 \\ 1 & 3 & 4 \end{vmatrix}$$

by hand.

Solution. Expand along the first row:

$$\begin{vmatrix} 4 & 3 & -5 \\ 1 & -4 & 1 \\ 1 & 3 & 4 \end{vmatrix} = 4\begin{vmatrix} -4 & 1 \\ 3 & 4 \end{vmatrix} - 3\begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix} + (-5)\begin{vmatrix} 1 & -4 \\ 1 & 3 \end{vmatrix}.$$

Evaluate each minor:

$$\begin{vmatrix} -4 & 1 \\ 3 & 4 \end{vmatrix} = (-4)(4) - (1)(3) = -19,$$

$$\begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix} = 4 - 1 = 3,$$

and

$$\begin{vmatrix} 1 & -4 \\ 1 & 3 \end{vmatrix} = 3 - (-4) = 7.$$

Therefore,

$$4(-19) - 3(3) + (-5)(7) = -76 - 9 - 35 = -120.$$

Hence,

$$\begin{vmatrix} 4 & 3 & -5 \\ 1 & -4 & 1 \\ 1 & 3 & 4 \end{vmatrix} = -120.$$

Example. Evaluate the same determinant by expanding along the second column:

$$\begin{vmatrix} 4 & 3 & -5 \\ 1 & -4 & 1 \\ 1 & 3 & 4 \end{vmatrix}.$$

Solution. The sign pattern along the second column is

$$-, +, -.$$

Therefore,

$$\begin{vmatrix} 4 & 3 & -5 \\ 1 & -4 & 1 \\ 1 & 3 & 4 \end{vmatrix} = -3\begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix} + (-4)\begin{vmatrix} 4 & -5 \\ 1 & 4 \end{vmatrix} - 3\begin{vmatrix} 4 & -5 \\ 1 & 1 \end{vmatrix}.$$

Compute the minors:

$$\begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix} = 4 - 1 = 3,$$

$$\begin{vmatrix} 4 & -5 \\ 1 & 4 \end{vmatrix} = 16 - (-5) = 21,$$

and

$$\begin{vmatrix} 4 & -5 \\ 1 & 1 \end{vmatrix} = 4 - (-5) = 9.$$

Thus,

$$-3(3) + (-4)(21) - 3(9) = -9 - 84 - 27 = -120.$$

As expected, the determinant is

$$-120.$$

Cramer's Rule for Three Variables

Consider the system

$$\begin{cases} a_1x + b_1y + c_1z = d_1, \\ a_2x + b_2y + c_2z = d_2, \\ a_3x + b_3y + c_3z = d_3. \end{cases}$$

Define

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}.$$

Then

$$D_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix},$$

$$D_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix},$$

and

$$D_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}.$$

Provided

$$D \neq 0,$$

Cramer's Rule gives

$$\boxed{x = \frac{D_x}{D}, \qquad y = \frac{D_y}{D}, \qquad z = \frac{D_z}{D}.}$$

Example. Solve the system using Cramer's Rule:

$$\begin{cases} x + 5y - z = 5, \\ 2x - 5y + z = 1, \\ -3x + 5y - 5z = 1. \end{cases}$$

Solution. The coefficient determinant is

$$D = \begin{vmatrix} 1 & 5 & -1 \\ 2 & -5 & 1 \\ -3 & 5 & -5 \end{vmatrix}.$$

Expanding along the first row,

$$\begin{aligned} D &= 1\begin{vmatrix} -5 & 1 \\ 5 & -5 \end{vmatrix} - 5\begin{vmatrix} 2 & 1 \\ -3 & -5 \end{vmatrix} - \begin{vmatrix} 2 & -5 \\ -3 & 5 \end{vmatrix} \\ &= 1(25-5) - 5(-10+3) - (10-15) \\ &= 20 + 35 + 5 \\ &= 60. \end{aligned}$$

Since

$$D = 60 \neq 0,$$

the system has a unique solution.

For x,

$$D_x = \begin{vmatrix} 5 & 5 & -1 \\ 1 & -5 & 1 \\ 1 & 5 & -5 \end{vmatrix}.$$

Evaluating,

$$D_x = 120.$$

Thus,

$$x = \frac{D_x}{D} = \frac{120}{60} = 2.$$

For y,

$$D_y = \begin{vmatrix} 1 & 5 & -1 \\ 2 & 1 & 1 \\ -3 & 1 & -5 \end{vmatrix}.$$

Evaluating,

$$D_y = 24.$$

Thus,

$$y = \frac{D_y}{D} = \frac{24}{60} = \frac{2}{5}.$$

For z,

$$D_z = \begin{vmatrix} 1 & 5 & 5 \\ 2 & -5 & 1 \\ -3 & 5 & 1 \end{vmatrix}.$$

Evaluating,

$$D_z = -60.$$

Therefore,

$$z = \frac{D_z}{D} = \frac{-60}{60} = -1.$$

Hence,

$$(x, y, z) = \left(2, \frac{2}{5}, -1\right).$$

Solving Equations Involving Determinants

Example. Solve for y:

$$\begin{vmatrix} -2 & 3 & 0 \\ y & 1 & 2 \\ -1 & 1 & 3 \end{vmatrix} = -53.$$

Solution. Expand along the first row:

$$\begin{vmatrix} -2 & 3 & 0 \\ y & 1 & 2 \\ -1 & 1 & 3 \end{vmatrix} = -2\begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} - 3\begin{vmatrix} y & 2 \\ -1 & 3 \end{vmatrix} + 0.$$

Compute the 2×2 determinants:

$$\begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} = 3 - 2 = 1,$$

and

$$\begin{vmatrix} y & 2 \\ -1 & 3 \end{vmatrix} = 3y + 2.$$

Therefore,

$$-2(1) - 3(3y + 2) = -53.$$

Simplifying,

$$-2 - 9y - 6 = -53,$$

so

$$-9y - 8 = -53.$$

Thus,

$$-9y = -45,$$

and therefore,

$$y = 5.$$

Using Technology

Most graphing calculators and computer algebra systems have a determinant command, often written as

$$\det(A).$$

Technology is useful for checking a determinant calculation. However, for small determinants, especially 2×2 and 3×3 matrices, you should also be able to compute the determinant by hand.

10.1: Sequences and Series

Sequences

A **sequence** is a function whose domain is the set of positive integers

$$1, 2, 3, \ldots$$

and whose range is a subset of the real numbers.
A sequence may be viewed as an ordered pattern of numbers:

$$a_1, a_2, a_3, \ldots$$

The notation

$$a_n$$

represents the nth term of the sequence.

A **finite sequence** has a finite number of terms; that is, it has both a first term and a last term.
For example,

$$2, 4, 6, 8, 10$$

is a finite sequence.

An **infinite sequence** has infinitely many terms and therefore has no last term.
For example,

$$2, 4, 6, 8, 10, \ldots$$

is an infinite sequence.
Example. Write the first four terms of the sequence

$$a_n = (-1)^n(3n).$$

Solution. Substitute

$$n = 1, 2, 3, 4.$$

For $n = 1$,

$$a_1 = (-1)^1(3) = -3.$$

For $n = 2$,

$$a_2 = (-1)^2(6) = 6.$$

For $n = 3$,

$$a_3 = (-1)^3(9) = -9.$$

For $n = 4$,

$$a_4 = (-1)^4(12) = 12.$$

Therefore, the first four terms are

$$-3,\ 6,\ -9,\ 12.$$

Example. Write the first four terms of the sequence

$$a_n = -\frac{3^n}{4n+1}.$$

Solution. Substitute

$$n = 1, 2, 3, 4.$$

We obtain

$$a_1 = -\frac{3}{5},$$

$$a_2 = -\frac{9}{9} = -1,$$

$$a_3 = -\frac{27}{13},$$

and

$$a_4 = -\frac{81}{17}.$$

Therefore, the first four terms are

$$-\frac{3}{5},\ -1,\ -\frac{27}{13},\ -\frac{81}{17}.$$

Example. For the sequence

$$a_n = \frac{(-1)^n}{(n+3)(n+6)},$$

find a_3 and a_6.

Solution. For $n = 3$,

$$a_3 = \frac{(-1)^3}{(3+3)(3+6)} = -\frac{1}{54}.$$

For $n = 6$,

$$a_6 = \frac{(-1)^6}{(6+3)(6+6)} = \frac{1}{108}.$$

Therefore,

$$a_3 = -\frac{1}{54}, \qquad a_6 = \frac{1}{108}.$$

Recursive Sequences

A **recursive sequence** is a sequence in which the first term, or first several terms, are given and each subsequent term is defined using one or more preceding terms.

Example. Suppose

$$a_1 = 3, \qquad a_2 = 4,$$

and

$$a_n = a_{n-1} - a_{n-2}, \qquad n \geq 3.$$

Find the first four terms.

Solution. The first two terms are given:

$$a_1 = 3, \qquad a_2 = 4.$$

For the third term,

$$a_3 = a_2 - a_1 = 4 - 3 = 1.$$

For the fourth term,

$$a_4 = a_3 - a_2 = 1 - 4 = -3.$$

Therefore, the first four terms are

$$3,\ 4,\ 1,\ -3.$$

Example. Find the first four terms of the recursively defined sequence

$$a_1 = 2, \qquad a_n = \frac{a_{n-1}}{n^2}, \qquad n \geq 2.$$

Solution. The first term is

$$a_1 = 2.$$

For $n = 2$,

$$a_2 = \frac{a_1}{2^2} = \frac{2}{4} = \frac{1}{2}.$$

For $n = 3$,

$$a_3 = \frac{a_2}{3^2} = \frac{1/2}{9} = \frac{1}{18}.$$

For $n = 4$,

$$a_4 = \frac{a_3}{4^2} = \frac{1/18}{16} = \frac{1}{288}.$$

Therefore, the first four terms are

$$2,\ \frac{1}{2},\ \frac{1}{18},\ \frac{1}{288}.$$

Factorials

For any positive integer n, the **factorial** of n is defined by

$$n! = n(n-1)(n-2)\cdots(3)(2)(1).$$

By definition,

$$0! = 1 \qquad \text{and} \qquad 1! = 1.$$

Example. Evaluate

$$5!.$$

Solution. By the definition of factorial,

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120.$$

Example. Simplify

$$\frac{11!\,4!}{2!\,15!}.$$

Solution. Write

$$15! = 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11!.$$

Therefore,

$$\frac{11!\,4!}{2!\,15!} = \frac{11!\,(4 \cdot 3 \cdot 2 \cdot 1)}{2(15 \cdot 14 \cdot 13 \cdot 12)11!}.$$

Cancel 11!:

$$= \frac{24}{2(15)(14)(13)(12)}.$$

Thus,

$$= \frac{12}{15 \cdot 14 \cdot 13 \cdot 12} = \frac{1}{15 \cdot 14 \cdot 13}.$$

Hence,

$$\frac{11!\,4!}{2!\,15!} = \frac{1}{2730}.$$

Series and Partial Sums

A **series** is the sum of the terms of a sequence.

A **finite series** is the sum of all the terms of a finite sequence.

The n**th partial sum**, denoted by

$$S_n,$$

is the sum of the first n terms of a sequence:

$$S_n = a_1 + a_2 + \cdots + a_n.$$

Summation Notation

The Greek capital letter sigma,

$$\sum,$$

is used to represent a sum.

In the expression

$$\sum_{i=1}^{n} a_i,$$

i is called the **index of summation**, 1 is the lower limit, and n is the upper limit.

It means

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n.$$

Example. Evaluate

$$\sum_{m=1}^{5} \frac{m}{2}.$$

Solution. Expand the sum:

$$\sum_{m=1}^{5} \frac{m}{2} = \frac{1}{2} + \frac{2}{2} + \frac{3}{2} + \frac{4}{2} + \frac{5}{2}.$$

Therefore,

$$= \frac{1+2+3+4+5}{2} = \frac{15}{2}.$$

Example. Expand and evaluate

$$\sum_{i=3}^{6} \left(2i + (i-3)!\right).$$

Solution. Substitute

$$i = 3, 4, 5, 6.$$

Thus,

$$\begin{aligned}\sum_{i=3}^{6} (2i + (i-3)!) &= (2(3) + 0!) + (2(4) + 1!) \\ &\quad + (2(5) + 2!) + (2(6) + 3!)\,.\end{aligned}$$

Since

$$0! = 1, \qquad 1! = 1, \qquad 2! = 2, \qquad 3! = 6,$$

we obtain

$$(6+1) + (8+1) + (10+2) + (12+6).$$

Therefore,

$$7 + 9 + 12 + 18 = 46.$$

Properties of Summation

Let c be a real number and n a positive integer.

$$\boxed{\sum_{i=1}^{n} c = cn}$$

That is, adding the same constant c a total of n times gives cn.

A constant factor can be factored out of a summation:

$$\boxed{\sum_{i=1}^{n} ca_i = c\sum_{i=1}^{n} a_i}$$

A summation can be distributed across a sum or difference:

$$\boxed{\sum_{i=1}^{n}(a_i \pm b_i) = \sum_{i=1}^{n} a_i \pm \sum_{i=1}^{n} b_i}$$

A summation can also be separated into smaller pieces:

$$\boxed{\sum_{i=1}^{m} a_i + \sum_{i=m+1}^{n} a_i = \sum_{i=1}^{n} a_i}$$

Useful Summation Formulas

The following formulas are useful when evaluating finite sums:

$$\boxed{\sum_{i=1}^{n} i = \frac{n(n+1)}{2}}$$

$$\boxed{\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}}$$

and

$$\boxed{\sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2.}$$

Example. Find the sum

$$\sum_{k=1}^{5}(4k+9).$$

Solution. Use the properties of summation:

$$\sum_{k=1}^{5}(4k+9) = 4\sum_{k=1}^{5} k + \sum_{k=1}^{5} 9.$$

Since

$$\sum_{k=1}^{5} k = \frac{5(5+1)}{2} = 15$$

and

$$\sum_{k=1}^{5} 9 = 5(9) = 45,$$

we have

$$4(15) + 45 = 60 + 45 = 105.$$

Therefore,

$$\sum_{k=1}^{5}(4k+9) = 105.$$

10.2: Arithmetic Sequences

Arithmetic Sequences

An **arithmetic sequence** is a sequence in which the difference between consecutive terms is constant. This constant is called the **common difference**.

Recursively, an arithmetic sequence can be written as

$$a_1 = a, \qquad a_n = a_{n-1} + d, \qquad n \geq 2,$$

where a is the first term and d is the common difference.

The common difference can be found by subtracting consecutive terms:

$$d = a_n - a_{n-1}.$$

Example. Find the first term and the common difference of the arithmetic sequence

$$3,\ 1,\ -1, \ldots$$

Solution. The first term is

$$a_1 = 3.$$

The common difference is

$$d = 1 - 3 = -2.$$

Checking the next two terms,

$$-1 - 1 = -2.$$

Therefore,

$$a_1 = 3, \qquad d = -2.$$

Example. Find the first term and the common difference of the arithmetic sequence

$$a_n = 2n + 8.$$

Solution. The first term is obtained by setting $n = 1$:

$$a_1 = 2(1) + 8 = 10.$$

To find the common difference, compute two consecutive terms:

$$a_2 = 2(2) + 8 = 12.$$

Thus,

$$d = a_2 - a_1 = 12 - 10 = 2.$$

Therefore,

$$a_1 = 10, \qquad d = 2.$$

The nth Term of an Arithmetic Sequence

For an arithmetic sequence with first term a_1 and common difference d, the nth term is

$$\boxed{a_n = a_1 + (n-1)d.}$$

This formula allows us to find any term of an arithmetic sequence without listing all of the preceding terms.

Example. Find the nth term of the arithmetic sequence if the initial term is

$$a_1 = 5$$

and the common difference is

$$d = 8.$$

Solution. Use

$$a_n = a_1 + (n-1)d.$$

Substituting $a_1 = 5$ and $d = 8$ gives

$$a_n = 5 + (n-1)8.$$

Simplifying,

$$a_n = 5 + 8n - 8 = 8n - 3.$$

Therefore,

$$a_n = 8n - 3.$$

Example. Find the fifth term of the sequence from the previous example.

Solution. Since

$$a_n = 8n - 3,$$

we have

$$a_5 = 8(5) - 3 = 37.$$

Therefore,

$$a_5 = 37.$$

Example. Find the indicated term of the arithmetic sequence:

$$\text{7th term of } -2, -7, -12, \ldots$$

Solution. The first term is

$$a_1 = -2.$$

The common difference is

$$d = -7 - (-2) = -5.$$

Using

$$a_n = a_1 + (n-1)d,$$

we obtain

$$a_7 = -2 + (7-1)(-5).$$

Thus,

$$a_7 = -2 - 30 = -32.$$

Therefore,

$$a_7 = -32.$$

Example. An arithmetic sequence satisfies

$$a_5 = 4, \qquad a_{17} = 28.$$

Find the first term and the common difference.

Solution. Using

$$a_n = a_1 + (n-1)d,$$

the two given terms produce

$$a_5 = a_1 + 4d = 4$$

and

$$a_{17} = a_1 + 16d = 28.$$

Thus, we have the system

$$\begin{cases} a_1 + 4d = 4, \\ a_1 + 16d = 28. \end{cases}$$

Subtracting the first equation from the second gives

$$12d = 24,$$

so

$$d = 2.$$

Substitute $d = 2$ into

$$a_1 + 4d = 4 :$$

$$a_1 + 8 = 4,$$

and therefore

$$a_1 = -4.$$

Thus,

$$a_1 = -4, \qquad d = 2.$$

Example. For the sequence in the previous example, find the 20th term.

Solution. We found

$$a_1 = -4, \qquad d = 2.$$

Using the nth-term formula,

$$a_{20} = a_1 + (20-1)d.$$

Therefore,

$$a_{20} = -4 + 19(2) = -4 + 38 = 34.$$

Thus,

$$a_{20} = 34.$$

Partial Sums of Arithmetic Sequences

Suppose an arithmetic sequence has first term a_1 and nth term a_n. The sum of the first n terms is called the n**th partial sum** and is denoted by S_n.

The nth partial sum of an arithmetic sequence is

$$\boxed{S_n = \frac{n(a_1 + a_n)}{2}.}$$

Equivalently,

$$S_n = \frac{n}{2}(a_1 + a_n).$$

Since

$$a_n = a_1 + (n-1)d,$$

the partial-sum formula may also be written as

$$\boxed{S_n = \frac{n}{2}\left[2a_1 + (n-1)d\right].}$$

Example. Compute the sum

$$30 + 35 + 40 + \cdots + 505.$$

Solution. This is an arithmetic series with

$$a_1 = 30, \qquad d = 5, \qquad a_n = 505.$$

First determine the number of terms. Using

$$a_n = a_1 + (n-1)d,$$

we have

$$505 = 30 + (n-1)5.$$

Thus,

$$475 = 5(n-1),$$

so

$$95 = n - 1$$

and therefore

$$n = 96.$$

Now use the partial-sum formula:

$$S_n = \frac{n(a_1 + a_n)}{2}.$$

Hence,

$$S_{96} = \frac{96(30 + 505)}{2}.$$

Therefore,

$$S_{96} = 48(535) = 25{,}680.$$

Thus,

$$30 + 35 + 40 + \cdots + 505 = 25{,}680.$$

Example. Compute the sum

$$\sum_{n=1}^{30}(2n-1).$$

Solution. The terms generated by

$$a_n = 2n - 1$$

form the arithmetic sequence

$$1,\ 3,\ 5, \ldots$$

The first term is

$$a_1 = 1,$$

and the 30th term is

$$a_{30} = 2(30) - 1 = 59.$$

Using the arithmetic-series formula,

$$S_{30} = \frac{30(a_1 + a_{30})}{2}.$$

Thus,

$$S_{30} = \frac{30(1+59)}{2} = 15(60) = 900.$$

Therefore,

$$\sum_{n=1}^{30}(2n-1) = 900.$$

10.3: Geometric Sequences and Series

Geometric Sequences

A **geometric sequence** is a sequence in which each term is obtained by multiplying the preceding term by the same nonzero constant, called the **common ratio**.

Recursively, a geometric sequence can be written as

$$a_1 = a, \qquad a_n = a_{n-1}r, \qquad n \geq 2,$$

where a_1 is the first term and r is the common ratio.

The common ratio can be found using

$$\boxed{r = \frac{a_n}{a_{n-1}}}$$

provided $a_{n-1} \neq 0$.

The nth term of a geometric sequence is

$$\boxed{a_n = a_1 r^{n-1}.}$$

Example. Determine the first term and the common ratio of the geometric sequence

$$2,\ 6,\ 18, \ldots$$

Solution. The first term is

$$a_1 = 2.$$

The common ratio is

$$r = \frac{6}{2} = 3.$$

Checking the next two terms,

$$\frac{18}{6} = 3.$$

Therefore,

$$a_1 = 2, \qquad r = 3.$$

Arithmetic, Geometric, or Neither

To determine whether a sequence is arithmetic or geometric:

- Check whether consecutive terms have a constant **difference**. If so, the sequence is arithmetic.
- Check whether consecutive nonzero terms have a constant **ratio**. If so, the sequence is geometric.
- If neither is constant, the sequence is neither arithmetic nor geometric.

Example. Determine whether each sequence is arithmetic, geometric, or neither.

(a) $1,\ 5,\ 10,\ 16, \ldots$

$$\text{(b)}\quad \frac{3}{5},\ -\frac{9}{25},\ \frac{27}{125},\ldots$$

Solution. (a) Compute consecutive differences:

$$5-1=4,$$

$$10-5=5,$$

and

$$16-10=6.$$

The differences are not constant, so the sequence is not arithmetic.

Now check the ratios:

$$\frac{5}{1}=5, \qquad \frac{10}{5}=2.$$

The ratios are also not constant. Therefore, the sequence is

$$\text{neither arithmetic nor geometric.}$$

(b) Compute consecutive ratios:

$$\frac{-9/25}{3/5}=-\frac{3}{5},$$

and

$$\frac{27/125}{-9/25}=-\frac{3}{5}.$$

Thus, the ratio is constant. Therefore, the sequence is geometric with

$$r=-\frac{3}{5}.$$

Example. Find a formula for a geometric sequence with

$$a_1=-3, \qquad r=2.$$

Solution. Use the nth-term formula

$$a_n=a_1r^{n-1}.$$

Substituting

$$a_1=-3 \qquad \text{and} \qquad r=2,$$

we obtain

$$a_n=-3(2)^{n-1}.$$

Therefore,

$$\boxed{a_n=-3\cdot 2^{n-1}.}$$

Example. Find the ninth term of the sequence from the previous example.

Solution. Using

$$a_n=-3\cdot 2^{n-1},$$

we have

$$a_9=-3\cdot 2^8.$$

Thus,

$$a_9=-3(256)=-768.$$

Therefore,

$$a_9=-768.$$

Partial Sums of Geometric Sequences

For a geometric sequence with first term a_1 and common ratio r, where

$$r \neq 1,$$

the sum of the first n terms is

$$\boxed{S_n = a_1 \frac{1 - r^n}{1 - r}.}$$

An equivalent form is

$$\boxed{S_n = a_1 \frac{r^n - 1}{r - 1}.}$$

Example. Compute the sum

$$-2 - 2(5) - 2(5)^2 + \cdots - 2(5)^{10}.$$

Solution. This is a geometric series with

$$a_1 = -2, \qquad r = 5.$$

The powers of 5 run from

$$5^0 \quad \text{through} \quad 5^{10},$$

so there are

$$11$$

terms.

Thus,

$$S_{11} = -2 \frac{1 - 5^{11}}{1 - 5}.$$

Therefore,

$$S_{11} = -2 \frac{1 - 5^{11}}{-4} = \frac{1 - 5^{11}}{2}.$$

Since

$$5^{11} = 48{,}828{,}125,$$

we have

$$S_{11} = \frac{1 - 48{,}828{,}125}{2}.$$

Thus,

$$S_{11} = -24{,}414{,}062.$$

Example. Find

$$\sum_{n=1}^{10} 2^n.$$

Solution. The terms are

$$2^1, 2^2, \ldots, 2^{10},$$

so

$$a_1 = 2, \qquad r = 2, \qquad n = 10.$$

Using

$$S_n = a_1 \frac{1-r^n}{1-r},$$

we obtain

$$S_{10} = 2\frac{1-2^{10}}{1-2}.$$

Therefore,

$$S_{10} = 2(2^{10}-1).$$

Since

$$2^{10} = 1024,$$

we obtain

$$S_{10} = 2(1023) = 2046.$$

Thus,

$$\sum_{n=1}^{10} 2^n = 2046.$$

Example. For a geometric sequence,

$$a_3 = 45 \qquad \text{and} \qquad a_6 = -\frac{243}{25}.$$

Find the common ratio, the first term, and the 13th term.

Solution. Using

$$a_n = a_1 r^{n-1},$$

we have

$$a_3 = a_1 r^2 = 45$$

and

$$a_6 = a_1 r^5 = -\frac{243}{25}.$$

Divide the second equation by the first:

$$\frac{a_6}{a_3} = r^3.$$

Thus,

$$r^3 = \frac{-243/25}{45}.$$

Simplifying,

$$r^3 = -\frac{243}{1125} = -\frac{27}{125}.$$

Therefore,

$$r = -\frac{3}{5}.$$

Now use

$$a_3 = a_1 r^2.$$

Then

$$45 = a_1 \left(-\frac{3}{5}\right)^2 = a_1 \frac{9}{25}.$$

Thus,

$$a_1 = 45 \cdot \frac{25}{9} = 125.$$

Now find the 13th term:

$$a_{13} = a_1 r^{12}.$$

Therefore,

$$a_{13} = 125\left(-\frac{3}{5}\right)^{12}.$$

Since the exponent is even,

$$a_{13} = 125\left(\frac{3}{5}\right)^{12}.$$

Hence,

$$a_{13} = \frac{531441}{1953125}.$$

Therefore,

$$\boxed{r = -\frac{3}{5}, \qquad a_1 = 125, \qquad a_{13} = \frac{531441}{1953125}.}$$

Infinite Geometric Series

An **infinite geometric series** has the form

$$a_1 + a_1 r + a_1 r^2 + \cdots + a_1 r^{n-1} + \cdots$$

or, in summation notation,

$$\sum_{n=1}^{\infty} a_1 r^{n-1}.$$

For an infinite geometric series, the behavior depends on the absolute value of the common ratio.

Example. Consider the infinite geometric series

$$1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots.$$

Find the common ratio and determine whether the terms are getting larger or smaller.

Solution. The common ratio is

$$r = \frac{1/3}{1} = \frac{1}{3}.$$

Since

$$|r| = \frac{1}{3} < 1,$$

the terms decrease in magnitude and approach 0.

Thus,

$$r = \frac{1}{3},$$

and the terms are getting smaller.

Example. Consider the infinite geometric series

$$2 + 4 + 8 + 16 + \cdots.$$

Find the common ratio and determine whether the terms are getting larger or smaller.

Solution. The common ratio is

$$r = \frac{4}{2} = 2.$$

Since

$$|r| = 2 > 1,$$

the terms increase in magnitude.

Therefore,

$$r = 2,$$

and the terms are getting larger.

Convergence of an Infinite Geometric Series

For the finite geometric sum,

$$S_n = a_1 \frac{1 - r^n}{1 - r}.$$

If

$$|r| < 1,$$

then

$$r^n \to 0 \qquad \text{as} \qquad n \to \infty.$$

Therefore,

$$S_\infty = \lim_{n\to\infty} S_n = \frac{a_1}{1 - r}.$$

Hence,

$$\boxed{\sum_{n=1}^{\infty} a_1 r^{n-1} = \frac{a_1}{1 - r}, \qquad |r| < 1.}$$

If

$$|r| \geq 1,$$

the infinite geometric series does not have a finite sum; the series **diverges**.

Example. Find the sum of the infinite geometric series

$$4 - \frac{1}{4} + \frac{1}{64} - \frac{1}{1024} + \cdots.$$

Solution. The first term is

$$a_1 = 4.$$

The common ratio is

$$r = \frac{-1/4}{4} = -\frac{1}{16}.$$

Check the next ratio:

$$\frac{1/64}{-1/4} = -\frac{1}{16}.$$

Since

$$\left|-\frac{1}{16}\right| < 1,$$

the series converges.
Use

$$S_\infty = \frac{a_1}{1-r}.$$

Thus,

$$S_\infty = \frac{4}{1-(-1/16)} = \frac{4}{17/16}.$$

Therefore,

$$S_\infty = 4 \cdot \frac{16}{17} = \frac{64}{17}.$$

Example. Compute

$$\sum_{i=1}^{\infty} 3(0.3)^i.$$

Solution. The first term corresponds to $i = 1$:

$$a_1 = 3(0.3) = 0.9.$$

The common ratio is

$$r = 0.3.$$

Since

$$|r| = 0.3 < 1,$$

the series converges.
Therefore,

$$S_\infty = \frac{0.9}{1-0.3} = \frac{0.9}{0.7} = \frac{9}{7}.$$

Thus,

$$\sum_{i=1}^{\infty} 3(0.3)^i = \frac{9}{7}.$$

Example. Determine whether

$$\sum_{i=1}^{\infty} 3^i$$

converges or diverges.
Solution. The first term is

$$a_1 = 3,$$

and the common ratio is

$$r = 3.$$

Since

$$|r| = 3 > 1,$$

the infinite geometric series diverges.

Therefore,

$$\sum_{i=1}^{\infty} 3^i$$

has no finite sum.

10.7: The Binomial Theorem

The Binomial Theorem

Consider the first few powers of a binomial:

$$(x+a)^1 = x + a,$$
$$(x+a)^2 = x^2 + 2ax + a^2,$$
$$(x+a)^3 = x^3 + 3ax^2 + 3a^2x + a^3,$$

and

$$(x+a)^4 = x^4 + 4ax^3 + 6a^2x^2 + 4a^3x + a^4.$$

Notice that the coefficients in these expansions follow a pattern. These coefficients can be generated using **binomial coefficients**.

Binomial Coefficients

If n and j are nonnegative integers with

$$0 \leq j \leq n,$$

then the binomial coefficient is defined by

$$\boxed{\binom{n}{j} = \frac{n!}{j!(n-j)!}.}$$

The notation

$$\binom{n}{j}$$

is read "n choose j."

Two useful facts are

$$\binom{n}{0} = 1 \qquad \text{and} \qquad \binom{n}{n} = 1.$$

Example. Evaluate

$$\binom{6}{3}.$$

Solution. Using the definition,

$$\binom{6}{3} = \frac{6!}{3!(6-3)!} = \frac{6!}{3!3!}.$$

Thus,

$$\binom{6}{3} = \frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} = 20.$$

Pascal's Triangle

The binomial coefficients can also be obtained from **Pascal's Triangle**:

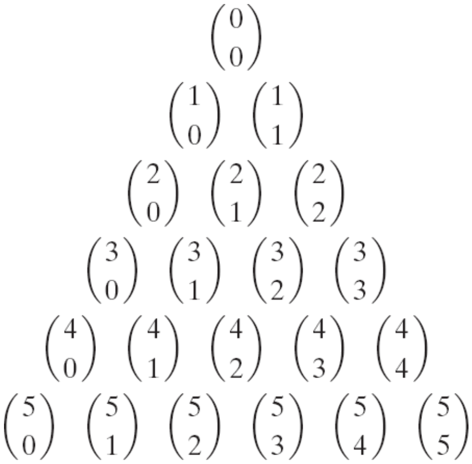


Figure 63: Pascal triangle written with binomial coefficients

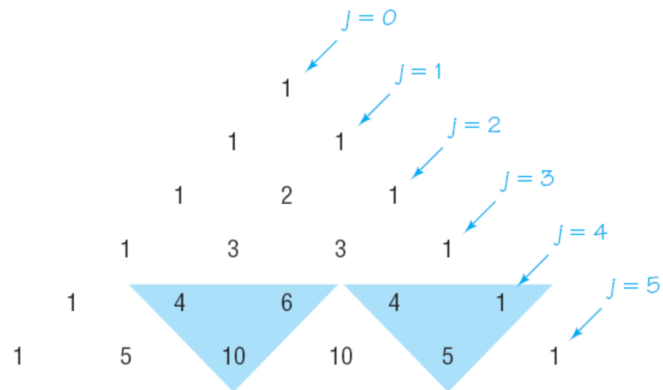


Figure 64: Pascal triangle numerical entries

Each interior entry is obtained by adding the two entries directly above it.
For example,

$$1,\ 4,\ 6,\ 4,\ 1$$

are the coefficients in the expansion of a fourth power, while

$$1,\ 5,\ 10,\ 10,\ 5,\ 1$$

are the coefficients in the expansion of a fifth power.

The Binomial Theorem

For any binomial $(a+b)$ and any nonnegative integer n,

$$\boxed{(a+b)^n = \sum_{j=0}^{n} \binom{n}{j} a^{n-j} b^j.}$$

Written term by term,

$$\begin{aligned}(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots \\ + \binom{n}{n-1}ab^{n-1} + \binom{n}{n}b^n.\end{aligned}$$

Notice that:

- the exponent on a decreases from n to 0;
- the exponent on b increases from 0 to n;

- the sum of the exponents in every term is n.

Example. Expand

$$(2x-3)^6$$

using the Binomial Theorem.

Solution. Using

$$(a+b)^n = \sum_{j=0}^{n} \binom{n}{j} a^{n-j} b^j,$$

take

$$a = 2x, \qquad b = -3, \qquad n = 6.$$

Therefore,

$$\begin{aligned}(2x-3)^6 = \binom{6}{0}(2x)^6 + \binom{6}{1}(2x)^5(-3) + \binom{6}{2}(2x)^4(-3)^2 \\ + \binom{6}{3}(2x)^3(-3)^3 + \binom{6}{4}(2x)^2(-3)^4 \\ + \binom{6}{5}(2x)(-3)^5 + \binom{6}{6}(-3)^6.\end{aligned}$$

Using the coefficients

$$1,\ 6,\ 15,\ 20,\ 15,\ 6,\ 1,$$

we obtain

$$\begin{aligned}(2x-3)^6 = 64x^6 - 576x^5 + 2160x^4 - 4320x^3 \\ + 4860x^2 - 2916x + 729.\end{aligned}$$

Finding a Particular Term

The general term in the expansion of

$$(a+b)^n$$

is

$$T_{j+1} = \binom{n}{j} a^{n-j} b^j.$$

Thus, the kth term is

$$\boxed{T_k = \binom{n}{k-1} a^{n-(k-1)} b^{k-1}.}$$

In other words,

$$j = k-1.$$

Example. Find the coefficient of the ninth term in the expansion of

$$(3x+1)^{12}.$$

Solution. For the ninth term,

$$k = 9,$$

so
$$j = k - 1 = 8.$$
The ninth term is
$$T_9 = \binom{12}{8}(3x)^{12-8}(1)^8.$$
Thus,
$$T_9 = \binom{12}{8}(3x)^4.$$
Since
$$\binom{12}{8} = \binom{12}{4} = 495$$
and
$$3^4 = 81,$$
we have
$$T_9 = 495(81)x^4.$$
Therefore,
$$T_9 = 40095x^4,$$
so the coefficient of the ninth term is
$$40095.$$

Example. Find the x^3y^4 term in the expansion of
$$(2x - 3y)^7.$$

Solution. The general term is
$$T_{j+1} = \binom{7}{j}(2x)^{7-j}(-3y)^j.$$
To obtain a term containing y^4, we need
$$j = 4.$$
Then
$$7 - j = 3,$$
so the corresponding power of x is x^3, as required.
Therefore,
$$T_5 = \binom{7}{4}(2x)^3(-3y)^4.$$
Now,
$$\binom{7}{4} = 35, \qquad (2x)^3 = 8x^3, \qquad (-3y)^4 = 81y^4.$$
Thus,
$$T_5 = 35(8)(81)x^3y^4.$$
Therefore,
$$T_5 = 22680x^3y^4.$$

Example. Find the first three terms of

$$(5a - 2b^3)^9.$$

Solution. The general term is

$$T_{j+1} = \binom{9}{j}(5a)^{9-j}(-2b^3)^j.$$

For the first term, $j = 0$:

$$T_1 = \binom{9}{0}(5a)^9 = 5^9 a^9.$$

Thus,

$$T_1 = 1953125a^9.$$

For the second term, $j = 1$:

$$T_2 = \binom{9}{1}(5a)^8(-2b^3).$$

Hence,

$$T_2 = 9(390625a^8)(-2b^3) = -7031250a^8b^3.$$

For the third term, $j = 2$:

$$T_3 = \binom{9}{2}(5a)^7(-2b^3)^2.$$

Since

$$\binom{9}{2} = 36, \qquad 5^7 = 78125, \qquad (-2)^2 = 4,$$

we obtain

$$T_3 = 36(78125)(4)a^7b^6.$$

Thus,

$$T_3 = 11250000a^7b^6.$$

Therefore, the first three terms are

$$1953125a^9 - 7031250a^8b^3 + 11250000a^7b^6.$$